



Technical University
of Denmark

Automated Gantry-Based Disassembly System for Remanufacturing

Bachelor Thesis

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Abstract

Disassembly in remanufacturing remains largely manual due to the high variability of end-of-life products, creating a need for flexible semi-automated solutions. A gantry-based disassembly system was developed incorporating a unified module capable of performing both fastener removal and axial press operations through a single toolhead, supported by camera-based screw localisation, laser-based contact detection, force monitoring, and passive automatic bit changing. Across the final ten full-system validation runs, the system achieved an operational success rate of 99.29% and a classification accuracy of 99.38%, both well within the 5% acceptance criteria. The results demonstrate that the unified module concept is mechanically and functionally feasible, though advancement toward industrial deployment will require structural redesign and validation on real end-of-life components.

1 Introduction

1.1 Background and Challenges in Remanufacturing

Remanufacturing is a production paradigm focused on recovering value from end of life (EoL) products through processes such as disassembly, reuse, and reassembly to “as new” condition. Instead of treating used products as waste, valuable components and materials are reintroduced into the value chain, reducing both environmental impact and resource consumption. The economic case for remanufacturing is well established. Studies have consistently shown that companies that remanufactured products can be produced at 40–65% of the cost of new equivalents while retaining equivalent performance [4]. As industries move towards more sustainable production systems, remanufacturing has gained increasing attention as a key enabler for circular economy principles [3].

A fundamental challenge in remanufacturing is the high variability and uncertainty associated with incoming products. Unlike traditional manufacturing, where components are standardised and well defined, products entering a disassembly process may differ significantly in geometry, condition, and assembly methods. Factors such as wear, corrosion, deformation, and unknown fastening conditions introduce additional complexity. As a result, designing automated systems for disassembly is inherently difficult, as such systems must be capable of handling a wide range of unpredictable scenarios while maintaining reliability.

1.2 Existing Approaches and Identified Research Gap

Existing approaches to automated disassembly

A range of approaches have been proposed to address the challenge of automating disassembly in remanufacturing contexts. A comprehensive overview is given in [9], while the most relevant categories for this project are outlined below.

The first is fully dedicated robotic cells, where a fixed robot is programmed to disassemble a specific product along a predetermined sequence. This approach has been demonstrated for relatively standardised products such as industrial electronic boards used in actuators and elevator control systems [8]. While such systems can achieve efficient operation for products with predictable geometry and disassembly sequences, their flexibility is limited when handling the large variation typically associated with EoL products in remanufacturing contexts.

The second is human robot collaboration (HRC), where robots handle repetitive or physically demanding subtasks while human operators manage uncertain or complex steps. This hybrid model improves flexibility and relies on safe and reliable human robot interaction [5].

The third is the use of a unified multi function module, where a single toolhead is designed to perform multiple distinct operations, for example both fastener removal and axial press operations, without requiring a tool change between them. Compared to dedicated single function tools, unified modules reduce cycle time and mechanical complexity

at the cost of stricter design constraints, since one assembly must simultaneously satisfy the requirements of functionally distinct operations. Despite this tradeoff, unified modules represent a promising direction for semi automated disassembly systems where flexibility and compactness are prioritised over peak performance in any single operation.

Existing work on unified modules and screw localisation

To date, work on unified modules has been narrow in scope. The closest example is the multi head tool developed by Chen et al. [1], which combines a screwdriver, hole saw, and angle grinder on a single robot wrist for LCD screen disassembly. While their system does include a non destructive unscrewing skill, the overall process is semi destructive. The hole saw and angle grinder cut through covers and fasteners, sacrificing those parts of the assembly to reach the components beneath. In a remanufacturing context, where the value of the process depends on recovering as many components intact as possible, a fully non destructive approach is preferable. The combination relevant to this setting, namely unscrewing and axial pressing integrated into a single gantry mounted toolhead, remains underexplored. This gap is the starting point for the present project.

Recent research has also investigated software based solutions for automated screw detection in disassembly systems. One notable example is the work by Yildiz and Wörgötter [12], who proposed a screw detection framework combining classical computer vision techniques with deep convolutional neural networks for automated disassembly applications. Their approach first applies a Hough Circle Transform to identify circular screw candidates within RGB images before classifying these candidates using an integrated deep learning model. The corresponding implementation was released publicly as an open source repository [11], enabling further experimentation and development within automated disassembly research. The work demonstrates the growing interest in vision based screw localization as a key enabling technology for autonomous disassembly systems.

1.3 Problem Statement and Project Objective

Problem statement

Because fully automated robotic cells struggle with the variability of EoL products, and human robot collaboration carries significant promise. Only utilizing skilled operators, results in low efficiency, high labour costs, and limited scalability. What is needed is a middle ground: a system that automates the repetitive and well defined portions of disassembly while leaving room for human intervention when conditions exceed its capabilities. The unified module approach, applied within a gantry based batch context, has not previously been investigated as a candidate for this middle ground, despite its potential to combine mechanical compactness with operational flexibility.

Project context and objective

This project is carried out as part of ongoing research at the national centre for Remanufacturing (REMANUFACTORY) at DTU, and addresses the gap identified above through the development of a semi automated gantry based system for batch based disassembly. Batch based disassembly refers to the process of dismantling groups of identical or similar products sequentially, in order to improve efficiency while preserving flexibility for varying EoL conditions. The project specifically investigates the unified module approach: a single module mounted on the gantry toolhead that is capable of performing both fastener removal (unscrewing) and axial press operations without requiring a tool change between the two types of operations. This choice is motivated by the desire to minimise cycle time in a batch setting, where the same operations must be performed repeatedly across many identical products. Designing one mechanical assembly to satisfy both rotational torque transmission and axial force application simultaneously introduces inherent trade-offs, which are discussed in Section 6.1.

The system uses sensor feedback to evaluate conditions during operation, for example verifying screwdriver bit engagement with a fastener head, or detecting whether a press force exceeds the threshold for autonomous operation. When conditions exceed the system's capabilities, the affected component is flagged for manual intervention rather than forcing an unreliable outcome. This enables a practical balance between automation and human adaptability, where repetitive and well defined tasks are handled by the machine while uncertain cases are escalated.

Problem formulation

The central question of this project is: how can a semi automated gantry based system for batch based disassembly be designed to reliably identify, locate, and interact with fasteners and components under uncertain conditions, while maintaining flexibility and cost effectiveness compared to fully automated solutions?

2 Methodology

This section describes the process used to develop and evaluate the unified module prototype. The work proceeded as a sequence of stages, beginning with a functional decomposition of the problem stated in section 2.4, followed by sensor research, concept generation, mechanical design, electronics and software implementation, calibration, and validation testing. In practice these stages were not strictly sequential.

2.1 Prototype Scope and Development Approach

Scope of validation

The prototype was designed and built to answer a specific question: can the unified module concept, together with the chosen sensors and the proposed control logic, function as a coordinated system in practice? The aim was therefore to demonstrate functional and behavioural feasibility, not to deliver an optimised final product. This has direct consequences for what the prototype is and is not intended to prove. Most structural parts were 3D printed in PLA, which makes the load capacity considerably lower than what the same geometry could achieve in metal. The strength of the prototype is therefore not a meaningful measure of the strength of the concept. To make the path from prototype to production more direct, the mechanical design was intentionally constrained so that the same geometry could be realised in 10 mm metal plates with minimal modification. This means that strength improvements can be introduced without changing the spatial relationships between sensors and actuators, and the control logic developed and tested on the prototype should remain valid through the transition. Small geometric tweaks may still be needed, but the sensor positions and the resulting measurement geometry are preserved by construction.

The validation effort accordingly focused on the aspects of the design that the prototype can legitimately test: sensor choice, sensor placement, mechanical kinematics of the unified motion, and the reliability of the control logic that coordinates them. Load testing and final mechanical optimisation are explicitly outside the scope of this work and are deferred to a future iteration.

Functional decomposition

Before any mechanical or electronic decisions were made, the problem stated in section 1.3 was broken down into the discrete functions the system needed to perform. Treating each of these as a separate function ensured that the later subsystem choices could be evaluated against a concrete list of requirements rather than against a general goal. The required system functions can be viewed in section 2.4.

Sensor research and selection

Each function in the decomposition was mapped to a sensing requirement, and candidate sensors were identified for each. The main feedback channels needed were distance (for detecting surface contact and bit engagement), axial force (for the press operation), motor stall (for screw removal), and image (for screw and press localisation and sizing). Candidate technologies were compared on the basis of measurement range, resolution, response time, cost, and integration effort.

Concept generation and down selection

With the required functions and the supporting sensors fixed, several mechanical concepts for the unified module were sketched. Sketches can be viewed in Appendix E. The central question at this stage was how to realise both rotational torque transmission and axial press capability through a single driven shaft, while leaving room for the sensors to observe contact and engagement without interference. Concepts were evaluated against criteria derived from the requirements, in particular the need for passive transition between the two operating modes, the need for unobstructed sensor lines of sight, and the requirement that the swap to 10 mm manufactured plates would be a small change to the system.

2.2 Mechanical Development

Mechanical design

The selected concept was developed into a full CAD model. The design was driven by three priorities. **First**, the geometry had to place each sensor where it could observe the relevant physical event without interference from moving parts. **Second**, the load path for both rotational and axial loads had to be continuous and predictable, so that the load cell reading reflected the actual axial force rather than the sum of several unrelated stresses. **Third**, the design had to be compatible with 10 mm metal plate construction, as discussed in Section 2.1, so that the same geometry could be reproduced in a stronger final version without affecting sensor placement or control logic.

Most structural components were 3D printed for the prototype, with metal hardware used only for a subsection of custom parts designed in the project. This choice was made deliberately to keep iteration time low: parts that failed could be redesigned and reprinted within a day, which was important given that several design issues only became visible after assembly and testing.

2.3 Calibration and Validation Procedure

Sensor calibration

The laser distance sensor and the load cell required offset and gain calibration against known references before they could be used by the control logic. The camera required a separate calibration procedure, both for intrinsic parameters such as lens distortion and for the relationship between pixel coordinates and gantry coordinates, so that screws detected in an image could be visited reliably by the toolhead. The full calibration procedures are documented in Appendix [A](#) and section [4.3](#).

Validation testing

Testing was carried out on dedicated test objects with known screw geometries and known press fit interfaces. Using purpose built test objects rather than real disassembly targets meant that any failure could be attributed to the system itself rather than to product variation, which is essential for evaluating the unified module concept on its own terms. For each function defined in the functional decomposition, a behavioural criterion was fixed before testing began. For example the axial force hard stop was required to terminate the press within a defined force threshold. Defining these criteria in advance rather than after the fact ensured that the testing outcome reflected the system's actual behaviour rather than a post hoc interpretation of it. Where a function failed to meet its criterion, the cause was diagnosed and either the control logic, the calibration, or in some cases the mechanical geometry was adjusted, and the test was repeated. This loop continued until each function passed its criterion or until the remaining issue was traced to a limitation outside the prototype's scope, in which case it was recorded as a finding rather than as a fault.

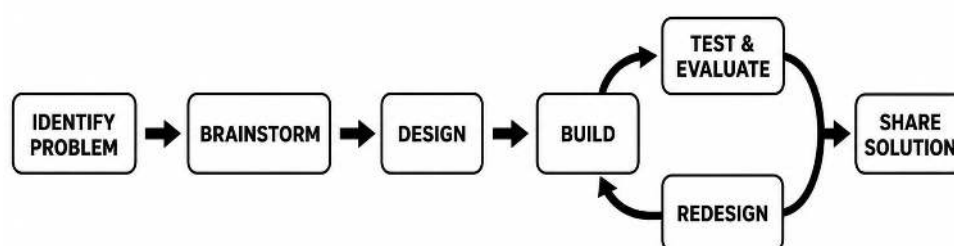


Figure 1: Sketch of the design iteration approach.

2.4 Deliverables

The full set of deliverable criteria evaluated in this work is listed below. Each item corresponds to a function from the decomposition above, and is referenced in the discussion section to determine whether the prototype met the criterion.

- **D1: Screw and press site localisation.** The camera based localisation must report machine frame coordinates for every screw and every press site within the workspace, with sufficient accuracy that the subsequent positioning step succeeds without manual correction.
- **D2: Screw size classification.** Each detected screw must be assigned to a size category from a single image captured at the fixed inspection height, and the classification must be correct for every screw in the validation batch.
- **D3: Toolhead positioning.** The gantry must position the driven shaft over a commanded coordinate with repeatability sufficient that the screwdriver bit aligns with the screw head on the first descent, without iterative search.
- **D4: Bit engagement detection.** The laser distance sensor must distinguish initial surface contact from the additional displacement caused by bit engagement, and the control logic must register engagement on every attempt across the validation batch.
- **D5: Screw removal.** Once engagement is detected, the screw motor must drive the fastener out of the workpiece while the Z axis follows the rising screw, and the sequence must complete without step loss or premature state machine advancement.
- **D6: Failure detection.** The system must detect both motor stall during removal and missed bit engagement during the descent, and must halt the active operation rather than continue blindly when either condition occurs.
- **D7: Axial press execution.** The unified module must transition passively from screw mode to press mode through the internal spring travel and deliver axial force through the same driven shaft, with no tool change or manual reconfiguration between the two operations.
- **D8: Axial force monitoring and hard stop.** The load cell must report continuous axial force during press operations, and a configurable hard stop force value must terminate the press immediately when exceeded, regardless of any other on-going condition.
- **D9: Passive bit changing.** The toolhead must release, store, and re engage bit adapters at the passive bit changer station using only the existing gantry axes and the screw motor, with hardcoded slot coordinates remaining valid across sessions.

- **D10: Operator logging and reporting.** The software must record the XY coordinate of every failed operation attempt, and must report the total count at the end of each batch run with a maximum classification error below 5%.
- **D11: Overall operation failure rate must be below 5%.**

3 Design Choices and Technical Decisions

3.1 Gantry System Design

The gantry platform used in this project was not designed or built from scratch. Instead, an existing in-house gantry was made available for the project. The platform had originally been constructed by engineers at DTU for use as a CNC welding machine, and was equipped with three stepper motors and their corresponding drivers for the X, Y, and Z axes. No software, sensors, or control logic were included. The platform was evaluated to determine whether it was suitable for the requirements of this project before any modifications were made. The platform is a Cartesian gantry with three orthogonal linear axes. The physical bed dimensions are approximately 770 mm in X, 480 mm in Y, and 120 mm in Z travel. However, due to the geometry of the unified module mounted on the toolhead, the usable workspace is reduced in all axes. The driven shaft is located inside the module body, meaning that when the module is positioned near the edge of the bed in any direction, the driven shaft cannot reach the outermost regions of the bed.

The linear motion in the X and Y axes is driven by lead screws with a pitch of 5 mm per revolution, and 4 mm per revolution for the Z-axis. These convert rotational motion from the stepper motors into controlled linear displacement. The motors were configured to operate at 400 steps per revolution, resulting in a theoretical linear resolution of 5 mm per 400 steps (0.0125 mm per step) in the X and Y directions, and 4 mm per 400 steps (0.01 mm per step) in the Z direction. This configuration was selected because 400 steps per revolution was the lowest setting available on the driver, and the resulting resolution at this setting was already sufficient for the intended use case. Higher microstepping settings would improve resolution further, but at the cost of reduced maximum speed and available torque. Since the achievable precision at 400 steps per revolution met the requirements of the application, the fastest available stepping configuration was selected. Experimental validation of positioning accuracy was therefore conducted only for the 400 steps per revolution configuration [4.2](#).

The mechanical structure was assumed to be sufficiently rigid for the forces expected during screw removal and press operations. A detailed structural verification was not performed within the scope of this project, and therefore the structural performance under load has not been quantitatively validated.

The Z-axis travel of 120 mm was noted as a possible limitation, as greater vertical range would improve the system's ability to handle products of varying height within the same batch. Extending the Z-axis travel would require modifications to the existing frame and lead screw assembly. Rather than redirecting effort toward frame modifications, the decision was made to accept the current Z-axis range and focus development resources on the unified module design and sensor integration, which represent the core contributions of this project. The available travel range was considered sufficient for the test objects and product geometries targeted in this prototype phase.

The gantry based Cartesian architecture was considered well suited to the project requirements. Since all three axes are decoupled and move independently, control is straightforward and precise positioning can be achieved without complex kinematic transformations. This is particularly important for the batch based disassembly application,

where the toolhead must repeatedly reach positions across a flat workspace with high repeatability.

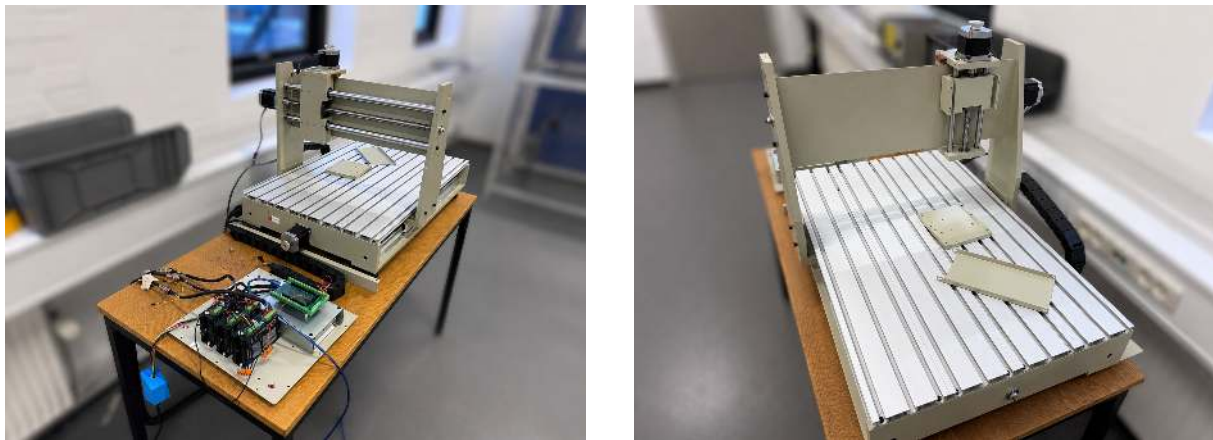


Figure 2: The received gantry platform.

To ensure repeatable positioning, the system requires a reliable method for establishing a consistent reference coordinate. Without such a reference, accumulated positioning errors between sessions would lead to unreliable operation. To fulfill this requirement, three snap-action microswitches (AV 32543 AT) were selected, one for each axis of the gantry system. These switches provide a simple, robust, and repeatable method for detecting a known mechanical reference position. The microswitches were mounted at the negative end of each axis. Upon system startup or reset, each axis moves in the negative direction until the corresponding microswitch is triggered. This procedure defines the machine origin at the corner of the workspace closest to all three switches, thereby establishing a consistent coordinate frame for all subsequent motion commands. This approach ensures repeatable initialization of the system and mitigates positioning drift between operating cycles. The Y and Z axis endstops are shown installed on the machine in figure 3.

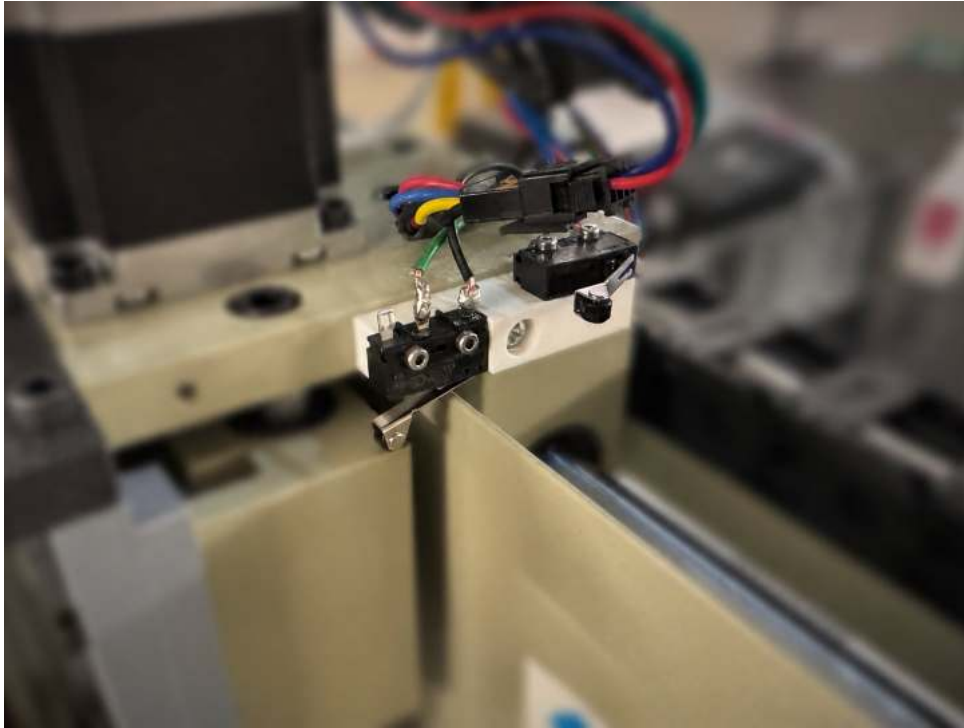


Figure 3: Y and Z axis endstops installed on the machine.

In figure 4 a sketch of the planned prototype can be viewed to give the reader a rough understanding of the layout of the prototype design.

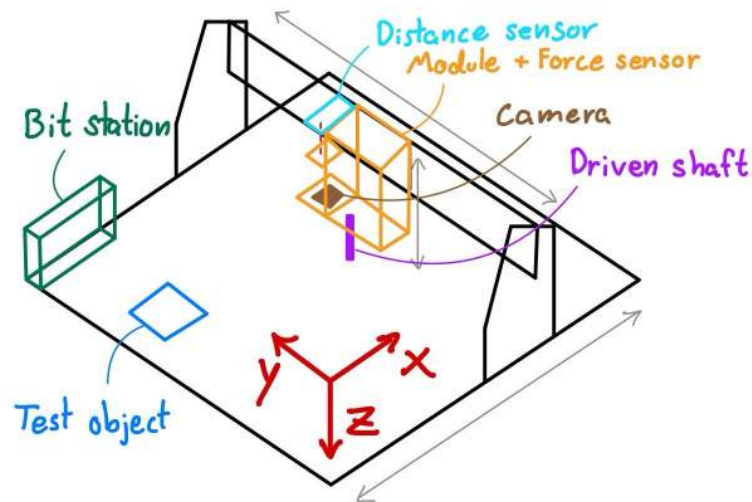


Figure 4: Sketch of prototype layout featuring the unified module, the automatic bit station and a test object.

3.2 Motor for screw operations

The screw removal operation imposes specific requirements on the actuation system. In contrast to the linear axes, the screw motor must be capable of delivering sufficient torque to overcome the breakaway torque of fasteners, while maintaining reliable positional control under varying and unpredictable resistance. Based on literature [6] and preliminary considerations, breakaway torques of up to approximately 14 Nm were considered as a worst-case scenario. Additionally, the system must be able to detect when a screw cannot be removed, in order to avoid mechanical damage and enable operator intervention.

To address these requirements, a T-Series NEMA 23 stepper motor was selected. The motor provides a peak torque of 3 Nm, which is below the highest breakaway torques reported in the literature [6]. Therefore, the motor selection should not be interpreted as sufficient for all realistic disassembly scenarios. Instead, it was chosen to demonstrate the proof of concept functionality of the system. The motor includes controlled screw rotation, sensor based monitoring, and detection of failure cases where the required torque exceeds the motor capability. For a fully industrial implementation, a higher torque solution would be required. This can be achieved by buying a stronger motor or applying a gear ratio to the output. The screw motor added to the system required a dedicated driver, a CL57T closed-loop stepper motor driver was chosen for this purpose. The CL57T operates on a 24 V supply and supports closed-loop control using encoder feedback from the motor, allowing the driver to detect and compensate for missed steps.

This capability is particularly important for the screw removal operation, where the motor may encounter varying resistance and must maintain reliable torque delivery without losing position. Furthermore, the CL57T features a built-in alarm output that is triggered when the motor accumulates excessive position error over a sustained period, indicating that the required torque cannot be delivered. This alarm signal is integrated into the system control logic to detect situations where a screw cannot be loosened within the motor's operational limits. When the alarm is triggered, the system halts the operation, logs the position of the stuck screw, and proceeds to the next task in the sequence, thereby enabling efficient handling of problematic fasteners through subsequent manual intervention.

3.3 Design of Test Object

To verify the functionality of the developed disassembly system, a modular test object was designed and manufactured. The purpose of the test object was to simulate multiple disassembly operations commonly encountered in remanufacturing applications, while simultaneously providing a controlled and repeatable testing platform for the developed hardware and software. The design of the test object was inspired by disassembly operations commonly encountered in automotive components, particularly car alternators. Operations such as screw removal and axial press operations are frequently required during alternator disassembly and therefore served as inspiration for the selected test scenarios. Figure 5 illustrates an example alternator with several relevant screw and press related operations highlighted.



Figure 5: Example of a car alternator with highlighted screw and press related disassembly operations that inspired the design of the test object.

Each test object was designed to accommodate four different operations:

- Unscrewing of three M4 Umbraco.
- Unscrewing of three M5 Umbraco.
- A removable press operation, where an object can successfully be pressed through the test object.
- A non-removable press operation, where the object is intentionally designed such that it cannot physically be pressed through the test object.

The inclusion of multiple screw sizes was intended to simulate realistic disassembly scenarios where products may contain different fastener types requiring different bits and operational parameters. Similarly, the two different press operations were included to evaluate both successful and unsuccessful press attempts. This allowed the system to verify whether it was capable of identifying cases where manual intervention would be required. A CAD section view of the two press operation geometries can be seen in Figure 6.

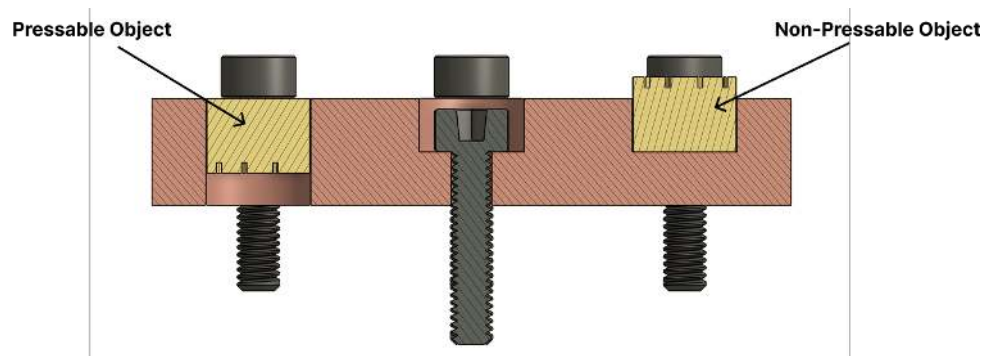


Figure 6: CAD section view of the two press operation geometries integrated into the test object. One geometry allows successful removal of the press object, while the other intentionally prevents complete removal in order to simulate a failed press operation requiring manual intervention.

To improve durability and repeatability, threaded heat inserts were integrated into the test object for all screw operations (Threaded heat inserts of the same kind can be seen in figure 17). The bolts were therefore threaded into metal inserts rather than directly into the 3D-printed material. This design choice was made because repeated tightening and untightening of the screws would otherwise gradually damage the printed threads, potentially introducing unwanted uncertainties and inconsistencies during repeated testing and verification experiments.

Each test object was positioned inside a dedicated tray, as illustrated in Figure 7. The tray served multiple purposes. Firstly, it collected the removable press objects after successful press operations. Secondly, it mechanically constrained the test object during screw removal operations, preventing the entire test object from rotating together with the screw during unscrewing. This ensured that the applied motor torque was transferred to the screw itself rather than causing unwanted movement of the test object.

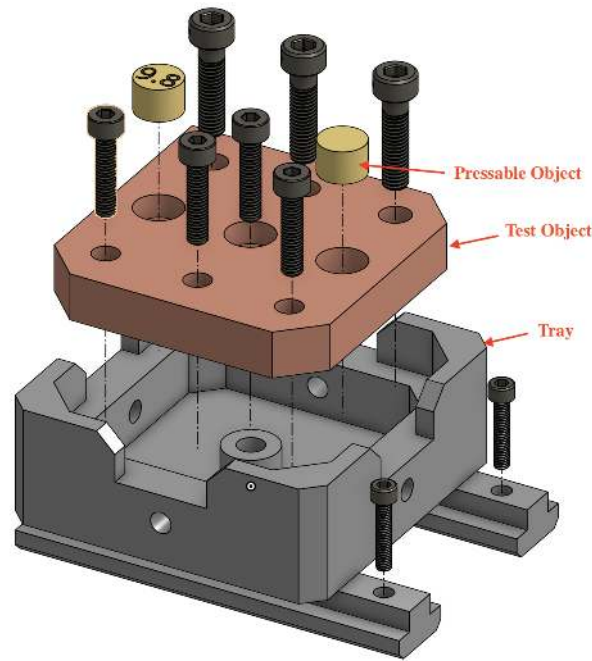


Figure 7: Exploded view of test object, tray, screws and press objects

The test object itself was mechanically attached to the tray using a single bolt located beneath the object. The tray slid into the aluminum extruded profile, that acts as the surface of the gantry. Two screws were then used to lock the item in place. This system allowed the trays to be securely fixed during operation while still enabling fast replacement or reconfiguration of the individual test objects when needed.

The trays were designed in a modular manner such that multiple trays could be mechanically attached together. This design approach was chosen to support future implementation of automatic batch disassembly. In the current prototype, each test object uses an individual tray. However, the design can easily be modified into a single larger fixture capable of accommodating multiple test objects simultaneously.

Such a configuration would support the long-term project goal of automatic batch disassembly, where an entire fixture containing multiple products or subassemblies could be transported into the machine using a conveyor system. The system would then automatically perform all required disassembly operations before the completed batch is removed and replaced with a new batch of products. In this way, the modular tray system acts as an intermediate step toward scalable semi-automated remanufacturing workflows.

The test objects were additionally designed to support repeated experimentation and subsystem verification. Since all operations were integrated into a single object, the same test platform could be used to evaluate screw removal, press operations, computer vision localization, and operation verification within a unified experimental setup.

3.4 Vision System and High-Level Control

To enable automated screw removal and press operations, the system requires the ability to detect operation targets within the workspace and relate these detections to the global machine coordinate system. In addition, the vision system must provide sufficient spatial resolution for reliable screw localization and geometric evaluation.

To fulfill these requirements, a borrowed Raspberry Pi 4 was integrated as the main computing platform for image processing and high-level control logic. A single Raspberry Pi Camera Module was mounted directly on the moving toolhead of the gantry, pointing downward toward the workspace. Due to the limited field of view of the camera relative to the size of the workspace, it is not possible to observe the entire workspace within a single image. Instead, the system performs sequential image acquisition by moving the camera to a set of predefined scan positions across the workspace. At each position, an image is captured and processed locally.

An alternative approach would have been to use a fixed overhead camera capable of capturing the entire workspace in a single image. While this would eliminate the need for sequential scanning, it would also reduce the achievable spatial resolution over the workspace area and increase sensitivity to calibration and perspective errors. By mounting the camera directly on the moving toolhead, a fixed geometric relationship between the camera and the driven shaft is maintained at all times. This simplifies coordinate transformations and improves local measurement accuracy, which was prioritized over acquisition speed for the present prototype.

As an initial approach, the open-source screw detection implementation by Yildiz and Wörgötter [11] was tested as a possible basis for the vision system. However, when applied to images from the present test setup, the detection performance was not sufficiently reliable. As shown in Figure 8, the method produced inconsistent results and was not able to reliably identify the screws in the test object.

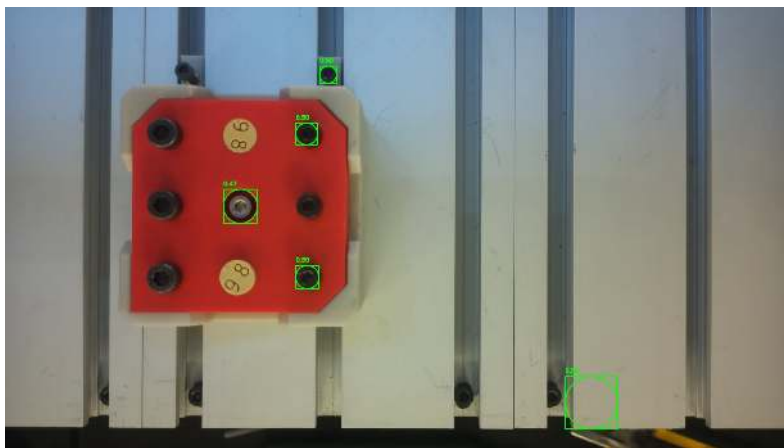


Figure 8: Result obtained when testing the open-source screw detection implementation on an image from the present test setup. The detection was not sufficiently reliable for direct use in the prototype system.

In addition, the processing time for a single image was relatively high even on a reasonably powerful desktop computer. Since the final system relies on a Raspberry Pi for onboard processing, the expected processing time would be even longer in the prototype system. It is likely that the detection performance and processing speed could have been improved further through additional tuning and retraining of the software. However, adapting and optimizing an existing framework was not considered a priority within the scope and timeline of the project. Instead, it was deemed more time efficient to develop a custom screw detection pipeline as quickly as possible, allowing development effort to shift earlier toward subsystem integration and physical testing. Selected concepts from the original approach, particularly the use of geometry based candidate detection prior to classification, were nevertheless used as inspiration for the final implementation.

3.4.1 Iterative Object Centering and Screw Localization

After moving to one of the predefined scan positions, the system captures an image of the workspace region beneath the camera. Since the target object is not necessarily centered within the initial image, an iterative centering procedure is performed before feature localization begins. Figure 9 provides an overview of the complete vision-based localization pipeline used by the system (See figure 57 for full page image). The procedure consists of two main stages. First, an iterative object centering loop is performed to position the target object near the center of the camera image. Once the object has been sufficiently centered, feature localization is performed to identify screw and press operation locations and transform their positions into the global gantry coordinate system.

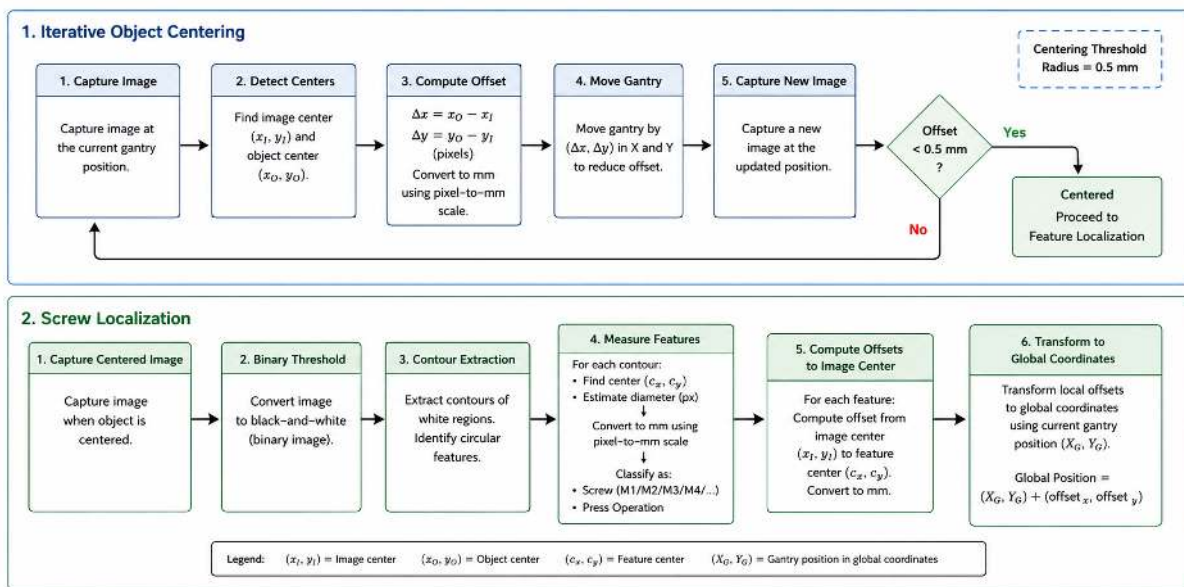


Figure 9: Overview of the vision based localization pipeline used by the system. The upper section illustrates the iterative object centering loop, while the lower section illustrates the screw and press operation localization process used to determine global feature coordinates.

First, the center of the captured image is determined. The center position of the detected target object is then estimated using the image processing pipeline. The offset between the image center and the object center is calculated in pixel coordinates and converted into physical distances using the calibrated pixel to millimeter scaling factors described in A.3.

The gantry then moves by the calculated offset in order to bring the target object closer to the center of the image. After the movement, a new image is captured and the process is repeated. Figure 10 illustrates the iterative centering procedure. The sequence shows how the target object progressively moves closer to the image center as the gantry updates the camera position based on the calculated positional offset after each image acquisition.

This iterative procedure continues until the remaining distance between the image center and the detected object center falls below a predefined threshold. For the present system, the object was considered sufficiently centered when the remaining positional offset was within a radius of 0.5 mm in the physical workspace.

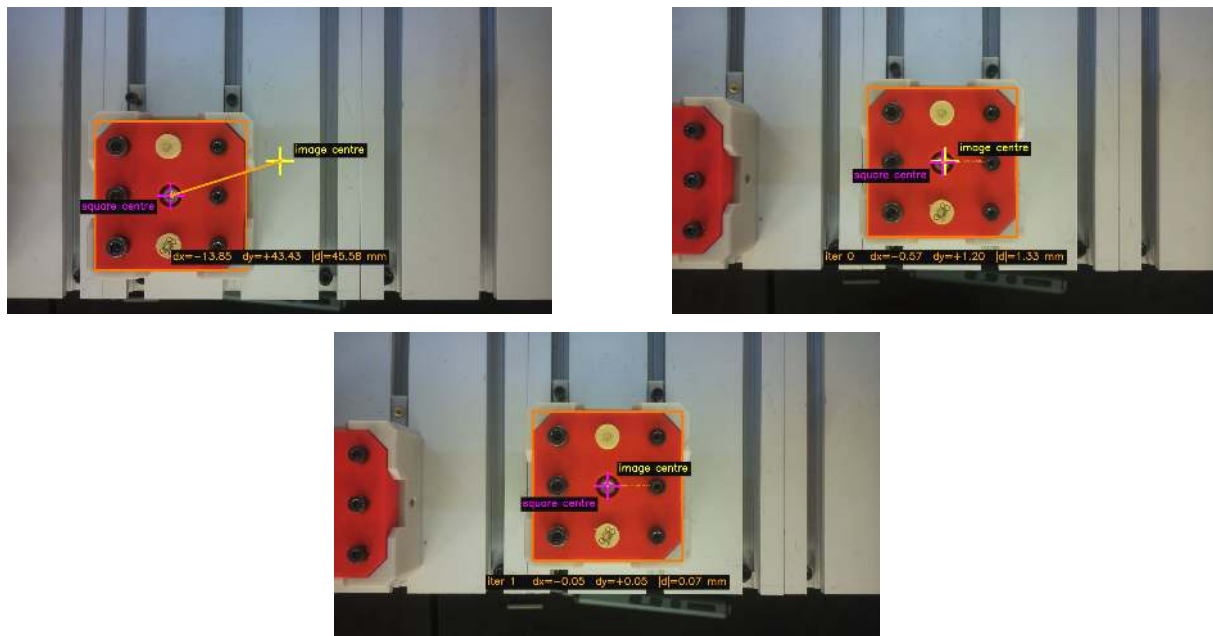


Figure 10: Iterative object centering procedure. After each image acquisition, the positional offset between the object center and the image center is calculated and used to move the gantry closer toward the target center.

Once the object had been centered, a new image was captured for feature localization. The image was converted into a binary black and white representation in order to simplify contour extraction. Figure 11 shows the binary contour image generated during the image processing stage. After thresholding and binary conversion, the screw holes and press-operation locations appear as bright circular regions, while the surrounding regions remain dark. This simplified representation allows contour extraction to be performed robustly.

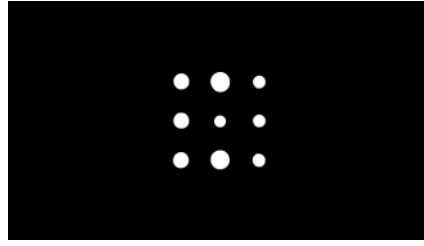


Figure 11: Binary contour image used for feature extraction. Screw holes and press-operation locations appear as bright circular regions after thresholding and binary image conversion.

Contour extraction was then applied to identify the boundaries of the detected circular features. From these contours, the center position and diameter of each feature were estimated in pixel coordinates.

Using the calibrated pixel to millimeter conversion factors [A.3](#), the estimated diameters were converted into physical dimensions. These dimensions were subsequently used to estimate the screw type by selecting the closest matching nominal screw head diameter among the predefined screw categories.

After contour extraction, the positional offset between the center of each detected feature and the image center was calculated. These offsets were converted into millimeters and transformed into global machine coordinates using the known gantry position at the time of image acquisition.

Figure 12 shows the interpreted localization result. Each detected feature is classified as either a screw location or a press operation location, while the estimated diameter is displayed for screw size estimation. The line between the image center and each detected feature center represents the offset used for conversion into global machine coordinates.

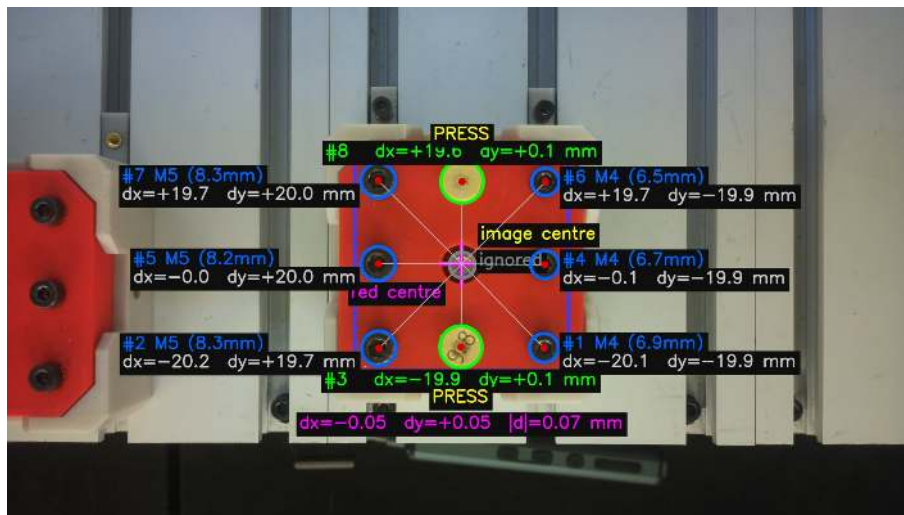


Figure 12: Processed localization result showing detected screw and press-operation features. The estimated diameter of each feature is displayed together with the calculated offset from the image center used for coordinate transformation into the global gantry coordinate system.

This procedure enabled the system to determine the estimated global position of each detected screw and press-operation target within the workspace, allowing subsequent autonomous positioning and interaction by the gantry system.

3.5 Linear Rail and Spring Configuration

Figure 13 is inserted to guide the readers understanding while reading about the geometric and mechanical designs described in the following subsections. For additional photos of the assembly from multiple angles see Appendix D

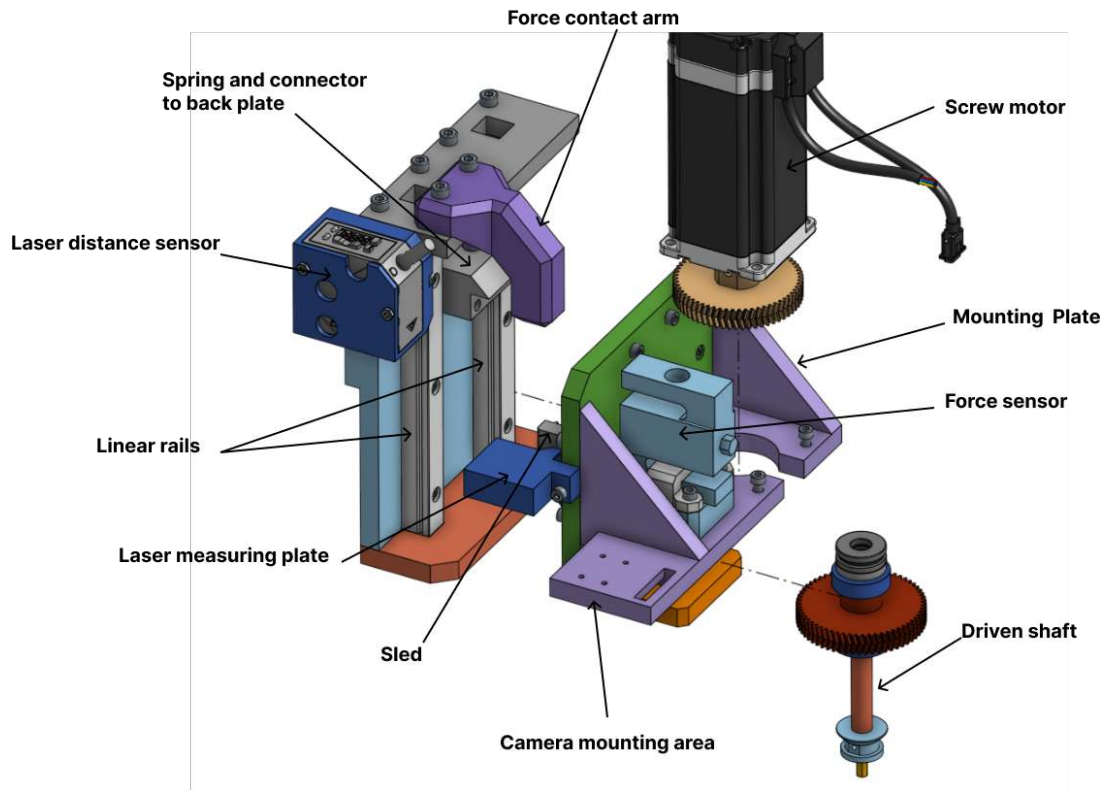


Figure 13: CAD render showing an exploded annotated view of the design that will be described in the following subsections.

A central mechanical requirement of the unified module was that the driven shaft should be able to move relative to the stationary laser sensor structure during contact with the workpiece. This relative motion was necessary to allow the system to detect contact and screw engagement using the laser distance sensor before transitioning to force controlled operation. To achieve this, the module was designed with a sliding mechanism based on two linear rails and two sleds. The linear rails and sleds used in the prototype were borrowed from the Additive Manufacturing Lab at DTU. Since suitable rails and two compatible sleds were available, no separate component selection process was carried out for this part of the design. This reduced both cost and development time, and made it possible to focus on validating the sensor integration and overall module concept rather than sourcing and comparing different linear guide systems.

The sleds allow the module, to slide up and down relative to the stationary frame. When the bit contacts a surface, continued downward motion of the Z axis causes the module to move upward along the linear rails. This motion compresses the spring and creates the relative displacement measured by the laser distance sensor. The sliding

mechanism therefore forms the mechanical basis for detecting contact, screw engagement, and the transition between position controlled motion and force controlled operation. This can be seen in figure 14.

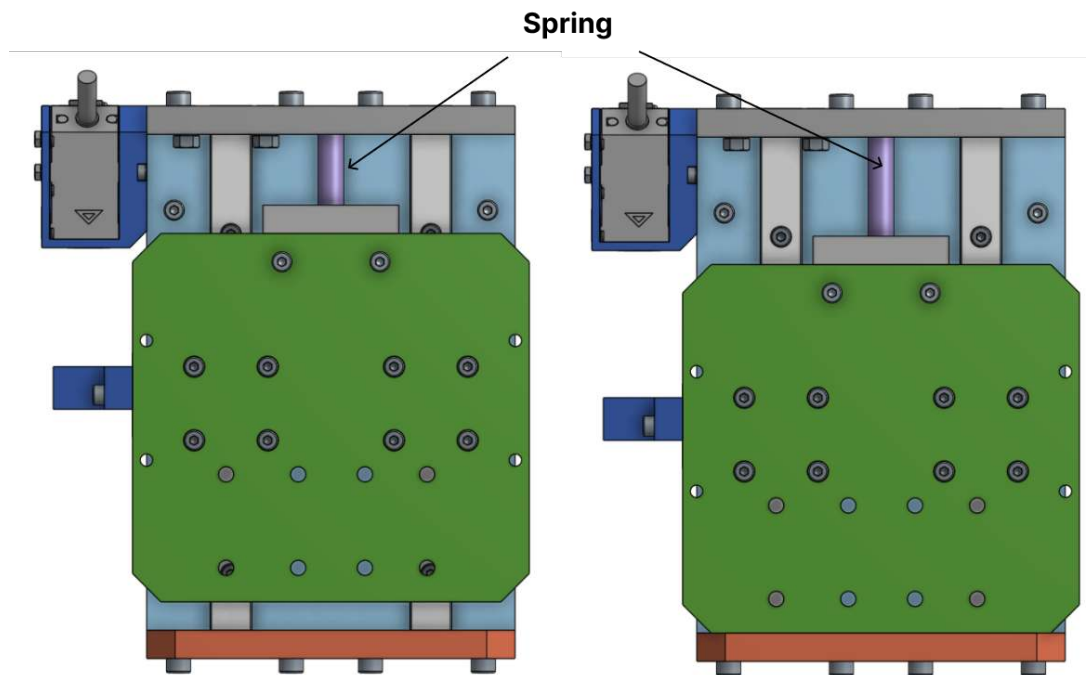


Figure 14: CAD render showing the spring compression as the back plate moves 10 mm.

Although the borrowed rail and sled configuration was sufficient for the prototype, it was not mechanically optimal. The rail and sled combination was relatively tall, which increased the distance between the applied pressing force and the Z axis support structure. During press operations, this creates a larger moment arm and therefore increases the bending moment acting on the Z axis and module structure. A lower profile rail and sled combination would reduce this moment arm and improve the mechanical stiffness of the system during pressing.

In addition, the prototype used one sled per rail. From a mechanical stiffness perspective, two sleds per rail would have been preferable, since this would provide better resistance against rotation and reduce play in the sliding mechanism under load. However, one sled per rail was considered sufficient for the purpose of validating the sensor design and demonstrating the functional principle of the compliant module. For a future version intended for higher pressing forces or industrial use, a more compact rail system with two sleds per rail should be considered.

3.6 Force Sensor Integration and Geometry

To enable controlled press operations and ensure safe interaction with the workpiece, the system requires the ability to measure axial forces applied through the tool. This is necessary both for monitoring process conditions and for preventing excessive loads that could damage the system or the components being processed. To fulfill these requirements, a load cell was integrated into the mechanical structure to measure axial forces along the load path. The selected sensor is an SS3G-250KG-C3 load cell, rated for loads up to 250 kg, operating on a supply voltage of 5 to 12 V. The analogue output of the load cell is conditioned and digitised using an HX711 24-bit analogue to digital converter module, which communicates with the Arduino Mega via a two wire serial interface. The Arduino Mega handles all sensor readings and motion control, while the Raspberry Pi is dedicated to image processing and screw detection tasks.

The force sensor is mounted in the load path of the unified module such that it is only engaged after the module has travelled the full 10 mm of spring displacement and reached a rigid hard stop. At this point, any additional axial force applied through the driven shaft is transferred directly through the module and into the force sensor. This sequential arrangement ensures that the force sensor measurement is not influenced by the spring compliance during the initial approach phase, and only becomes active when the system transitions to force controlled operation. This lets the system probe and read laser sensor outputs without having to load the force into the force sensor at the rest of the module. More on this in Section 4.2.1.

However, the force measured by the load cell is not equal to the total axial force experienced by either the driven shaft or the press object. Before the module reaches the force sensor arm, the entire moving module must first translate upward along the linear rails while compressing the spring. During this motion, the driven shaft and the press object already experience the force required to both support the module mass against gravity and compress the spring.

As a result, the actual axial force applied at the tool tip is given by

$$F_{\text{tool}} = F_{\text{sensor}} + mg + ky_1$$

where F_{sensor} is the measured load cell force, m is the mass of the moving module, g is the gravitational acceleration, k is the spring stiffness, and y_1 is the vertical displacement of the module before contact with the force sensor arm occurs.

The sensor capacity of 250 kg was selected to cover the expected range of axial forces encountered during press operations, including cases involving tightly fitted or resistant components. During operation, the measured force is continuously monitored, and motion is halted if a predefined threshold is exceeded. This makes the force sensor a critical component for ensuring safe and reliable operation.

3.7 Laser Distance Sensor Integration and Geometry

To enable reliable detection of surface contact and screw engagement, the system requires a method for measuring the relative displacement between the moving module and the stationary frame. This measurement must be sufficiently sensitive to detect small changes in position associated with initial contact, screw bit engagement, and potential drop-in events during alignment.

To fulfill this requirement, a borrowed ZX1-LD50A81 2M laser distance sensor was integrated into the system. The selected sensor was an available component with a measurement range of 40 mm to 60 mm, providing sufficient resolution (0.002 mm) within this interval for detecting small displacement changes.

Rather than selecting a sensor to match a predefined mechanical layout, the geometry of the unified module was designed around the sensor's operating range. The sensor is mounted on the stationary frame and directed toward a reflective target plate attached to the moving sled. The geometry can be seen in figure 15. In the neutral position, the distance between the sensor and the target plate is approximately 59 mm. Leaving a comfortable 1 mm gap to the sensors end of operating range.

During operation, when the driven shaft makes contact with a surface, the module is displaced upward along the linear guide rails against the spring. This movement reduces the measured distance from approximately 59 mm to 49 mm at full compression, resulting in a usable displacement range of 10 mm entirely within the sensor's reliable operating window.

This measurement serves as the primary feedback signal for detecting contact and engagement events. By monitoring changes in the measured distance, the system can identify when the tool makes contact with a surface, distinguish between free motion and constrained motion, and detect characteristic displacement changes associated with screw engagement. The laser sensor therefore plays a central role in guiding the operations. To ensure the lasers accuracy the laser was calibrated using an experimental setup described in Appendix A.2.

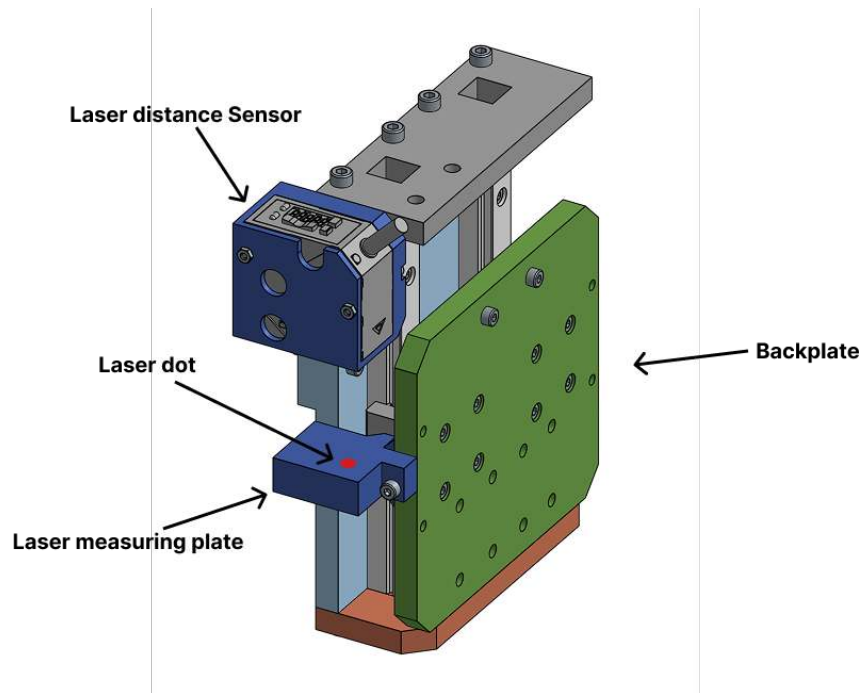


Figure 15: CAD render showing the laser sensor mounted in a blue mount on the stationary frame and the target plate (also blue) on the moving sled, illustrating the neutral distance

To enable both sleek mounting of the laser housing onto the module as well as fixing the laser sensor within the housing, the setup in figure 16 was used. Here we see three screws connecting the housing to the module. The access holes in the wall of the housing must be countersunk to enable the laser sensor to fit into the housing. The housing is mounted onto the module using the holes in the thin wall of the housing. Then the laser can be put into the housing and locked in place using two long bolts.

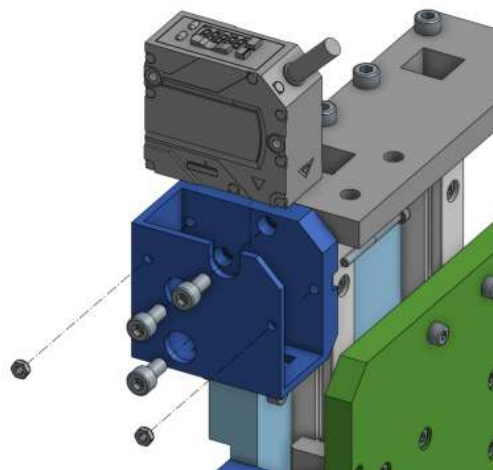


Figure 16: Laser housing assembly showing the screws and bolts needed to fix the laser and its housing.

3.8 Drivetrain Configuration and Mechanical Integration

During press operations, significant axial forces are applied through the driven shaft. If the driven shaft were coupled directly to the motor shaft, these axial forces would be transmitted directly into the motor bearings, which are not designed to withstand sustained axial loading. This would risk mechanical damage to the motor and reduce the reliability and lifespan of the drivetrain. To prevent this, the driven shaft is physically separated from the motor axis, with rotational motion transmitted between them via a gear pair. This decoupling ensures that axial forces applied through the driven shaft are redirected into the module structure rather than into the motor shaft, fully protecting the motor bearings regardless of the magnitude of the applied press force.

An additional benefit of using a gear pair rather than a direct coupling is that the gear ratio can be adjusted in future iterations of the design simply by changing the gear sizes, without requiring big modifications to the drivetrain layout. This gives the system flexibility for future development, for example if a higher output torque or a different rotational speed is needed for a different product type or fastener size.

For the current prototype, a gear ratio of 1:1 was selected using a pair of Herringbone gears with a centre distance of 60 mm between the motor axis and the driven shaft axis. Since the prototype gears are 3D printed, herringbone gears were used instead of standard spur gears. Herringbone gears engage gradually along the tooth face rather than making full contact instantaneously, which reduces noise, distributes load more evenly across the tooth width, and eliminates the axial thrust forces that standard helical gears produce. For a 3D printed prototype this comes at no additional cost or manufacturing complexity, since the herringbone geometry is simply included in the print file. In a final production design using machined metal gears, herringbone gears would introduce additional manufacturing cost and complexity compared to standard spur gears, and whether they are necessary would depend on the load and speed requirements of the final system. For the purposes of this prototype, the herringbone geometry was considered a practical and beneficial choice given the manufacturing method.

Mounting Plate and Backplate Structure

The main structural load carrying element of the drivetrain assembly is the mounting plate shown in Figure 13. Most drivetrain components, including the screw motor, force sensor, driven shaft assembly, camera mount, and bearing supports, are directly mounted to this plate. The mounting plate itself is attached to the backplate, shown in green in Figure 13, which connects the drivetrain assembly to the remainder of the unified module structure.

During press operations, axial forces are transferred from the bit through the driven shaft and into the mounting plate structure. Since the force application point is offset from the backplate connection, significant bending moments can be introduced into the mounting plate during operation. To improve structural stiffness and reduce deflection under load, two triangular reinforcement geometries were integrated between the mounting plate and the backplate. These reinforcements increase the effective moment resistance of the structure and allow the applied loads to be distributed more effectively into the backplate through the larger contact surfaces between the components. The 3D-printed

part can be viewed in figure 17.

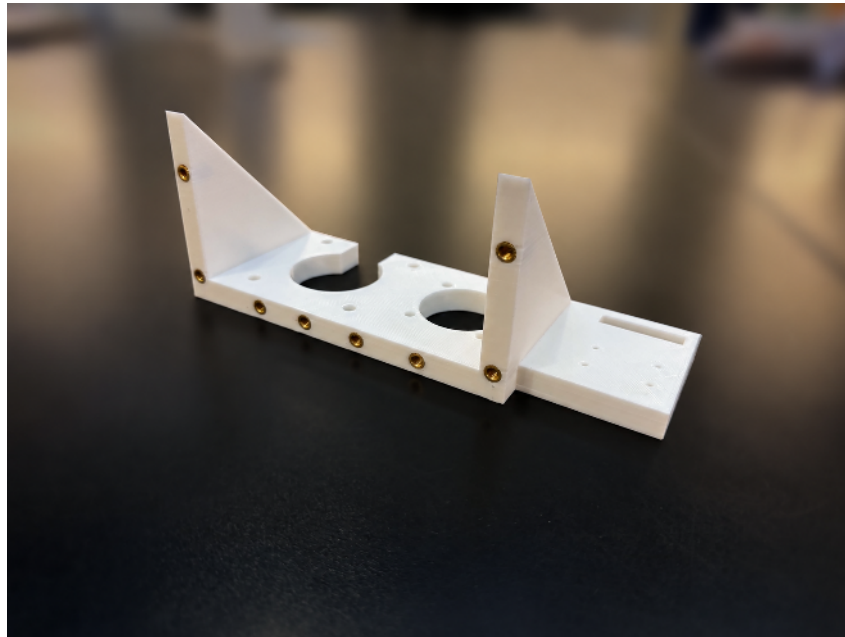


Figure 17: 3D-Printed mounting plate with threaded heat inserts.

It can additionally be observed in Figure 13 that the mounting plate contains an open slot around the screw motor and motor gear region. This geometry was intentionally introduced to simplify assembly of the drivetrain. The herringbone gear can be mounted onto the motor shaft outside the structure and subsequently slid into position through the open slot before the motor is fully fastened to the mounting plate. Without this opening, assembly and disassembly of the gear and motor system would be significantly more difficult due to the limited clearance between the drivetrain components.

3.9 Driven shaft and Bit Adapter Geometry

The driven shaft is the mechanical link between the gear transmission and the screwdriver bit. It transmits rotational torque from the gear pair down to the bit while also allowing the spring compliance mechanism to function along the Z axis. The driven shaft was derived from a Wera 1/4 inch hexagonal bit extension (model 05160979001), 200 mm in length. The lower end of the extension retains its original female 1/4 inch hex socket, which accepts standard screwdriver bits directly.

3.9.1 Original Design Approach

The original design for the driven shaft assumed that the Wera extension was made from standard, non hardened steel. Under this assumption, the upper hexagonal driven shaft section was to be machined down to a round rod of diameter 6.3 mm with a tolerance of $+0.019/+0.01$ mm. This would have allowed precision bushings to be fitted around the driven shaft using an H7/n6 interference fit, relying on the friction of the fit alone to prevent the bushings from migrating axially along the driven shaft. Additionally, the plan required a threaded hole to be drilled into the top end of the driven shaft. This thread would have been used to fasten a circular disc, referred to throughout as the contact circle, to the top of the driven shaft. The contact circle would then bear against the thrust bearing, transmitting axial loads from the Z axis press operation into the module frame. A schematic of this original design approach is shown in Figure 18.

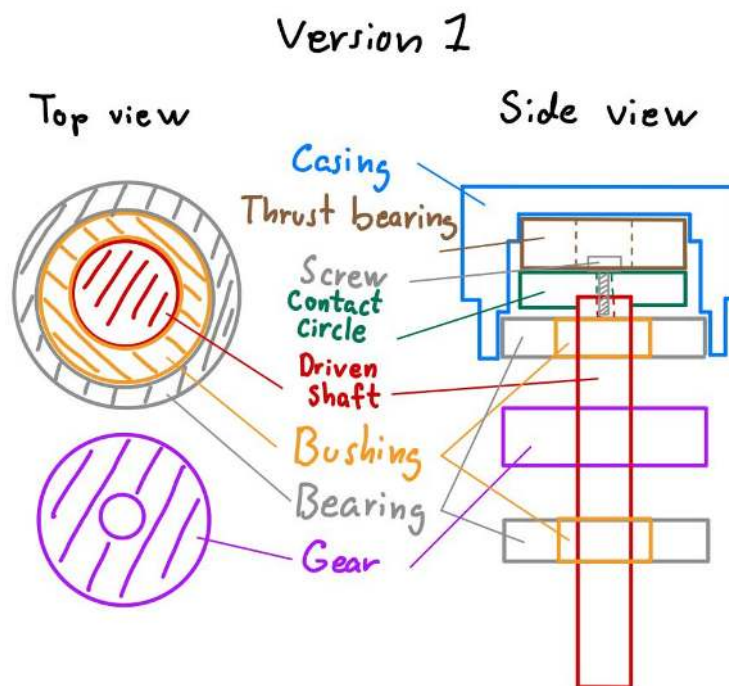


Figure 18: Schematic of the original driven shaft design, showing the planned machined round section, H7/n6 bushing fit, and threaded contact circle connection at the top end.

The technical drawing of casing component shown in Figure 18 can be viewed in Appendix B.1. The component was manufactured using a CNC machine at the DTU workshop in Building 427. To verify the assumption that the driven shaft was not made from hardened steel, a physical inspection was arranged with the workshop staff at DTU Skylab prior to ordering the component. All four members of staff present tested the surface of the driven shaft by filing its edge. Based on their assessment, the steel was confirmed to be non hardened, and a new unit identical to the tested sample was ordered for machining.

3.9.2 Discovery of Hardened Steel and Design Rework

When the ordered driven shaft was brought to the machine shop for drilling, the drill bit was unable to penetrate the surface of the driven shaft. The material proved to be hardened steel, despite the earlier assessment. The machine used during this attempted drilling is shown in Figure 19.



Figure 19: The lathe used during the attempted thread cutting operation on the top end of the driven shaft.

This result made the original design unachievable and necessitated a full rework of the driven shaft assembly. The revised design retained the same functional requirements: a thrust bearing to absorb axial load, two radial bearings to constrain the driven shaft radially, and the contact circle at the top of the driven shaft aligned centrally to interface with the thrust bearing. A schematic of the reworked design is shown in Figure 20.

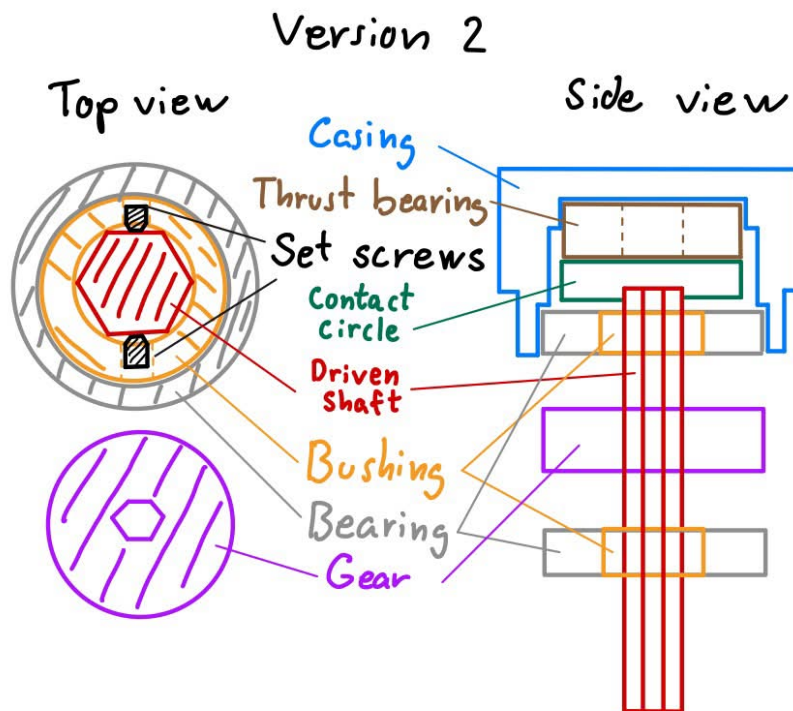


Figure 20: Schematic of the reworked driven shaft design, accommodating the hardened hexagonal steel driven shaft with set screw retained bushings and a press fitted plate.

Bushing Retention In the original design, the bushings were to be held in position by an H7/n6 interference fit on the turned round section of the driven shaft. Since the driven shaft could not be machined, a different approach was required. The inner diameter of each bushing was instead made equal to the largest diagonal of the hexagonal driven shaft cross section, allowing the bushing to slide over the hex profile. Two set screws per bushing were then used to clamp against the flat faces of the hex driven shaft, locking each bushing at the correct axial position.

To illustrate the physical implementation of the revised driven shaft assembly, Figure 21 shows an exploded view of the main components, including the driven shaft, bushings, bearings, herringbone gear, and contact circle. The figure highlights how the individual components are assembled around the hardened hexagonal driven shaft to achieve both radial support and axial force transmission.

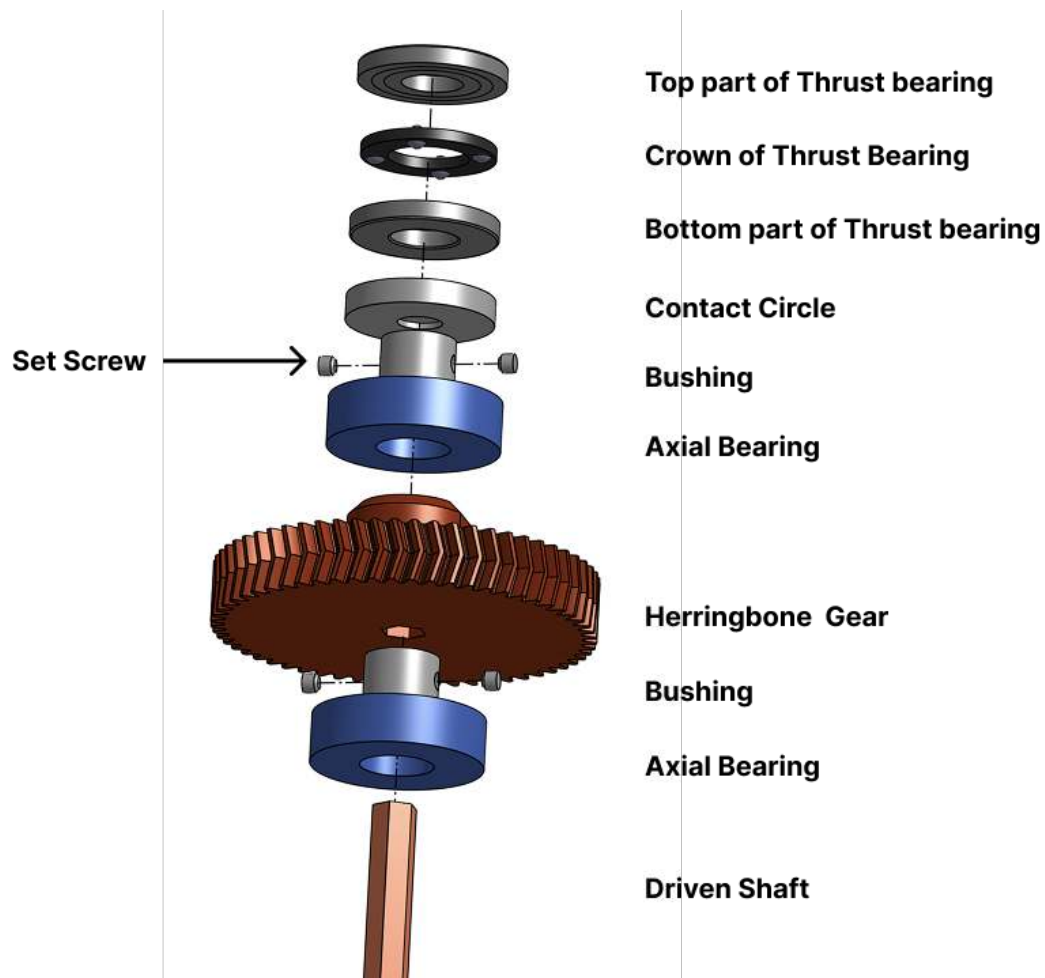


Figure 21: Exploded view of the revised driven shaft assembly showing the hardened hexagonal driven shaft, bushings, radial bearings, herringbone gear, and contact circle used for axial force transfer into the thrust bearing system.

The set screws available in stock had a minimum length of 5 mm. However, the outer surface of each bushing carries an radial bearing, which imposed a maximum set screw length of 2.5 mm to avoid interference with the bearing inner race. The 5 mm screws were therefore shortened to the required length by grinding them on an automatic belt grinding machine, as shown in Figure 22.



Figure 22: The automatic belt grinding machine used to reduce the set screws to the required 2.5 mm length.

With the set screws prepared, the bushings were positioned at the correct locations along the driven shaft. The herringbone gear was placed between them, and the radial bearings were then fitted onto the outer surface of each bushing using an H7/r6 interference fit. One bearing was seated with moderate hand force, while the other required a pulling tool to achieve full seating, as shown in Figure 23.



Figure 23: The pulling tool used to press the radial bearing fully onto the bushing outer diameter.

The original intent for the contact circle was to include a 1 mm hexagonal recess machined into one face, into which the tip of the hex driven shaft would be placed, providing positive rotational alignment between the contact circle and the driven shaft. This feature was not manufacturable with the equipment available at DTU Skylab. As an alternative, the contact circle was produced with a circular central bore sized for a press fit onto the hexagonal driven shaft tip. While this does not provide geometric keying in the same way as a hex recess, the press fit delivers a connection sufficiently rigid for the loads involved within the constraints of the project timeline. The total usable length of the driven shaft from the bit connection point upward is 91 mm.

To illustrate the physical implementation of the revised driven shaft assembly, Figure 24 shows the assembled driven shaft together with the herringbone gear, contact circle, and the two radial bearings used to support the shaft. The upper radial bearing is mounted directly on the shaft, while the lower radial bearing is press fitted into a 3D printed housing connected to the backplate.

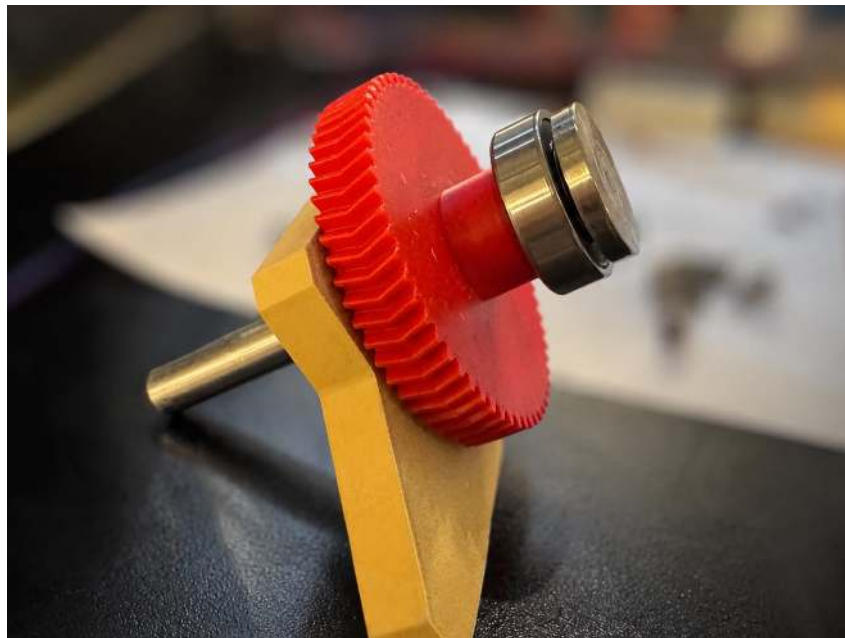


Figure 24: Assembled driven shaft system showing the driven shaft, herringbone gear, contact circle, and upper radial bearing. The lower radial bearing is mounted inside the 3D printed support structure connected to the backplate and is therefore not directly visible in the image.

The use of two radial bearings serves two purposes. First, the additional support helps reduce shaft deflection during press operations, where axial loading of the tool can introduce bending moments on the driven shaft. Second, the bearings help absorb the radial forces generated by the meshing herringbone gears. During torque transmission, the gear teeth apply forces onto each other that tend to push the two gears apart, as illustrated in Figure 25. Supporting the driven shaft at two locations therefore improves the rigidity of the drivetrain and reduces unwanted lateral movement of the shaft during operation.

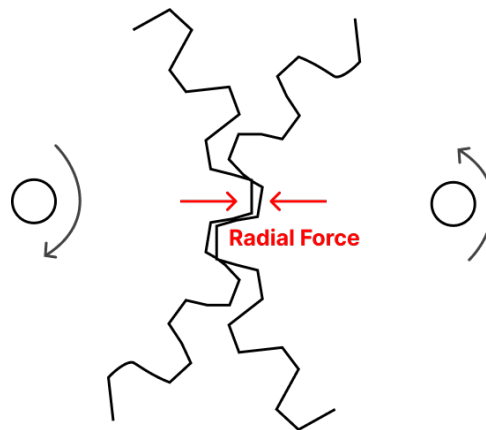


Figure 25: Illustration of the radial forces generated between two meshing gears during torque transmission. The radial force component acts outward along the line connecting the gear centres, tending to push the gears apart.

Although herringbone gears eliminate the axial thrust forces associated with standard helical gears, the radial separating forces remain and must still be supported by the drivetrain structure and bearings. The lower bearing housing additionally includes an integrated shoulder at the top of the bearing seat. This shoulder acts as an assembly stop, ensuring that the lower bearing is seated at the intended depth during insertion into the housing. The full assembled unified module can be viewed in [figure 26](#).

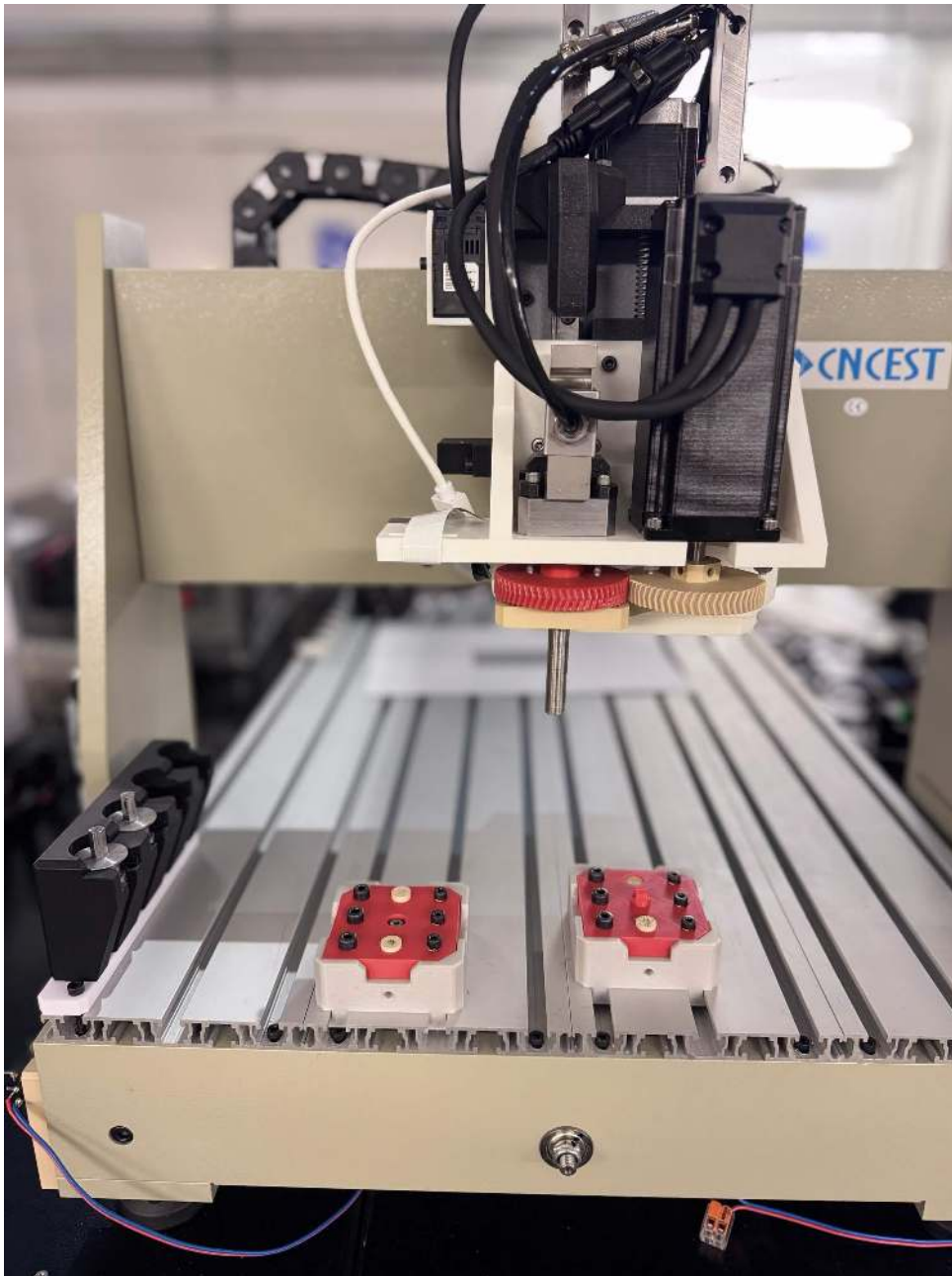


Figure 26: Full assembly of the produced module.

For additional photos of the CAD assembly from more angles see [Appendix D](#).

3.10 Bit Changer Station

A key requirement established during the design phase, detailed in Section 2.4, was that the bit changing operation must be performed without any additional actuator beyond the existing gantry axes. This constrained the design to rely entirely on geometry and gravity to release, store, and re-engage bit adapters.

The bit adapter is a custom CNC-machined steel component that sits between the driven shaft and the screwdriver bit. It serves three functions: connecting to the driven shaft above it, holding the screwdriver bit below it, and providing the geometric features needed for passive retention in the station. At the top, the adapter has a male hexagonal section that inserts into the female 1/4 inch hex socket of the driven shaft and is held in place by the magnetic retention built into the socket. The adapter is light enough that the magnetic retention force reliably supports its own weight during operation, yet that same force is low enough to be overcome passively by the station retention geometry during a bit release.

At the bottom, the adapter features a female socket with a set screw, into which the screwdriver bit is inserted and locked. This allows any standard bit to be pre-loaded into an adapter and stored in the station ready for use. The middle section of the adapter has an I-beam profile when viewed from the side, creating a gap between upper and lower flanges that the station retention arms engage into. In addition, the adapter features a conical outer profile on the bottom part of the upper part of the I, intended to self-centre as it is lowered into a station slot. The bit adapter can be viewed in figure 27. The technical drawing of the bit adapter that was sent to the workshop is included in Appendix B.2.

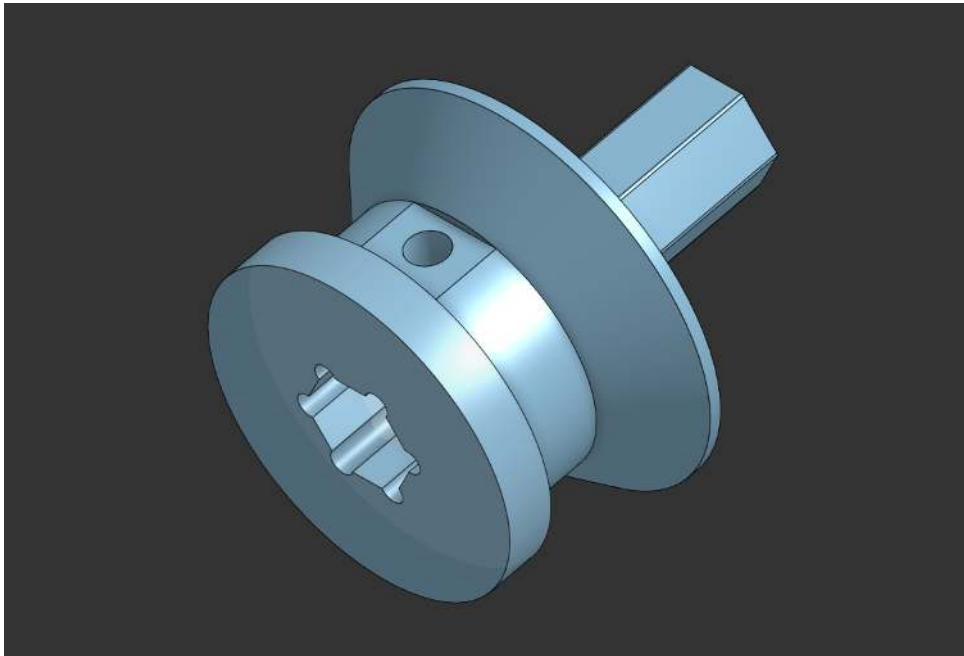


Figure 27: CAD rendering of the bit adapter. The male hex upper section connects to the driven shaft socket. The female lower socket with set screw retains the screwdriver bit. The I-beam mid-section profile provides passive retention in the station slots.

3.10.1 Station Design Iterations

The station was developed through three design iterations, each motivated by mechanical observations from physical evaluation of the previous version. The first version was 3D printed and designed to hold six bit adapters. Each slot featured a conical entry geometry intended to guide the adapter into the correct XY position as it was released from the driven shaft, working together with the matching conical profile on the adapter body to funnel it toward the centre of the slot. The retention arms engaged the I-beam gap of the adapter to resist upward pull, decoupling the adapter from the driven shaft during a bit release. No additional active components were required at any stage of the operation.

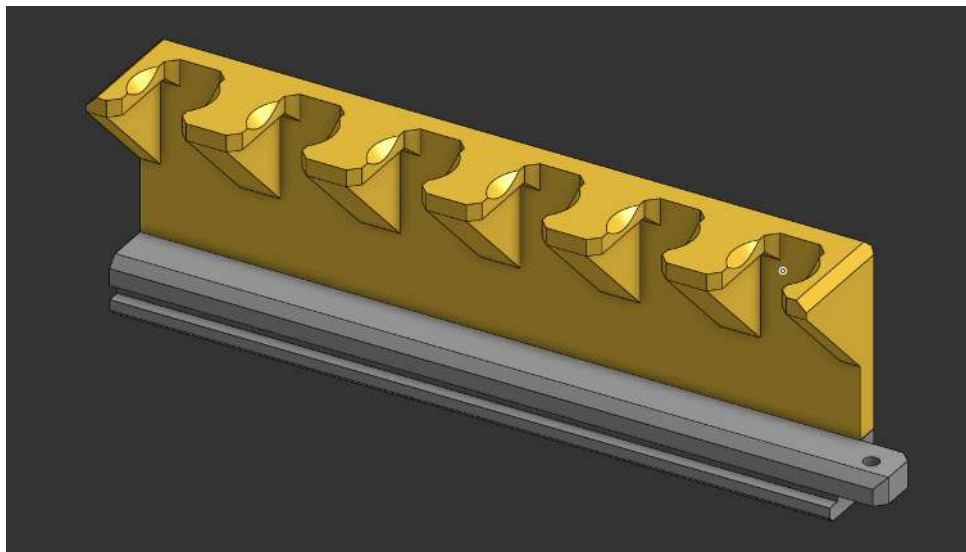


Figure 28: CAD rendering of the bit changer station version 1. The six slots rely on the dual conical geometry of the slot and adapter body to self-centre during lowering. The geometry provided minimal lateral support during the drop phase.

Initial physical evaluation indicated that the open slot geometry provided insufficient lateral guidance immediately after release. Once the adapter was released from the driven shaft and allowed to fall into the slot, it could come to rest in misaligned positions, reducing the consistency of subsequent re-engagement.

Based on these observations, it became clear that the adapter required additional physical guidance during the drop and seating phases. Version 2 therefore introduced extended lateral slot walls intended to constrain the adapter motion more during release and seating. The number of slots was also reduced from six to four, since no more than four custom adapters were produced during the project. Version 2 can be viewed in [figure 29](#).

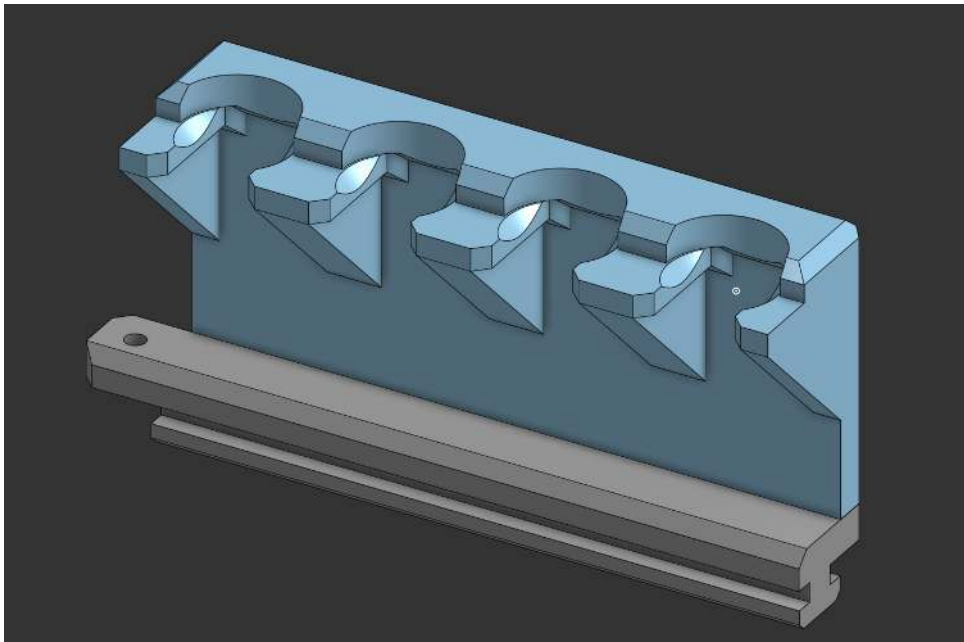


Figure 29: CAD rendering of the bit changer station version 2. The added curved lateral walls and retention hooks provide significantly more guidance during the adapter drop phase compared to version 1.

The added lateral guidance improved the seating consistency of the adapter compared to version 1. It in fact improved it enough that the machine could pick and place the bit adapters with 100% reliability. But storing the bit adapters by hand without care, could result in a failed machine pickup. Indicating that further constraint of the adapter position within the lower slot region was required.

Version 3 was therefore designed to further reduce the positional freedom of the adapter after seating. The lower slot body was revised to more closely conform to the adapter profile, reducing the possibility of lateral movement after release. The conical geometry was retained on both the station slot and the adapter body, creating a dual-cone self-centring mechanism intended to improve alignment consistency during both release and re-engagement. This improved design is shown in figure 30.

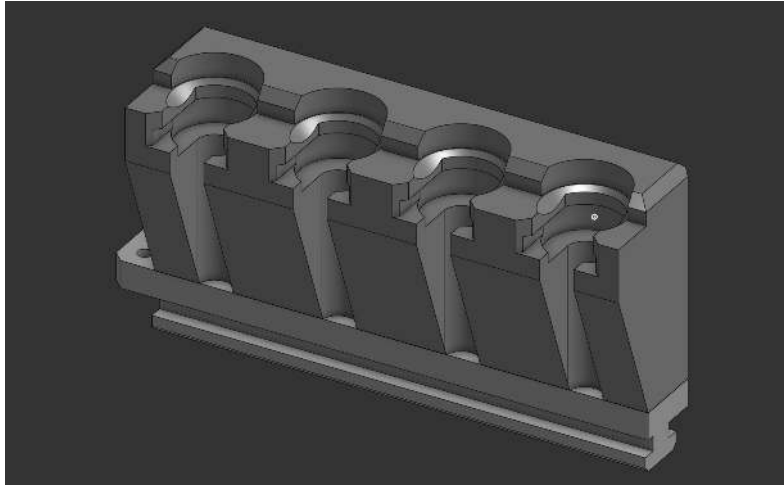


Figure 30: CAD rendering of the bit changer station version 3. The tighter lower slot profile provides more complete guidance throughout the adapter drop and seating phases.

The final station geometry enabled fully passive tool exchange using only the existing gantry axes and the rotational degree of freedom of the driven shaft. Figure 31 illustrates the sequence used when picking up a bit adapter from the station, while Figure 32 illustrates the sequence used when returning an adapter to the station after operation.

Grab Bit

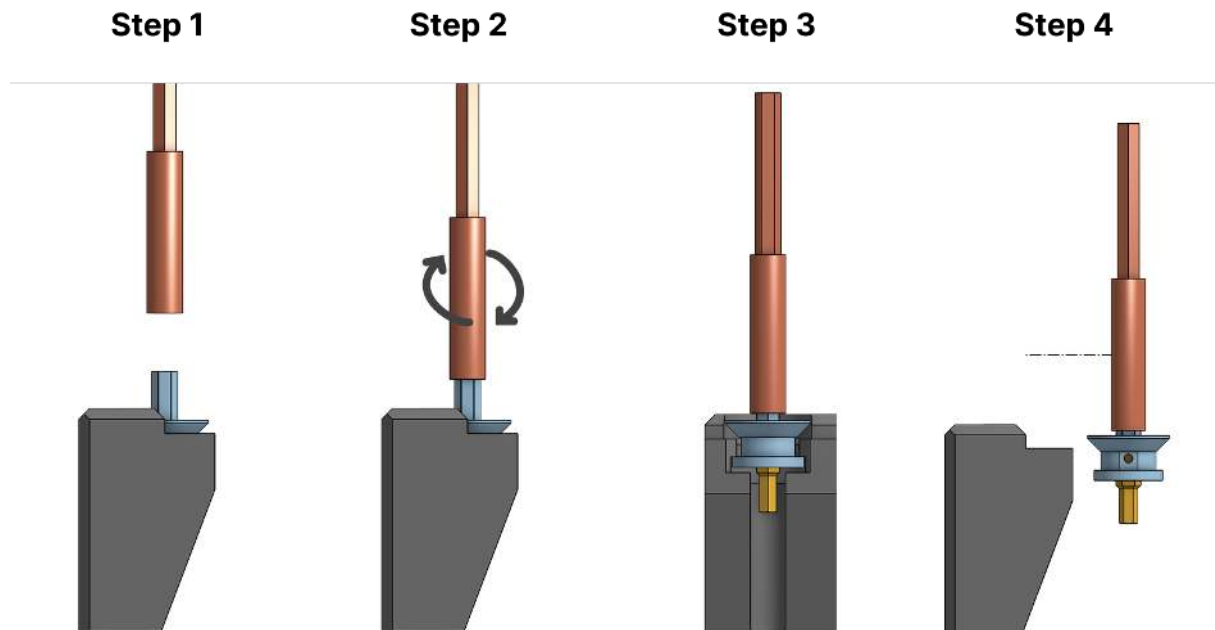


Figure 31: Sequence illustrating passive pickup of a bit adapter from the final bit changer station geometry. The driven shaft aligns with the adapter, lowers into the slot, magnetically engages the adapter, and lifts the adapter out of the station.

Place Bit

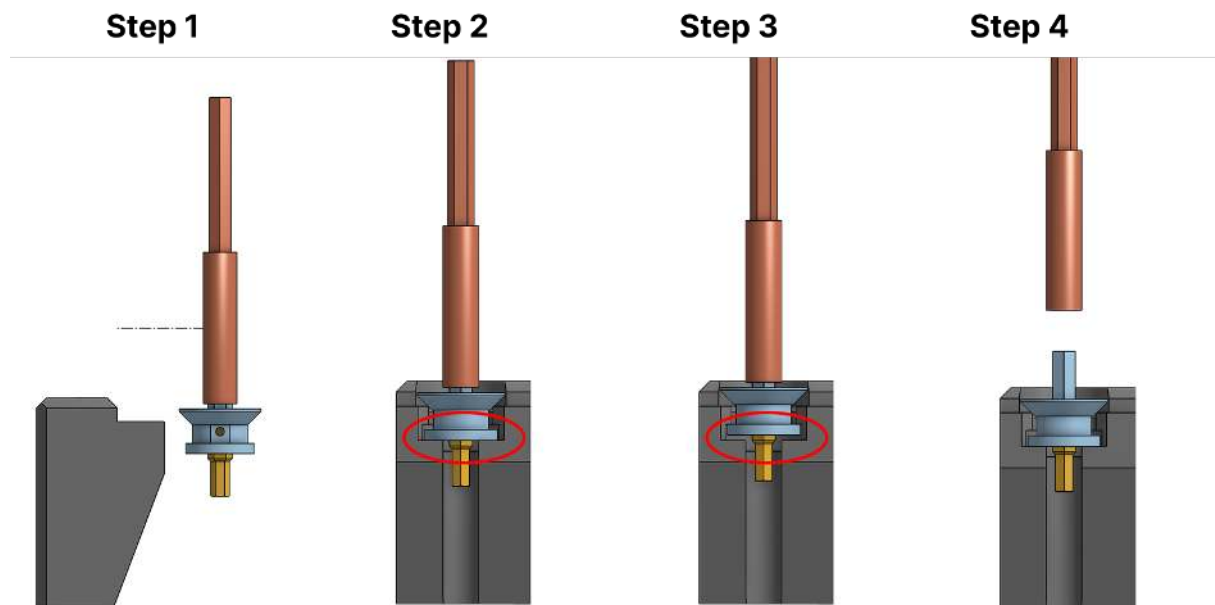


Figure 32: Sequence illustrating passive placement of a bit adapter into the final bit changer station geometry. The driven shaft lowers the adapter into the slot, the retention geometry overcomes the magnetic holding force, and the driven shaft retracts while leaving the adapter seated in the station.

Detailed validation of the station performance, including repeated pick and place testing and re engagement reliability, is presented in Section 4.4. To fully constrain the bit station when installing it on the aluminum extruded surface frame of the gantry, the following approach was used. First, the bit station geometry was connected to a sliding part designed to fit into the aluminum frame (like the one used on the test objects described in section 3.3). Then a pre-existing threaded hole on the gantry was utilised to lock the bit station in the machines x axis. Hereby fully fixing the station. An exploded view of the bit station assembly including the locking screw is shown in figure 33.

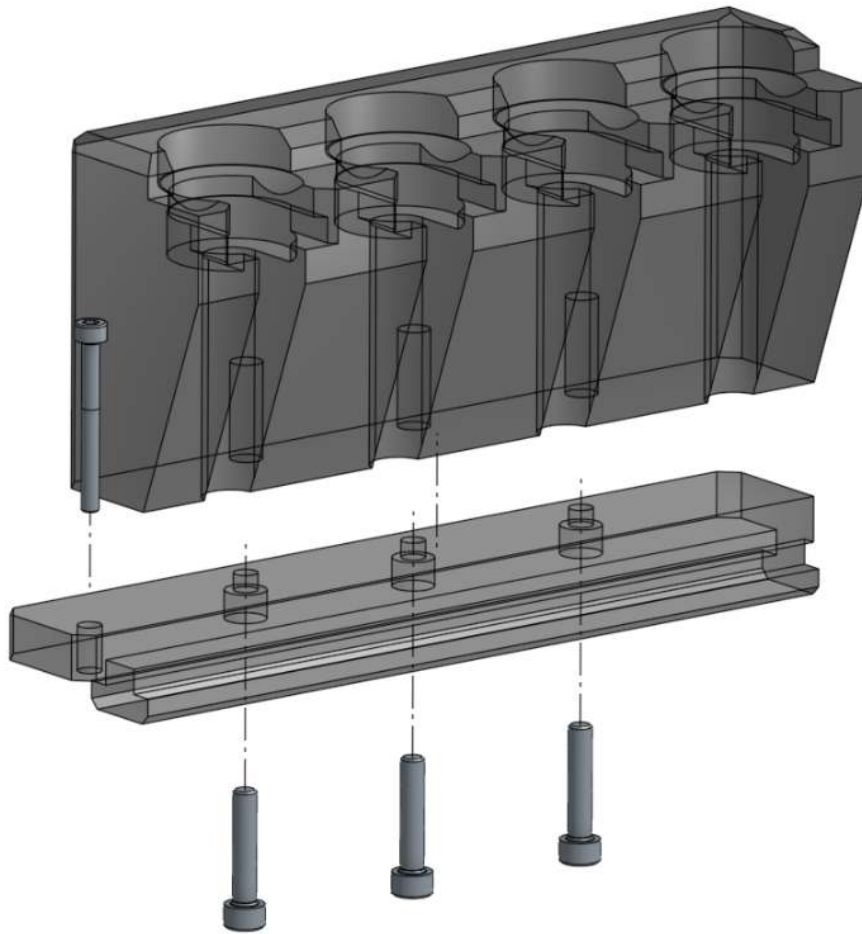


Figure 33: Exploded view of the finalized design of the bit station.

3.11 Power Supply Upgrade

The gantry platform was originally equipped with a 24 V, 5 A power supply, providing a maximum output of 120 W. An assessment of the power requirements of the full system, including the three existing gantry axis motors, the added screw motor, the motor drivers, the Raspberry Pi, and all sensors, indicated that this supply was clearly insufficient for reliable operation.

The three gantry axis motors are NEMA 23 stepper motors (model 23HS22 1504S), rated at 1.5 A per phase with two phases each. At peak draw, each motor can consume up to 3 A, giving a combined peak draw of approximately 9 A for the three gantry axes alone. The added screw motor, a T Series closed loop NEMA 23 rated at 3 Nm, draws approximately 4 A peak. The Raspberry Pi, sensors, and associated electronics contribute an additional estimated 2 A. The total estimated peak system draw is therefore approximately 15 A at 24 V, corresponding to approximately 360 W. The original 5 A supply could provide a third of this, making simultaneous operation of all four motors

impossible without risk of voltage drop or supply shutdown. To address this, the power supply was replaced with a Mean Well HRP 600 24 switch mode supply, providing 24 V at up to 25 A (600 W). This gives a peak current headroom of approximately 10 A above the estimated system demand, ensuring stable voltage during simultaneous motor operation. The additional headroom also provides capacity for future expansion of the system without requiring further power supply changes.

3.12 Data flow and code base

The system is controlled through a distributed software and hardware architecture, where the Raspberry Pi handles high-level decision making and image processing, while the Arduino Mega handles low-level motion control and sensor acquisition. An overview of the data flow between the main system components is shown in Figure 34. The complete code base developed for the project is available in the project GitHub repository [7]. This repository contains the Raspberry Pi code for image processing and high-level sequencing, as well as the Arduino code used for motion control, sensor reading, serial communication and any raw data not included in the report. The code consists of 13283 lines of code in total. 4936 lines of code for the main script. The remaining lines of code not used in the main script are for testing, analyzing and calibration.

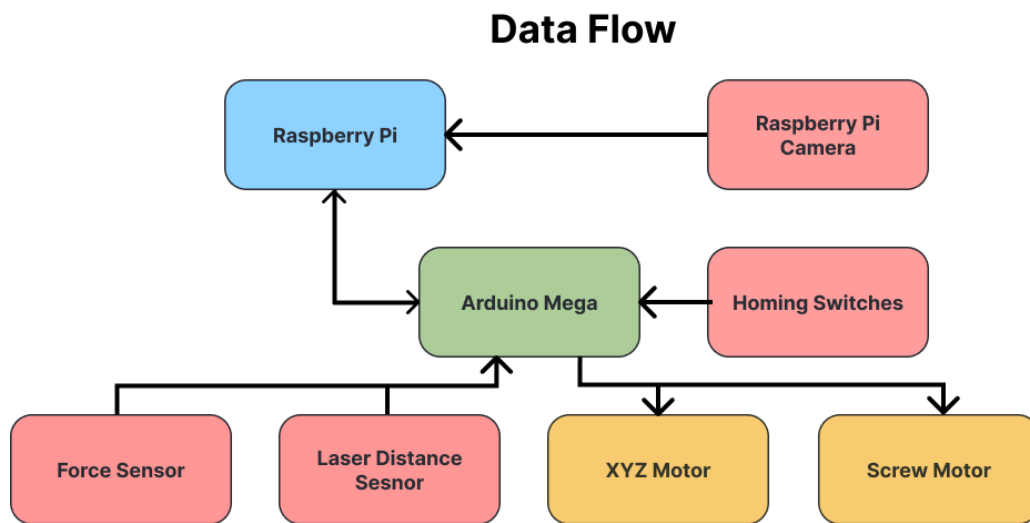


Figure 34: Data flow between the Raspberry Pi, Arduino Mega, sensors, camera, motors, and homing switches.

The Raspberry Pi functions as the main computing unit of the system. It communicates with the Raspberry Pi Camera for image acquisition and performs the computer vision routines described in Section 3.4. The Raspberry Pi is also responsible for deciding which operations must be performed and in which order. Motion commands are then sent from the Raspberry Pi to the Arduino Mega through serial communication. The Arduino Mega acts as the real time control layer for the machine. It receives high level commands from the Raspberry Pi and translates these into motor movements for the X,

Y , and Z axes, as well as the screw motor. The Arduino Mega also receives input from the homing switches, laser distance sensor, and force sensor. These sensor readings are used during homing, screw removal, press operations, and safety monitoring. The main code file follows a sequential workflow. First, the system performs a sweep of predefined camera positions across the workspace. This is detailed further in figure 9.

For each bit type, the machine moves to the bit changer station and picks up the correct bit. The detected operations requiring that bit are then ordered using a travelling salesman algorithm [2][10], so that the system follows an efficient route through the workspace rather than simply processing the targets in the order they were detected. After all operations requiring the current bit have been completed, the system returns the bit to the bit changer station and collects the next required bit. This procedure continues until all detected operations have been attempted. Examples of the generated travelling salesman paths for bit 1 operations, bit 2 operations, press operations, and the final verification sequence for the pressed objects can be viewed in Figure 35.

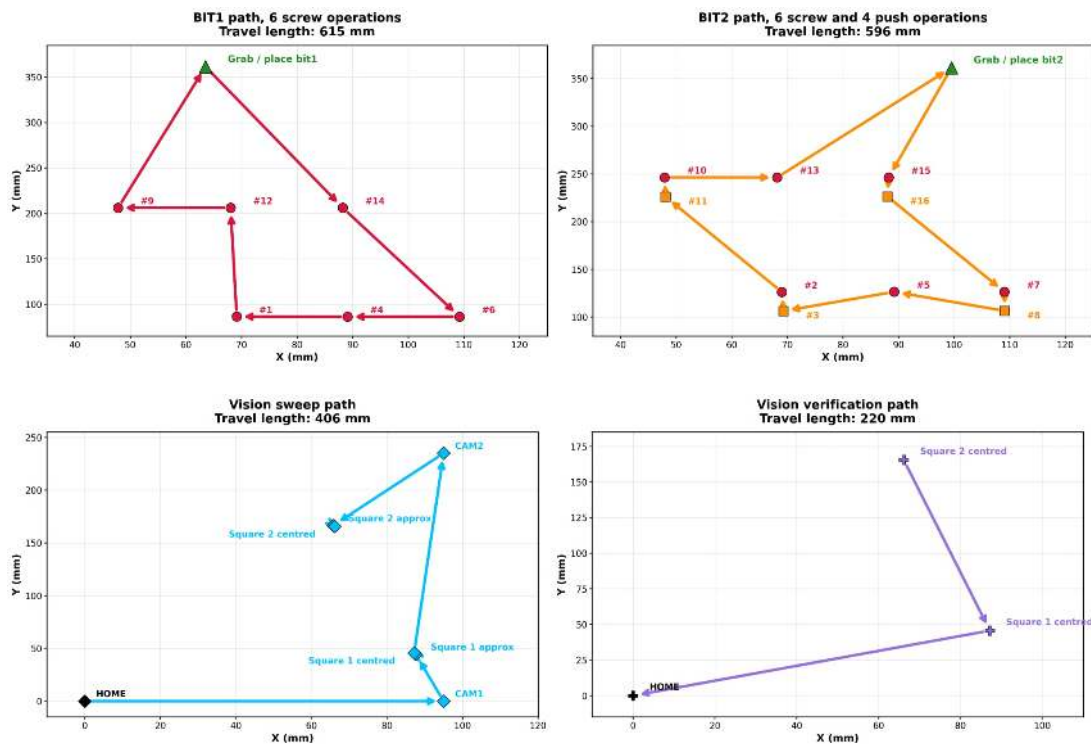


Figure 35: Examples of travelling salesman paths generated for bit 1 operations, bit 2 operations, vision sweep, and the final verification sequence used to inspect whether the press objects had been successfully removed.

After the screw and press operations have been completed, the camera is used for final verification of the press operations, as described in Section 4.5. The camera moves to the centre of each test object and captures an image of the press operation areas. These images are used to determine whether the press objects have been successfully removed or whether they are still present.

Finally, the system homes the gantry and generates a report describing the outcome of the performed operations. This report indicates which operations were completed

successfully and which operations failed or require manual intervention. In this way, the software does not only control the disassembly process, but also provides traceability of the final result. A safety related feature is also included in the main control logic. If the laser distance sensor produces unnatural readings, for example due to noise or sensor disturbance, the automatic sequence is interrupted. The system then prompts the operator to inspect the laser sensor and correct any potential issue before the process continues. This prevents unreliable sensor data from being used for contact detection or press operation decisions.

4 Subsystem verification

4.1 Verification of Automated Screw Removal

The purpose of this test was to verify whether the unified module could autonomously perform the complete screw removal sequence using the integrated laser feedback and rotational control logic. The test specifically focused on verifying automatic surface detection, screw engagement detection, pitch estimation, continuous screw removal, and detection of when the screw had been fully removed from the threaded insert. The test was not intended to evaluate the maximum achievable unscrewing torque of the system. Instead, the objective was to verify the functional behaviour of the mechanical design, sensor integration, and software coordination during automated screw removal.

The test object consisted of multiple identical screws mounted at different locations within the workspace. The screws were threaded into metallic heat inserts embedded within the test object, ensuring realistic metal threads during the removal process. For the present tests, an M4 umbraco screw with a length of 10 mm was used. The screws were tightened manually by hand in order to avoid introducing excessive torque requirements unrelated to the verification objective.

Prior to the test, the correct screwdriver bit corresponding to the screw type was manually mounted in the driven shaft. Automatic bit changing was therefore not included in this verification. Similarly, camera-based screw localization was not used during this test. Instead, the XY coordinates of each screw were hardcoded within the control software. The results of this initial unscrew test are shown in Table 2.

4.2 Axis Repeatability Test

To evaluate the positional repeatability of the gantry system, a repeatability test was conducted for the X -, Y -, and Z -axes. The purpose of the test was to determine whether the system could repeatedly perform the same commanded displacement after repeated homing cycles. For each test iteration, the system first performed the homing procedure in order to establish the global machine coordinate system. After homing, the distance between a fixed non-moving reference surface on the gantry structure and a moving component attached to the corresponding axis was measured manually using a caliper. This measurement served as the initial reference position.

The gantry was then commanded to move exactly 20 mm in the positive direction of the tested axis. After the movement had been completed, the distance between the same two reference surfaces was measured again. The measured displacement was subsequently calculated as the difference between the second and first measurements:

$$\Delta x = x_{\text{final}} - x_{\text{initial}}$$

Ideally, the calculated displacement should equal the commanded movement of 20 mm for every test iteration. The procedure was repeated 10 times for each axis indepen-

dently. The measured results are summarized in Table 1. The table shows the calculated displacement values for all three axes.

Table 1: Measured axis displacements during repeatability testing.

Test	ΔX [mm]	ΔY [mm]	ΔZ [mm]
1	19.99	20.00	19.99
2	19.98	19.99	20.03
3	20.00	20.00	20.05
4	20.00	19.98	20.09
5	20.00	20.00	20.04
6	20.00	20.00	19.98
7	19.99	19.99	20.03
8	20.01	20.01	20.02
9	20.01	19.99	20.01
10	20.00	19.99	20.01

The results show that all three axes repeatedly achieved displacement values very close to the commanded 20 mm movement. The largest observed deviation from the target value was 0.09 mm for the Z-axis during test 4, while most measurements deviated by less than 0.03 mm. All three axes showed highly consistent displacement measurements across the repeated tests. Small deviations from the ideal 20 mm displacement were observed, but these deviations were very limited in magnitude and most likely measurement errors and not gantry in-precision. Overall, the results indicate that the gantry systems lead values for each of the three axis where correctly input into the control software and that the control logic worked as expected.

4.2.1 Automated Unscrewing Procedure

At the start of each test sequence, the system first performed the homing procedure to establish the global machine coordinate system. The gantry then moved to one of the predefined screw coordinates before lowering the unified module in the Z-direction toward the workpiece surface. An overview of the complete automated unscrewing logic implemented in the software is shown in Figure 36.

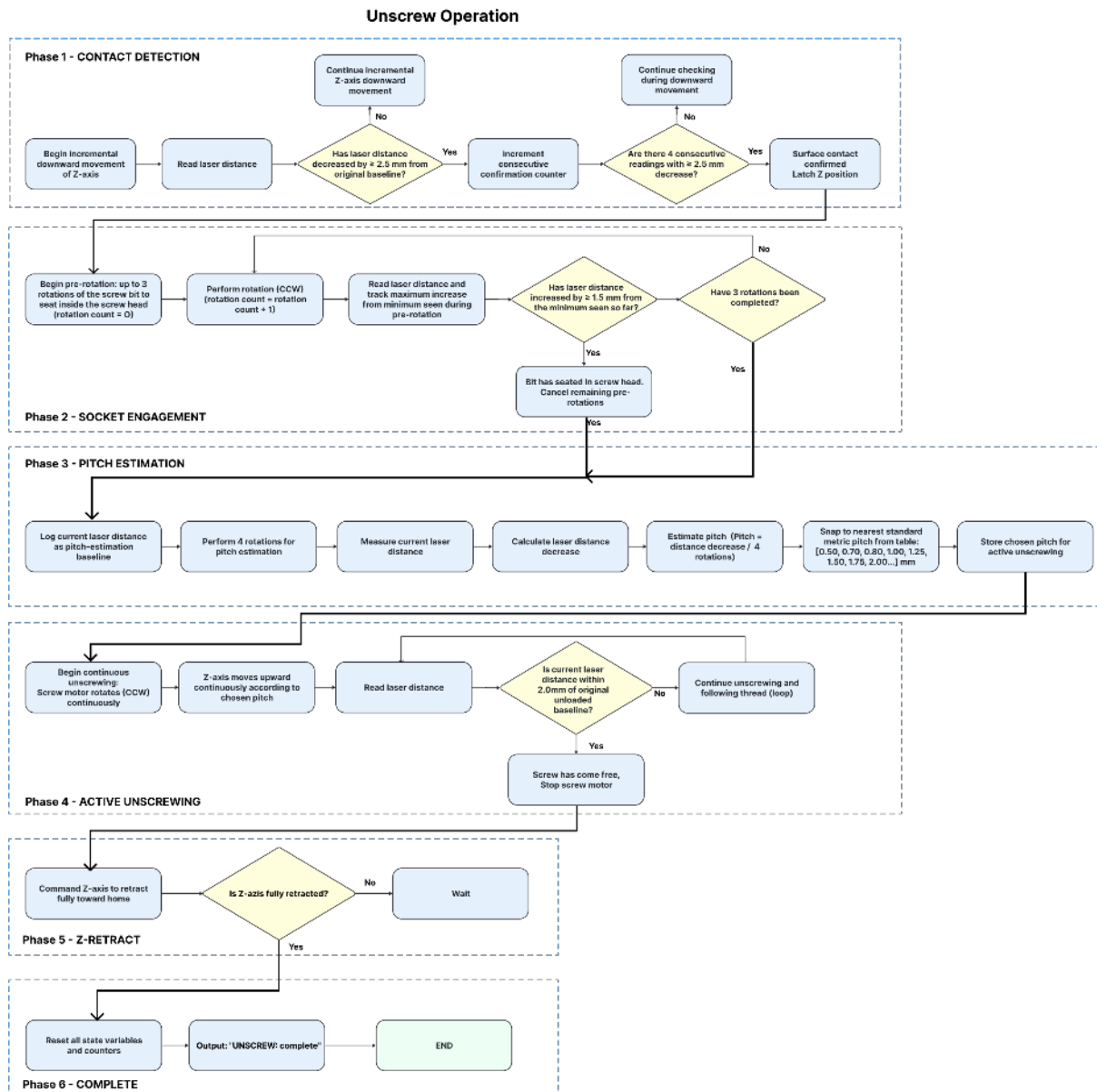


Figure 36: Flowchart of the implemented automated unscrewing logic, including surface contact detection, screw engagement verification, pitch estimation, continuous unscrewing, and operation termination.

A larger version of the flowchart is provided in Appendix C.1, where the figure has been divided into two images to improve readability. Surface contact was detected using the laser distance sensor described in section 3.7. Contact was defined as a measured decrease of 2.5 mm in the laser distance. After contact had been detected, the screw motor performed three initial rotations to ensure engagement between the screwdriver bit and the screw head before pitch estimation was started. If the laser sensor measured a vertical displacement increase during these rotations, the system interpreted this as successful screw engagement, as the bit has now fallen into the socket of the screw, and proceeded directly to pitch estimation, instead of completing the full three rotations. If no increase in distance is measured during the pre-rotations, then the pitch estimation would begin after the three rotations (see figure 37).

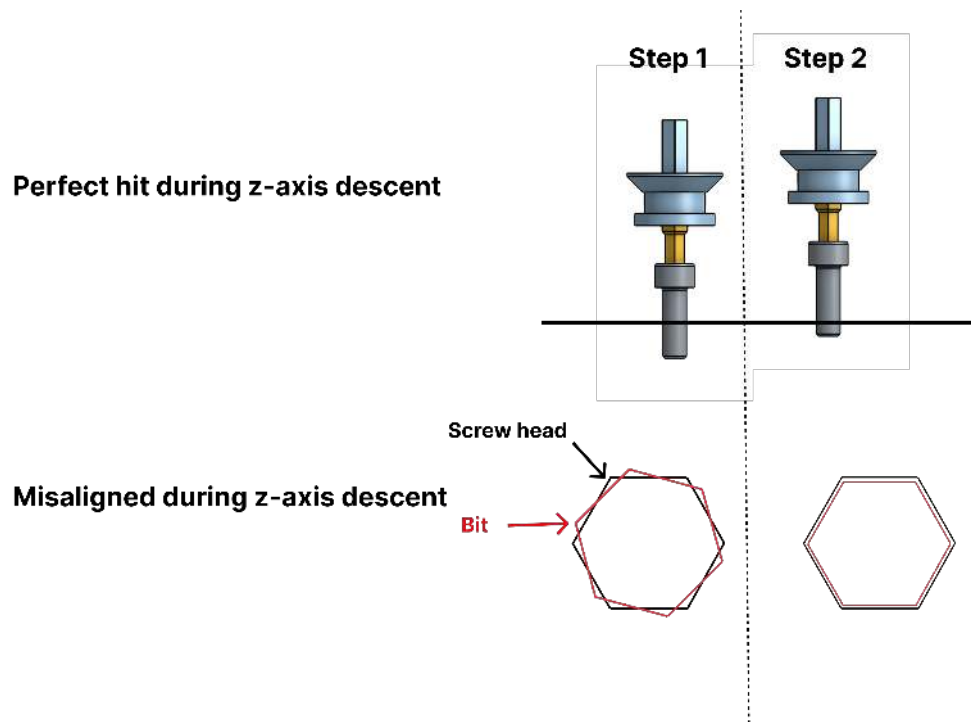


Figure 37: Illustration showing that when the bit aligns perfectly with the screw no socket drop is measured in the laser, the pre-rotations result in unscrewing of the screw followed by the pitch estimation. When initial misalignment occurs a drop-in can be measured by the laser and pitch estimation can start early.

If during the 4 pitch-estimation rotations no decrease in the measured distance occurred the system would interpret this as a missed engagement event, indicating that the screwdriver bit had not entered the screw head correctly. In this case, the operation was aborted and the tool retracted from the surface without continuing the unscrewing sequence. Following successful engagement, the screw motor continued rotating while the system simultaneously estimated the screw pitch based on the measured vertical displacement of the module during rotation. This estimated pitch was then used to coordinate the upward motion of the Z-axis during continued screw removal.

As the screw approached complete removal from the threaded insert, a characteristic behaviour was observed where the screw periodically dropped slightly downward between thread engagements while still retained by the screwdriver bit. Once this condition was reached the continued upward movement of the z axis resulted in an increased distance in the sensor reading letting the screw motor know that the screw was fully unscrewed. Hereby stopping rotations and retracting the module in the z axis.

4.2.2 Results and Observations

Early in the project a recurring error was undermining the reliability of the unscrew process. An analysis of the problem was conducted. To investigate what caused the error a specialized script was written to plot the laser distance sensor values alongside the time

it took to do the full unscrew cycle. In figure 38 the measurements taken during an unscrew operation can be seen. Here the results of a successful unscrew operation can be seen. It is clearly seen when the bit contacts the top of the screw and when the bit aligns into the head of the screw. Then a period of decreasing distance measured as the pitch of the screw is estimated. Followed by a constant distance because the z axis moves up with the screw (called active unscrewing). Finally, the screw head moves up no longer, and the distance measured goes back to the baseline as the z axis continues to rise.

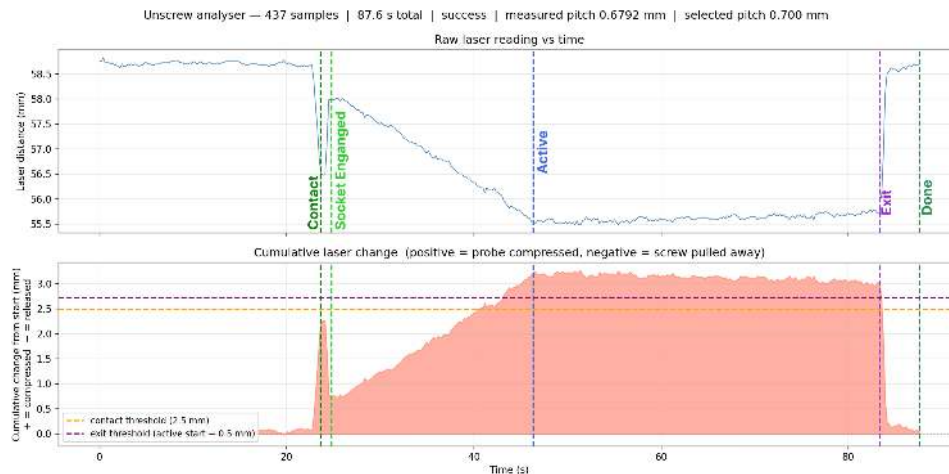


Figure 38: Results from a successful unscrew operation.

The problem destroying the 100% reliability, can be viewed in figure 39. As seen in the figure, sometimes the laser would output a random big decrease in distance. If this spike occurred during the probing phase (moving down towards the screw) of an unscrew operation the system would conclude that contact had been achieved and start the unscrew process. Then once the pitch estimation failed the process failed.

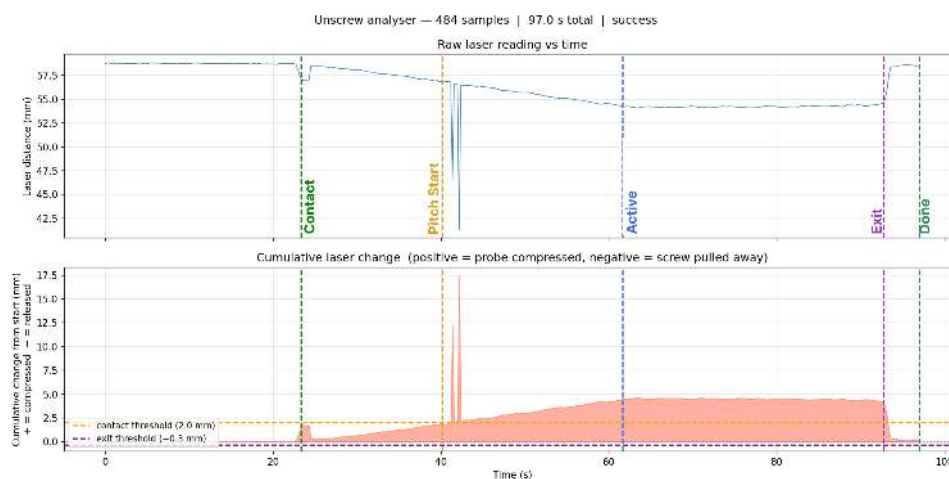


Figure 39: Results from an unscrew operation showing the sensors output spike.

To fix it, all measurements that deviated more than 2 mm from the previous reading were omitted. A spike of 2 mm from one reading to the next would never happen organically since measurement happens multiple times every second. A verification test was conducted to evaluate the fix and in general the autonomous unscrewing procedure and provide data for further improvements. The test was repeated 20 times, as shown in Table 2. The system achieved a 100% success rate in removing the screws and a 95% success rate in estimating the screw pitch correctly. It should be noted that the operation was still completed successfully in the trial where the pitch was estimated incorrectly.

Unscrew #	Unscrew Fail	Unscrew Success	Pitch Measured	Pitch Estimated	Actual Pitch	Pitch Fail	Pitch Success
1	0	1	0.6426	0.7	0.7	0	1
2	0	1	0.6489	0.7	0.7	0	1
3	0	1	0.6228	0.7	0.7	0	1
4	0	1	0.6551	0.7	0.7	0	1
5	0	1	0.6238	0.7	0.7	0	1
6	0	1	0.6170	0.7	0.7	0	1
7	0	1	0.6428	0.7	0.7	0	1
8	0	1	0.2992	0.5	0.7	1	0
9	0	1	0.6364	0.7	0.7	0	1
10	0	1	0.6363	0.7	0.7	0	1
11	0	1	0.6109	0.7	0.7	0	1
12	0	1	0.6491	0.7	0.7	0	1
13	0	1	0.6551	0.7	0.7	0	1
14	0	1	0.6493	0.7	0.7	0	1
15	0	1	0.6175	0.7	0.7	0	1
16	0	1	0.6232	0.7	0.7	0	1
17	0	1	0.6427	0.7	0.7	0	1
18	0	1	0.6175	0.7	0.7	0	1
19	0	1	0.6679	0.7	0.7	0	1
20	0	1	0.6361	0.7	0.7	0	1

Table 2: Results from the repeated unscrewing verification test. The test was repeated 20 times. All screws were successfully removed, while the screw pitch was correctly estimated in 19 out of 20 trials.

The incorrect pitch estimate occurred because the screwdriver bit required more than one rotation to fully engage with the screw head. Based on this observation, the initial engagement phase was changed from one to three full rotations before pitch estimation was allowed to begin. This increased the likelihood that the bit was properly aligned with the screw before the pitch was estimated. A scale of 1.05 was also added to the measured pitch before the pitch was estimated as a consistent pattern was that the pitch was measured too low.

The verification tests showed that the system was capable of autonomously detecting surface contact, identifying successful screw engagement, estimating the screw pitch, coordinating Z axis motion during unscrewing, detecting failed engagement attempts, and determining when the screw had been fully removed from the threaded insert. The system

was also capable of detecting failed screw engagement attempts. If surface contact was detected, but no characteristic vertical displacement change occurred during the initial rotational verification step, the system correctly identified the event as a failed engagement attempt and aborted the procedure without continuing the unscrewing sequence. This was tested by deliberately calling the `unscrew` command at locations where no screw was present.

Across the repeated trials, the automated unscrewing procedure operated reliably when the predefined XY coordinates were sufficiently accurate. Under these conditions, the system consistently succeeded in engaging and removing the screws without manual intervention. The experiments therefore demonstrated that the combined mechanical design, laser feedback system, and control logic were capable of autonomously performing the intended screw removal procedure under the tested operating conditions.

4.3 Verification of Camera Based Screw Localization

The purpose of this test was to verify that the camera based localization procedure could determine the global machine coordinates of screws and press locations with sufficient accuracy for the gantry to subsequently position the tool above the detected target. This verification is directly related to deliverable D1, where the vision system must report usable machine frame coordinates for every screw and press site in the workspace.

The validation was performed on the red test component used throughout the development which can be viewed in figure 40. The component contained six screw positions and two press positions arranged on a known 20 mm grid. The reference position of a screw was measured using the machine `position` command (see git repository at [7]), and the remaining operation positions were derived from the known grid spacing. This provided a fixed ground truth layout against which the vision based estimates could be compared.



Figure 40: Test object used for validation of the camera based screw localization. The object contains six screw locations and two press locations with known relative spacing, allowing the detected machine coordinates to be compared against reference positions.

For each validation run, the system first moved the camera to the approximate inspection position (known in the code as CAM1), captured an image, and used the red square detection to iteratively centre the component in the image. Once the component was centred (within 0.5mm of the center), the screw and press locations were detected using contour extraction. The detected pixel coordinates were then transformed into machine coordinates using the calibrated camera model and the camera to toolhead offset. See Appendix A for details as to how this was done.

Running a validation on all eight operations showed that, the errors varied across the component. The RMS (Root Mean Squared) combined error was 1.085 mm, with a maximum combined error of 1.681 mm. The mean error was also position dependent: the lower right region of the component showed a negative X bias, while the left side showed larger positive Y errors. This indicated that the error could not be corrected reliably by changing only the constant camera offset.

A second validation was performed later. This run showed the same general behaviour, with an RMS combined error of 1.208 mm and a maximum combined error of 1.729 mm. The repeated position dependent pattern suggested that the remaining error was caused by perspective distortion from the camera mounting angle and lens distortion, rather than random detection noise. This was supported by the low standard deviation within each operation position, meaning that the system was repeatable, but systematically biased depending on where the feature appeared in the image.

To reduce this error a homography calibration was introduced to map pixel positions to machine coordinates. This allowed the coordinate conversion to account for the fact that the effective millimetre per pixel value changes across the image when the camera is not perfectly normal to the work surface. After applying the homography calibration, the vision validation was repeated. The final validation run used 10 iterations and included all eight operations. The homography calibration reported an RMS error of 0.0887 mm and a maximum calibration error of 0.2574 mm over 25 calibration points. The resulting localization accuracy improved substantially. Across all detected operations, the RMS combined validation error was reduced to 0.247 mm, and the maximum combined error was reduced to 0.497 mm.

Table 3: Summary of camera localization validation before and after calibration.

Trial #	Operations tested	RMS combined error [mm]	Max combined error [mm]
1	80	1.085	1.681
2	80	1.208	1.729
3	80	0.247	0.497

The results show that the original vision pipeline was able to detect the correct features, but that the coordinate transformation was not sufficiently accurate before homography calibration. The main issue was not detection repeatability, but systematic spatial distortion across the image. After introducing the updated camera calibration and homography based coordinate mapping, the localization error was reduced to below 0.5 mm for all measured operations in the final validation run. This accuracy was considered sufficient for the subsequent screw removal and press operation tests, since the remaining error was small compared to the screw head diameter and the mechanical compliance available during bit engagement.

4.4 Verification of Automatic Bit Changing

The purpose of this verification was to evaluate whether the bit changer station could reliably support passive release, storage, and re-engagement of bit adapters without the use of dedicated actuators. The verification focused primarily on evaluating alignment consistency, adapter seating behaviour, and repeatability of the pick-and-place sequence.

4.4.1 Test Procedure

The verification was performed using the bit changer station geometries described in 3.10.1. During the tests, bit adapters containing preloaded screwdriver bits were placed within the station slots. The XY coordinates of each slot were hardcoded within the control software. For a visual representation of the grab and place sequences see figures 31 and 32. Below, the testing procedure is described.

At the beginning of each test sequence, the gantry first performed the homing procedure before moving to the predefined XY coordinates corresponding to the desired tool location. The driven shaft was then lowered in the Z-direction to a predefined height slightly below the contact height of the bit adapter. This caused the adapter to contact the driven shaft and pressed the module slightly upward along the linear rails, compressing the spring by a small amount.

At this stage, the male hexagonal section of the bit adapter and the female hex socket of the driven shaft were not necessarily rotationally aligned. To ensure alignment, the screw motor performed two full rotations while the driven shaft remained pressed against the adapter. During this rotation, the adapter geometry could align with the socket geometry, allowing the adapter to enter the driven shaft socket. After the alignment rotations, the driven shaft was lowered further to ensure that the magnetic retention inside the socket attached the adapter to the driven shaft. Following successful attachment, the Z-axis was raised slightly before the gantry moved laterally out of the station slot, thereby extracting the tool from the station.

To return the tool to the station, the gantry moved to the predefined slot position and lowered the adapter into the station geometry. The Z-axis was then raised while the station retention geometry constrained the adapter from moving upward together with the driven shaft. Once the upward retention force exceeded the magnetic holding force between the adapter and the driven shaft socket, the adapter was released and remained seated within the station slot.

Following release, the gantry again moved to the predefined pickup height, performed the two alignment rotations, and attempted re-engagement with the adapter. This sequence was repeated 20 times for each station version in order to evaluate the consistency and repeatability of the passive pick-and-place mechanism.

Table 4: Repeated bit changing verification tests for three station versions.

Attempt	Version 2		Version 3.1		Version 3.2	
	Fail	Success	Fail	Success	Fail	Success
1	0	1	1	0	0	1
2	0	1	0	1	0	1
3	0	1	0	1	0	1
4	0	1	0	1	0	1
5	0	1	0	1	0	1
6	0	1	0	1	0	1
7	0	1	0	1	0	1
8	0	1	0	1	0	1
9	0	1	1	0	0	1
10	0	1	0	1	0	1
11	0	1	1	0	0	1
12	0	1	0	1	0	1
13	0	1	1	0	0	1
14	0	1	1	0	0	1
15	0	1	0	1	0	1
16	0	1	0	1	0	1
17	0	1	0	1	0	1
18	0	1	1	0	0	1
19	0	1	1	0	0	1
20	0	1	0	1	0	1
Success rate	100%		65%		100%	

4.4.2 Version 2 Results

Version 1 seen in figure 28 was tested by hand and quickly concluded that it did not provide enough support for the bits to even qualify for a machine test. The version 2 station geometry (seen in figure 29) demonstrated that the passive release and re-engagement concept was mechanically feasible. As shown in Table 4, version 2 achieved a 100% success rate across 20 repeated attempts.

However, the successful operation of version 2 depended strongly on the initial manual placement of the bit adapter within the station. During testing, the adapter had to be placed very accurately by hand in order to achieve reliable operation. This geometry provided only limited lateral support after release, meaning that vibrations from the gantry motion or small disturbances could cause the adapter to shift position inside the slot.

This introduced a practical limitation for real operation. If the adapters needed to be loaded, replaced, or reorganized by an operator, the station required unnecessarily careful manual handling to maintain reliable alignment. The geometry therefore demonstrated that the passive concept itself was feasible, but also revealed that additional mechanical constraint of the adapter was necessary in order to make the station robust against vibrations and less sensitive to operator placement accuracy. During testing, bending of the station structure was also observed, particularly during repeated insertion and extraction

cycles. Although this did not prevent successful operation under controlled conditions, it further motivated the development of a revised geometry with improved structural support and constraint. These observations motivated the development of version 3.1.

4.4.3 Version 3.1 Results

Version 3.1 was developed to improve the positional stability of the adapter inside the station. The geometric constraints securing the adapter were extended further downward, providing a secondary support plane in addition to the upper retention geometry. This was intended to reduce lateral movement of the adapter after release and improve the consistency of subsequent re-engagement attempts.

Physical testing showed that the revised geometry significantly improved the positional stability of the adapter compared to version 2. However, despite the improved guidance, the repeated verification tests only achieved an overall success rate of approximately 65%, as shown in Table 4.

The primary reason for the reduced success rate was not insufficient guidance, but rather that the clearance of the station geometry had become too tight. In multiple tests, the adapter was unable to fall fully into the slotted geometry after release because the fit between the adapter and the station walls was overly constrained. In some cases, the adapter became partially wedged inside the slot geometry.

When this occurred, the magnetic retention force of the driven shaft was no longer strong enough to overcome the frictional grip between the adapter and the station during pickup. As a result, the driven shaft aligned successfully with the adapter, but the adapter itself remained mechanically locked inside the station slot, causing the pickup operation to fail. The tests therefore demonstrated that increasing geometric guidance alone was not sufficient. The station geometry also required sufficient clearance to ensure that the adapter could move and self-align freely during both release and pickup operations. These observations motivated the development of version 3.2.

4.4.4 Version 3.2 Results

Version 3.2 was developed as a direct refinement of version 3.1. The overall geometric concept was retained, since the additional support and guidance introduced in version 3.1 had clearly improved the positional stability of the adapter. The primary modification in version 3.2 was therefore an increase in the mechanical clearances between the adapter and the surrounding station geometry. The tighter regions responsible for the sticking behaviour observed in version 3.1 were revised to provide additional freedom of movement while still maintaining sufficient guidance to constrain the adapter position after release.

Repeated verification testing of version 3.2 demonstrated a significant improvement in reliability. As shown in Table 4, the system achieved a 100% success rate across the repeated pick-and-place experiments. The adapter consistently seated correctly after release, and the driven shaft was able to repeatedly align with, attach to, extract, and return the adapter without failed pickup events. The tests therefore demonstrated that the

combination of increased geometric guidance together with sufficiently large mechanical clearances was necessary for reliable passive bit changing. The final geometry successfully balanced these two competing requirements, providing both stable positioning and sufficient freedom for consistent self-alignment during autonomous operation.

4.5 Verification of Press Operation

The purpose of the press operation verification was to evaluate whether the system was capable of performing a press operation and subsequently determining whether the operation was successful or unsuccessful. Two validation experiments were conducted. In the first test, the system attempted to press out an object that could physically be removed from the surrounding material. In the second test, the system attempted to press an object that could not be removed. For both cases, 10 consecutive tests were performed.

The implemented software logic for the press operation is illustrated in Figure 41. The press operations were first localized using the vision system described previously. After localization, the gantry moved the press tool above the target position in the X - and Y -directions. A slow incremental downward movement in the Z -direction was then initiated. During this movement, the laser distance sensor monitored the displacement of the spring-loaded mechanism.

Initially, the system moved downward until the laser sensor measured a decrease in distance of 2.5 mm relative to the baseline measurement. This decrease was interpreted as surface contact. After surface contact, the system continued moving downward in order to apply a pressing force to the object.

Three stop conditions were implemented:

- A further decrease in measured laser distance of 15 mm.
- A measured force exceeding 5 N.
- A downward movement to absolute 75 mm in the Z -direction.

If any of these conditions were reached, the downward movement was terminated.

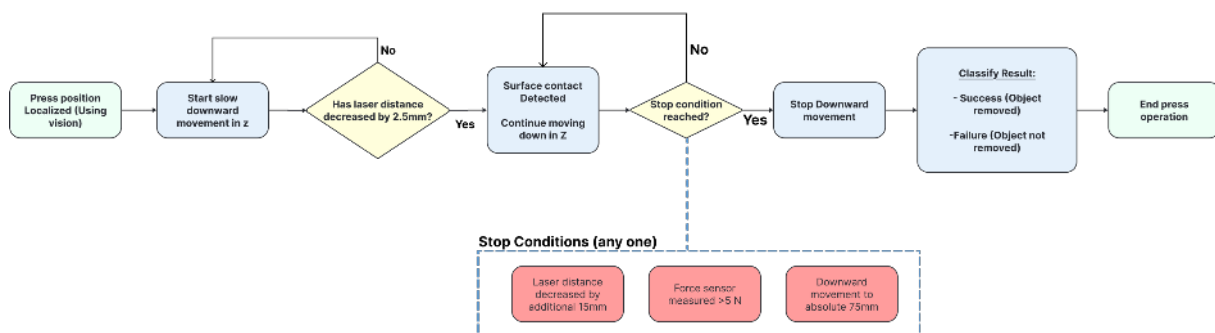


Figure 41: Flowchart of the press logic

For each test, the system logged relevant operational data to a CSV file, including the software status messages, measured force values, operation duration, and whether the operation was classified as successful or unsuccessful. This made it possible to evaluate the repeatability and reliability of the implemented press logic.

For the removable press object, all 10 test cycles were classified as successful. The measured operation duration remained highly consistent at approximately 23.7 s for all tests, while the measured forces remained low throughout the experiments. However, during testing it was observed that the removable test object could be pressed out before the laser sensor detected the intended 2.5 mm displacement. This occurred because the removable test object was only loosely constrained within the surrounding material. As a result, the spring force and the weight of the module alone was sufficient to displace the object before significant resistance was encountered.

Because of this limitation, computer vision was used as the final verification method for removable press objects. After the press operation had been completed, the camera moved to the centre of each test object and captured an image of the press area. The vision system then evaluated whether the press object was still present or whether it had been removed. The results for the removable press object are shown in Table 5.

Table 5: Validation results for the removable press object.

Parameter	Result
Number of tests	10
Successful press operations	10/10
Vision-confirmed removals	10/10
Average maximum force	0.07 N
Maximum measured force	0.11 N

For the non-removable press object, the object could not be displaced. In these cases, the spring mechanism compressed instead, which caused the force sensor to stop the operation. The system therefore classified the operation as unsuccessful. The results for the non-removable press object are shown in Table 6.

Table 6: Validation results for the non-removable press object.

Parameter	Result
Number of tests	10
Correctly rejected press operations	10/10
Vision-confirmed object still present	10/10
Average maximum force	10.4 N
Maximum measured force	12.67 N

The results show that the system was able to distinguish between removable and non-removable press objects under the tested conditions. For the removable object, the final verification depended on computer vision, since the object was removed before the laser-based contact detection could reliably identify the pressing event. For the non-removable

object, the force and displacement thresholds were sufficient to stop the operation and classify the attempt as unsuccessful.

This limitation is mainly caused by the simplified test object. In a realistic industrial application, press-fitted or corroded components would typically be held more firmly in place. Under such conditions, the spring mechanism and weight of the module alone would not be expected to press the object out. Instead, the preload would engage the force transfer structure, allowing the rigid mechanical structure of the system to apply the pressing force.

5 System integration and results

5.1 Full System Testing Methodology

Full system testing was performed using a dedicated validation script, `main_validation.py`, which executes the complete production pipeline and appends a post run accuracy assessment to each run. Each run follows the sequence described in section 3.12. The system prints a per operation result table and prompts the operator to enter two ground truth values: (i) the total number of operations that actually failed, and (ii) the number of operations the system placed in the wrong outcome category (said *fail* but the operation succeeded, or said *pass* but the operation failed). Two performance metrics are derived from these inputs:

- **True fails:** unexpected failures after accounting for the two press buttons that are permanently blocked by design and therefore always expected to fail:

$$\text{true_fails} = \text{actual_fails} - 2.$$

- **Classification accuracy:**

$$a = \frac{N_{\text{operations}} - N_{\text{misclassified}}}{N_{\text{operations}}}.$$

All run metrics including per category counts, true fails, misclassifications, and classification accuracy are appended to a persistent CSV file for analysis. The test items used across all 21 runs was 2 test objects with 2 press operations and 6 screw operations on each, resulting in 12 screw operations (M4 and M5) and 4 press operations. Of the 4 press operations, 2 are permanently blocked by the board design and are always expected to yield a *fail* outcome. They are included in *actual_fails* but excluded from *true_fails*. The acceptance criteria for the final validation phase were a classification error rate below 5% and an operations error rate below 5% as described in section 2.4.

5.2 Full System Test Results

A total of 21 full system runs were conducted. The first 11 runs formed an iterative development phase in which encountered failures drove successive changes to the control software. The subsequent 10 runs were performed consecutively after the bug fixing phase.

5.2.1 Initial Runs (1–11)

Table 7 presents the results of the first eleven runs. Press Fails is the raw count of press operations reported as failed by the vision system. 2 failures are always expected by design (two permanently blocked buttons), so values above 2 indicate additional unexpected press failures.

Run	Screw Fails	Press Fails	True Fails	Misclassified	Op. Success (%)	Class. Accuracy (%)
1	0	2	0	0	100.00	100.00
2	0	2	1	1	92.86	93.75
3	0	2	0	0	100.00	100.00
4	2	2	2	0	85.71	100.00
5	0	2	0	0	100.00	100.00
6	0	2	0	0	100.00	100.00
7	0	2	0	0	100.00	100.00
8	5	4	7	0	50.00	100.00
9	2	3	3	0	78.57	100.00
10	0	2	0	0	100.00	100.00
11	8	4	10	0	28.57	100.00
Avg	1.55	2.45	2.09	0.09	85.06	99.43

Table 7: Summary of operation failures and classification performance across validation runs.

Seven of the eleven runs completed without any true failures. The dominant failure mode across the remaining four runs was erratic output from the laser distance sensor. In run 4, two M5 screw operations aborted due to serial communication timeouts. In run 8, a sudden sensor instability event caused five M5 screw operations to abort. Two press operations on the same square were also affected, resulting in seven true fails in total. Run 9 followed the same pattern on a smaller scale, with two screw operations and one press operation failing, yielding three true fails. Run 11 produced the largest failure event of the entire testing phase: the laser baseline reading settled at approximately 56.1 mm across all screw positions (even though the true distance was ≈ 59 mm.) which was well below the expected ≈ 59 mm. Which caused all eight screw operations to be aborted and both additional press operations to fail, resulting in ten true fails. Run 2 produced one misclassification in which one failed operation was incorrectly reported as successful by the system.

Following these runs, several measures were introduced into the control software to address the laser sensor behaviour, including improved baseline validation, spike filtering logic, and operator guided recovery prompts for out of range readings. These changes to the control logic resulted in the following improvements.

5.2.2 Final Ten Runs (12–21)

Table 8 presents the results of the ten consecutive runs conducted after the bug fixing phase.

Table 8: Full system test results, runs 12–21. Press Fails of 2 in every run reflects the two always-expected blocked-button failures. No additional press failures occurred.

Run	Screw Fails	Press Fails	True Fails	Misclassified	Op. Success (%)	Class. Accuracy (%)
12	0	2	0	0	100.00	100.00
13	0	2	0	0	100.00	100.00
14	0	2	0	0	100.00	100.00
15	0	2	0	0	100.00	100.00
16	0	2	0	0	100.00	100.00
17	0	2	0	1	100.00	93.75
18	0	2	0	0	100.00	100.00
19	0	2	0	0	100.00	100.00
20	1	2	1	0	92.86	100.00
21	0	2	0	0	100.00	100.00
Avg	0.20	2.00	0.10	0.10	99.29	99.38

Eight of the ten runs completed with zero true fails and 100 % classification accuracy. In run 17, the system reported an M5 screw operation as a failure after a pitch measurement step failed on retry. The operator confirmed the operation had in fact succeeded, recording this as a misclassification (false negative). In run 20, the laser baseline reading dropped to 41.3 mm on a single M5 screw operation, causing it to abort. The operator confirmed the failure was real, and the system’s classification was correct (no misclassification). No unexpected press failures occurred in any of the ten runs. Table 9 summarises the aggregate performance across the 10 final runs.

Table 9: Aggregate performance metrics for the final ten runs. The true fail rate and operation success rate are computed over the 140 candidate operations (14 per run $\times$ 10 runs, excluding the 2 permanently blocked buttons per run). The classification error rate and accuracy are computed over all 160 operations (16 per run $\times$ 10 runs).

Metric	Count	Rate	Criterion
True fails	1	0.71 %	< 5 %
Misclassifications	1	0.63 %	< 5 %
Op. success rate	—	99.29 %	> 95 %
Classification accuracy	—	99.38 %	> 95 %

6 Discussion

6.1 Mechanical limitations of the unified module design

The unified module developed in this project successfully demonstrates the feasibility of the design concept, but its mechanical layout has several limitations that became apparent during assembly and testing. These limitations did not prevent operation of the prototype, but they constrain the maximum reliable load the system can sustain and would need to be addressed before the design could be advanced toward a production capable platform. Each limitation is described below.

The force sensor described in Section 3.6 is mounted such that the axial reaction from the driven shaft is transmitted into the top fixed plate of the module housing. However, the contact point between the sensor and the plate lies at a significant distance from any point at which the plate is rigidly fastened to the surrounding frame. As a result, the entire axial reaction force is first carried by the probe in bending, and then transferred into the frame through only two fastening screws. Under axial press loads approaching the sensor's 2450 N upper range, this load path creates a substantial bending moment on the top plate and a load on the two fasteners. A sketch of the issue described can be seen in figure 42.

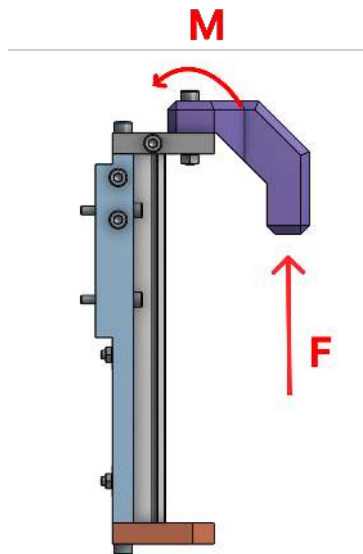


Figure 42: CAD rendering illustrating the force hitting the force sensor probe.

Two consequences follow. First, plate deflection under load reduces the accuracy of the force measurement, since the sensor experiences relative motion of its mounting surface rather than pure compression, which introduces a load dependent measurement error that the calibration in Section A.1 cannot correct. Second, the two carrying screws are loaded well beyond what would be expected of an ordinary fastening role, increasing the risk of fastener fatigue or gradual loosening over repeated press cycles. A more robust

arrangement would route the reaction force directly into a rigidly anchored structural member, eliminating the bending path through the plate entirely.

The driven shaft is not located along the central axis of the module housing, but offset laterally to accommodate the gear pair described in Section 3.8. During an axial press operation, the applied force passes downward through the driven shaft, while the reaction is distributed across the surrounding module structure. Because the driven shaft is not coaxial with the geometric centre of the module, the resulting force couple introduces a tilting moment equal to the product of the applied axial force and the lateral offset between the driven shaft axis and the centre of structural support. With 30 mm offset from the middle (in the y direction) of the module, and at peak press loads the tilting moment loads the linear guide rails in a direction they are not optimised to resist. The practical consequences are that the asymmetric loading accelerates wear of the linear guide bearings on the loaded side and in worst cases can stop the movement of the sled on the rails. See figure 43 for an illustration.

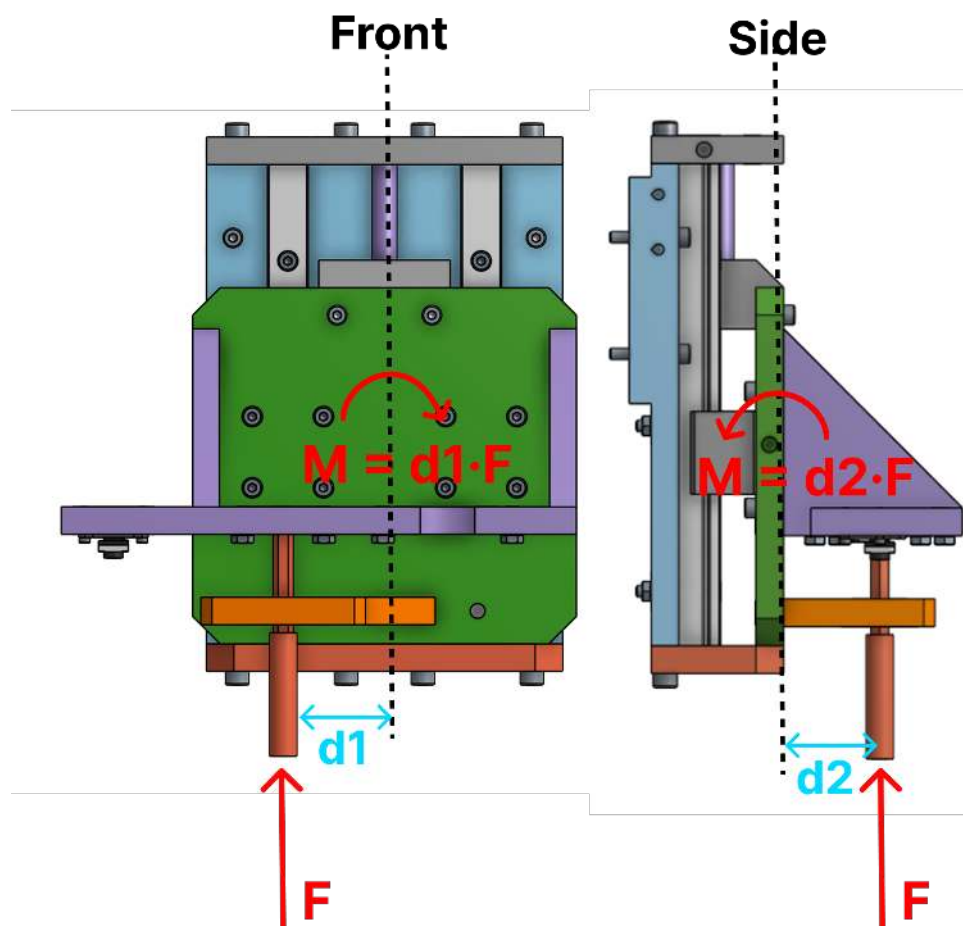


Figure 43: Uneven forces illustrated.

A coaxial arrangement, in which the driven shaft passes through the geometric centre of the module, would eliminate the tilting moment entirely around the x-axis. As described in Section 3.9.2, the original design intent for the contact circle at the top of the driven shaft included a 1 mm hexagonal recess machined into one face, which would

have provided positive geometric keying between the contact circle and the hex tip of the driven shaft. Since this feature could not be manufactured with the equipment available at DTU Skylab, the contact circle was instead produced with a circular bore and press fitted onto the hex tip. The press fit relies entirely on friction to maintain concentricity and resist relative rotation. Although the contact circle is not part of the primary torque transmission path during screw removal, since torque flows directly from the gear pair through the hex driven shaft to the bit, the contact circle still rotates with the driven shaft and is in contact with the rotating race of the thrust bearing. The friction inside the bearing applies a small parasitic torque to the contact circle to driven shaft joint on every rotation. Over many cycles, this can cause the contact circle to walk on the driven shaft, gradually losing concentricity. The consequence is increased radial runout at the thrust bearing interface, which manifests as vibration during operation and accelerated, uneven wear of the bearing race. The intended hex recess solution would have provided positive geometric resistance to this slip and is the preferred long term fix.

The two bushings that locate the driven shaft radially were originally intended to be retained by an H7/n6 interference fit on a machined round section of the driven shaft. Because the driven shaft turned out to be hardened (Section 3.9.2), this approach could not be implemented and was replaced with set screws clamping against the flats of the hexagonal driven shaft cross section. Two consequences follow from this. First, the clamping action is concentrated at two points around the driven shaft circumference rather than distributed uniformly, which leads to small asymmetries in the bushing concentricity and contributes to radial runout of the driven shaft that is then carried through to the gear mesh and the screwdriver bit. Second, set screws are inherently susceptible to loosening under cyclic loading and vibration, particularly when the clamping element is steel against steel without a thread locking compound. Loss of clamping force would allow the bushing to migrate axially along the driven shaft, displacing the gear and the radial bearing it carries, and ultimately destroying the meshing geometry of the gear pair. This was observed during prototype testing, it therefore represents a latent failure mode that would need to be eliminated before extended operation.

The gear pair described in Section 3.8 is manufactured by 3D printing. While the herringbone geometry was selected specifically because it suits additive manufacturing well, the resulting gears are made from a polymer with significantly lower tensile strength, hardness, and creep resistance than machined metal alternatives. Under the 3 Nm peak torque of the screw motor, the loaded teeth experience contact stresses high enough to cause measurable plastic deformation over time, particularly at the tooth root where bending stresses concentrate. The consequences are progressive loss of tooth profile accuracy, increasing backlash between the two gears, and eventual tooth failure under sustained operation. For the prototype, this is acceptable because the duty cycle is low and the gears can be reprinted at negligible cost. For any system intended to operate for production realistic durations, the gears would need to be replaced with machined metal versions, and the choice between herringbone and spur geometry revisited in light of the higher manufacturing cost of herringbone in metal. The redesign approaches that would address each of these limitations are discussed in the outlook section (Section 8).

6.2 Results of the validation tests

The subsystem verification tests and full system integration runs, together provide a clear picture of whether the prototype met the deliverables defined in Section 2.4. Deliverables **D1** and **D2**, concerning screw and press site localisation and screw size classification, were addressed through the camera validation in Section 4.3. Before homography calibration the RMS localisation error was around 1.1 mm, with systematic position dependent bias indicating the source was perspective distortion rather than random noise. After introducing homography based coordinate mapping the RMS error dropped to 0.247 mm with a maximum of 0.497 mm across all detected operations. Screw size classification from contour diameter estimation was correct across the validation batch. Both deliverables can therefore be considered met.

Deliverable **D3**, toolhead positioning repeatability, was confirmed in Section 4.2, where repeated x, y and z movements were tested. The machine proved to move precise enough to satisfy the deliverable.

Deliverables **D4** and **D5**, bit engagement detection and screw removal, were evaluated across 20 repeated unscrewing trials in Section 4.1. The system achieved a 100% screw removal success rate and correctly estimated pitch in 19 of 20 trials, with the one incorrect estimate still resulting in a successful removal. The laser based contact and engagement detection logic proved reliable once spike filtering was introduced, and the system correctly coordinated Z-axis motion with screw motor rotation throughout the removal sequence. Both deliverables are met.

Deliverable **D6**, failure detection, was verified both actively and deliberately. The system correctly identified missed engagement events when the unscrew command was issued at locations with no screw present, halting the operation rather than continuing blindly. Motor stall detection via the CL57T alarm output was not tested in this prototype and the deliverable is therefore partly met.

Deliverables **D7** and **D8**, axial press execution and force monitoring with hard stop, were confirmed in Section 4.5. For the removable press object all 10 trials succeeded, verified by computer vision since the object displaced before the laser threshold was reached. For the non removable object all 10 trials were correctly halted and classified as unsuccessful, with the force sensor triggering the stop at an average of 10.4 N. The passive transition between screw mode and press mode required no manual reconfiguration. Both deliverables are met, though it is noted that for very loosely constrained objects the force sensor may not engage before displacement occurs, making vision based post verification a necessary complement rather than an optional one. For a real press operation more force than the weight of the moving part of the module and the springs resistance to compression is needed. Therefore pure distance sensor output combined with the force sensor would be sufficient to determine the success or failure of an operation. Which would leave the final vision pass as an optional confirmation.

Deliverable **D9**, passive bit changing, was verified through three station design iterations in Section 4.4. Version 3.2 achieved a 100% success rate across 20 repeated pick and place cycles, demonstrating that the combination of geometric guidance and sufficient clearance enables reliable autonomous bit exchange using only the existing gantry axes and screw motor. Not to mention that no fails were observed during the full system

validation test in the bit station. This deliverable is met.

Deliverable **D10**, operator logging and classification accuracy, and **D11**, overall operation failure rate, were evaluated across the 10 final full system runs in Section 5.2. Across 160 total operations the system recorded a true failure rate of 0.71% and a misclassification rate of 0.63%, both well within the 5% acceptance criteria defined for each. Both deliverables are met.

It should however be noted that the test objects were designed to validate the sensor logic, the control architecture, and the mechanical kinematics of the unified module and for that purpose they are appropriate. The results do demonstrate convincingly that the underlying sensing approach is sound, laser based contact and engagement detection, camera based localisation with homography correction, and force monitored pressing, each behaved consistently and predictably across repeated trials. The challenge of transferring this performance to real industrial disassembly targets remains open, and is identified as an important next step in Section 8.

7 Conclusions

This project set out to investigate whether a semi-automated gantry-based system, built around a unified module combining fastener removal and axial press operations in a single toolhead, could be made sufficiently reliable for batch-based disassembly. The results are strongly encouraging.

All eleven deliverables defined in Section 2.4 were met or substantially met. Camera-based screw localisation, after the introduction of homography calibration, achieved an RMS positional error of 0.247 mm with a maximum of 0.497 mm across all detected operations. Gantry repeatability was confirmed to within 0.09 mm across all three axes. Automated screw removal succeeded in every one of the twenty repeated trials. Passive bit changing reached a 100% success rate with the final station geometry. Across the ten consecutive final full-system runs, encompassing 160 operations in total, the true failure rate was 0.71% and the misclassification rate was 0.63% both comfortably within the 5% acceptance threshold.

The project must be assessed with appropriate honesty about its scope. Structural load testing was not performed, and all validation was carried out on purpose-built 3D-printed components with controlled screw torques and predictable press-fit resistances conditions considerably more favourable than those encountered in actual end-of-life industrial products, where surface corrosion, damaged fastener heads, unknown tightening torques, and geometric variability are the norm rather than the exception. Additionally, the mechanical limitations identified in section 6.1 mean that a focused redesign of the unified module is required before press operations can be validated at realistic industrial loads. Until the results of that redesign are available, a complete declaration of success for the unified module concept would be premature.

What this project does establish, clearly and with repeatable data, is that the chosen sensors, control architecture, and localisation pipeline each performed reliably within their tested scope, and the system demonstrated that the fundamental premise, a single gantry-mounted toolhead capable of both fastener removal and axial press operations, is behaviourally feasible. This project therefore represents a successful first step in validating this direction, providing a solid foundation from which the next iteration can be developed with confidence.

8 Outlook

The prototype demonstrates that the design concept is feasible, but several development paths remain before the system could be advanced toward production deployment. These are presented below in order of priority, beginning with the mechanical changes that address the limitations identified in Section 6.1, followed by validation, capability extensions, and finally improvements to the operational environment.

The most urgent area of future work is a coordinated mechanical redesign of the unified module to resolve the limitations described in Section 6.1. Five issues should be addressed in this redesign. **First**, the force sensor load path should be reworked so that the axial reaction is routed directly into a rigidly anchored structural member rather than through the arm connected to the top plate, eliminating the bending path and removing the load on the two carrying screws. **Second**, the driven shaft should be relocated to the geometric centre of the module, in order to eliminate the tilting moment introduced by the current 30 mm offset. **Third**, the contact circle at the top of the driven shaft should be manufactured with the originally intended 1 mm hexagonal recess to provide geometric keying against the hex tip of the driven shaft, replacing the friction press fit currently in use. **Fourth**, the bushings should be redesigned to remove the dependency on set screws for retention. The most direct solution is to source or machine a driven shaft that is not hardened, allowing the original H7/n6 interference fit on a turned round section to be implemented as originally planned. **Fifth**, the 3D printed herringbone gears should be replaced with machined metal versions, with the choice between herringbone and spur geometry revisited in light of the higher manufacturing cost of herringbone in metal. Since these changes touch nearly every element of the torque transmission path, the redesign should be approached as a single coordinated effort rather than as a sequence of independent fixes.

Instead of testing on purpose built test objects, future work should therefore include testing on representative industrial items drawn from the product families that the system is intended to disassemble. This would provide a realistic assessment of the camera based screw detection performance under varied surface finishes and lighting conditions, the reliability of the engagement detection logic on worn or partially corroded fastener heads, and the robustness of the press operations against component tolerances that differ from those of the test objects. The results would also inform the dimensioning of the redesigned module described above, since the actual breakaway torques and press forces encountered in industrial products are the loads the redesigned structure must be capable of transmitting.

The current vision system reliably estimates screw size, as described in Section 3.4. Full autonomous classification of bit type, distinguishing for example between Phillips, hex, slotted, and Torx heads across the full range of sizes the system is intended to handle, was outside the scope of this project but represents a natural extension. This would close the remaining gap between the prototype and the full system requirement of fully autonomous screw identification, and is a prerequisite for processing batches in which the fastener type is not known in advance. Implementation would likely involve revisiting the camera resolution and lighting setup, where the additional image detail required for confident head type discrimination would be necessary.

At present, the only possible indication that the screw motor has reached its practical torque limit is the failed pitch estimation due to no rise in the unscrewing process which therefore provides only a binary indication of whether the operation has succeeded or failed. It does not provide a continuous measured estimate of the torque required during a successful screw removal.

A useful improvement would be to add torque sensing to the screw motor drive system. The CL57T and NEMA 23 stepper motor combination already uses encoder feedback internally to detect position error and maintain closed loop motion, but the available driver outputs do not provide a direct measurement of the torque applied during screw removal. An external torque sensor could therefore be integrated into the driven shaft assembly as an additional measurement channel. This would enable direct measurement of the torque required to rotate the screw during operation, providing more accurate information about screw resistance, tightening condition, and potential failure or jamming events.

This would allow the system to collect more information during each removal attempt, rather than only recording whether the operation failed or succeeded. Over time, this data could be used to identify components that require unusually high removal torque. If such components are consistently damaged or unsuitable for reuse, the system could use the measured torque requirement as a decision parameter for whether disassembly is worthwhile. In addition, current based feedback could be used as a protective control input, allowing the rotation speed to be reduced when the motor approaches its practical torque limit.

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Appendices

A Sensor Calibration

A.1 Force Sensor Calibration

Purpose

The calibration was performed in order to verify the measurements of the force sensor and to obtain readings in newtons. This is necessary because each individual sensor exhibits slightly different output characteristics, meaning a standard factory calibration factor is not sufficient to rely on the sensor's output with confidence.

Methodology

The test was performed using an initial example calibration value of -1264.4 , which was used to convert the raw sensor output into force units during data collection. A Bowflex SelectTech 552i adjustable dumbbell, with 15 weight settings ranging from 2.3 kg to 23.8 kg, was used to apply known loads to the sensor. To verify the mass of each dumbbell setting, every weight configuration was weighed individually on a KERN HGN 100K200IP precision scale shown in figure 44. The scale has a resolution of 0.2 kg. The purpose of weighing the dumbbell settings was not to obtain highly precise reference masses, but rather to confirm that the dumbbell settings were not significantly off from their labelled values before the calibration began. In that context the scale resolution is sufficient, as even at the lowest setting of 2.3 kg, where the relative uncertainty is at its largest at approximately 8.7%, any gross discrepancy between the labelled and actual mass would have been clearly detected.



Figure 44: Control weighing of the dumbbell settings.

Each of the 15 available weight settings was applied to the force sensor five times, with the sensor tared before every individual measurement. During each measurement the dumbbell was placed as centrally as possible on the sensor surface to minimise off axis loading effects. This yielded a total of 75 data points covering the full range of the dumbbell, providing a thorough characterisation of the relationship between the sensor's raw output and the applied load in newtons.

Test Setup

A 3D printed fixture from a separate project was used to raise the force sensor off the table surface, allowing the dumbbell weights to rest solely on the sensor without contact with the table. A photograph of the test setup is shown in figure [45](#)



Figure 45: Picture of the setup used to calibrate the force sensor.

Raw Data

Table 10: Raw force sensor measurements.

Weight	Raw meas. 1	Raw meas. 2	Raw meas. 3	Raw meas. 4	Raw meas. 5	Abs. raw avg.
2	$-5,75 \cdot 10^3$	$-5,69 \cdot 10^3$	$-5,73 \cdot 10^3$	$-5,73 \cdot 10^3$	$-5,74 \cdot 10^3$	$5,74 \cdot 10^3$
3	$-8,89 \cdot 10^4$	$-8,89 \cdot 10^4$	$-8,88 \cdot 10^4$	$-8,89 \cdot 10^4$	$-8,88 \cdot 10^4$	$8,89 \cdot 10^4$
4	$-1,18 \cdot 10^5$	$-1,18 \cdot 10^5$	$-1,18 \cdot 10^5$	$-1,18 \cdot 10^5$	$-1,18 \cdot 10^5$	$1,18 \cdot 10^5$
5	$-1,49 \cdot 10^5$	$-1,49 \cdot 10^5$	$-1,49 \cdot 10^5$	$-1,49 \cdot 10^5$	$-1,49 \cdot 10^5$	$1,49 \cdot 10^5$
7	$-1,77 \cdot 10^5$	$-1,77 \cdot 10^5$	$-1,77 \cdot 10^5$	$-1,77 \cdot 10^5$	$-1,77 \cdot 10^5$	$1,77 \cdot 10^5$
8	$-2,09 \cdot 10^5$	$-2,09 \cdot 10^5$	$-2,08 \cdot 10^5$	$-2,08 \cdot 10^5$	$-2,09 \cdot 10^5$	$2,09 \cdot 10^5$
9	$-2,38 \cdot 10^5$	$-2,37 \cdot 10^5$	$-2,38 \cdot 10^5$	$-2,37 \cdot 10^5$	$-2,38 \cdot 10^5$	$2,38 \cdot 10^5$
10	$-2,69 \cdot 10^5$	$-2,70 \cdot 10^5$	$-2,70 \cdot 10^5$	$-2,69 \cdot 10^5$	$-2,69 \cdot 10^5$	$2,70 \cdot 10^5$
11	$-2,99 \cdot 10^5$	$-2,99 \cdot 10^5$	$-2,99 \cdot 10^5$	$-2,98 \cdot 10^5$	$-2,99 \cdot 10^5$	$2,99 \cdot 10^5$
14	$-3,59 \cdot 10^5$	$-3,58 \cdot 10^5$	$-3,58 \cdot 10^5$	$-3,58 \cdot 10^5$	$-3,59 \cdot 10^5$	$3,59 \cdot 10^5$
16	$-4,19 \cdot 10^5$	$-4,19 \cdot 10^5$	$-4,18 \cdot 10^5$	$-4,19 \cdot 10^5$	$-4,19 \cdot 10^5$	$4,19 \cdot 10^5$
18	$-4,79 \cdot 10^5$	$-4,78 \cdot 10^5$	$-4,79 \cdot 10^5$	$-4,79 \cdot 10^5$	$-4,79 \cdot 10^5$	$4,79 \cdot 10^5$
20	$-5,38 \cdot 10^5$	$-5,39 \cdot 10^5$	$-5,38 \cdot 10^5$	$-5,38 \cdot 10^5$	$-5,38 \cdot 10^5$	$5,38 \cdot 10^5$
23	$-5,98 \cdot 10^5$	$-5,99 \cdot 10^5$	$-5,99 \cdot 10^5$	$-5,99 \cdot 10^5$	$-5,99 \cdot 10^5$	$5,99 \cdot 10^5$
24	$-6,29 \cdot 10^5$	$-6,30 \cdot 10^5$	$-6,30 \cdot 10^5$	$-6,30 \cdot 10^5$	$-6,30 \cdot 10^5$	$6,30 \cdot 10^5$

Data Processing

The raw sensor output was recorded while the sensor was operating under the example calibration value of -1264.4 . Because the sensor was tared before each measurement, the offset term in the linear model is physically guaranteed to be zero at zero applied mass a tared sensor reads exactly zero by definition. The fit is therefore constrained to pass through the origin, and the relationship between the raw output y_{raw} and the applied mass m is modelled as

$$y_{\text{raw}} = a_1 \cdot m. \quad (1)$$

The five repeated measurements at each weight setting were averaged to reduce noise, and the absolute value of the product of the average raw output and the example calibration value was computed to recover the underlying raw sensor signal:

$$\bar{y}_{\text{abs}} = |\bar{y}_{\text{avg}} \cdot (-1264.4)|. \quad (2)$$

An origin constrained least squares fit was applied to the true mass versus $\bar{y}_{\text{abs}}$ data, computed as

$$a_1 = \frac{\sum_i m_i \bar{y}_{\text{abs},i}}{\sum_i m_i^2} = 26860.97, \quad (3)$$

yielding the calibration slope a_1 . Figure 46 shows the resulting fit alongside the averaged measurements.

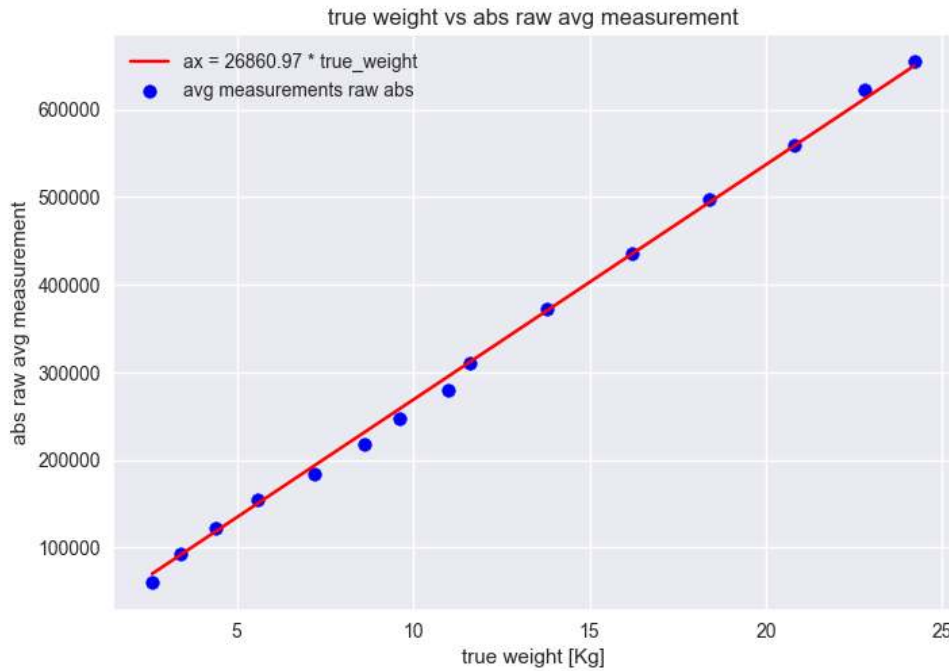


Figure 46: Absolute average raw sensor output as a function of true applied mass. The red line shows the origin constrained fit with slope $a_1 = 26860.97$.

To assess the accuracy of the initial example calibration value of -1264.4 , the expected force in newtons was computed from the true mass as $F_{\text{expected}} = m \cdot g$ where $g = 9.81 \text{ m/s}^2$, and compared against the sensor output obtained using that example value. An origin constrained fit of the form

$$F_{\text{measured}} = a_2 \cdot F_{\text{expected}} \quad (4)$$

yielded a slope of $a_2 = 2.17$. A slope of unity would indicate a perfect calibration, so a value of 2.17 confirms that the example value of -1264.4 was substantially incorrect, producing readings approximately 2.17 times larger than the true applied force. Figure 47 shows this comparison.

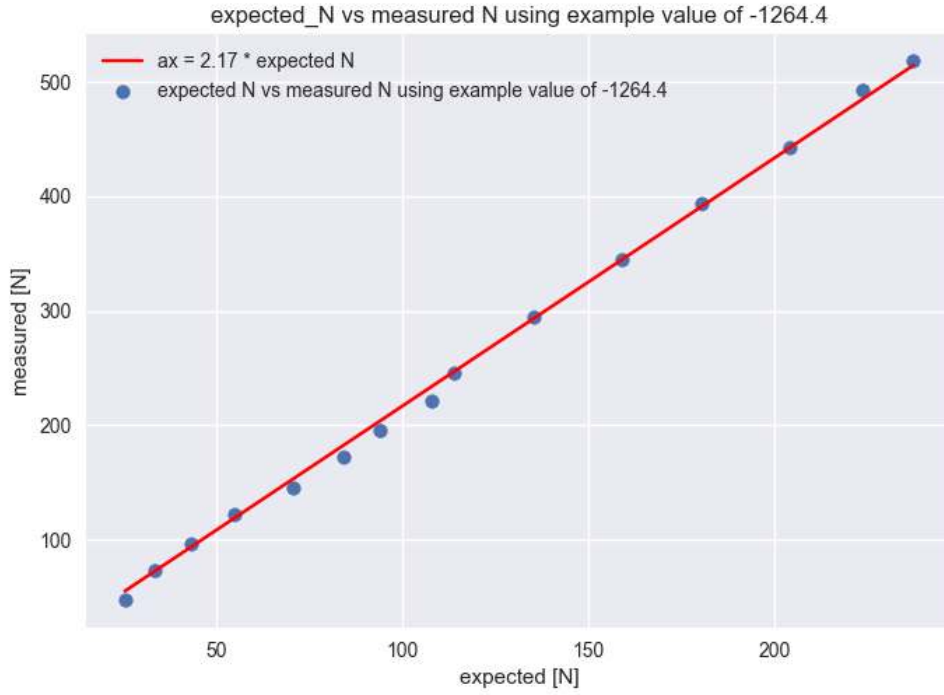


Figure 47: Expected force in newtons versus sensor output using the example calibration value of -1264.4 . The fitted slope of $a_2 = 2.17$ shows the example value overestimates the true force by a factor of approximately 2.17.

The correct calibration slope $a_1 = 26860.97$ was then written into the firmware. An independent verification test was performed by placing each of the 15 weight settings on the sensor with the corrected calibration value active, collecting a fresh set of measurements not used in the derivation of a_1 . The resulting sensor output is plotted against the expected force in Figure 48, where the red line represents the ideal one to one relationship. The residuals between this independent measurement and the expected force were computed as

$$\epsilon_i = F_{\text{measured},i} - F_{\text{expected},i}, \quad (5)$$

yielding a mean residual of $\bar{\epsilon} = -0.72$ N and a standard deviation of $\sigma_\epsilon = 2.75$ N.

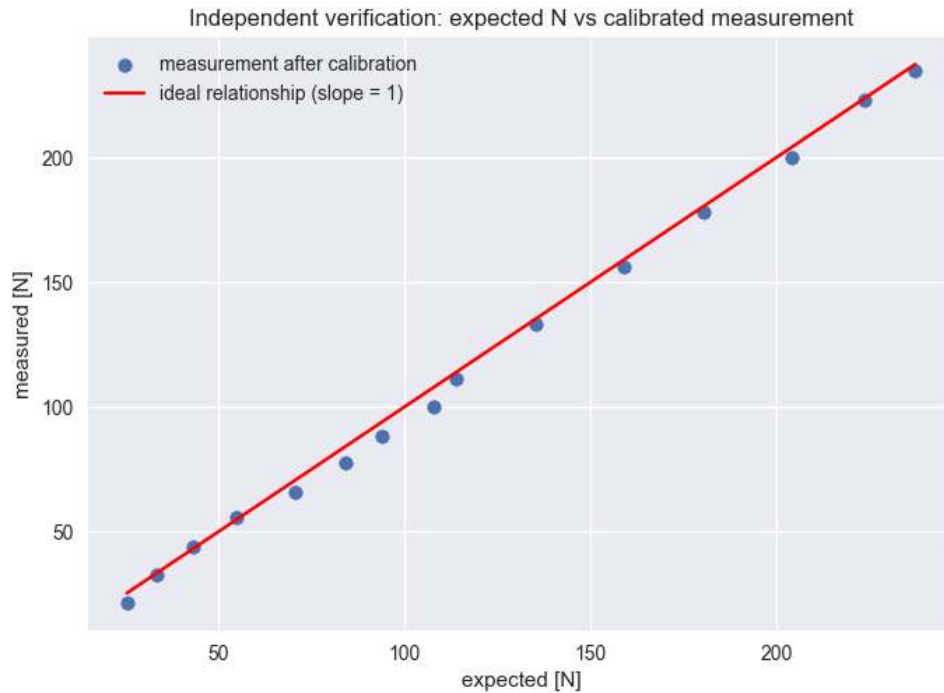


Figure 48: Independent verification of the force sensor calibration. Each weight setting was applied once with the correct calibration value active in the firmware. The red line shows the ideal one to one relationship.

Conclusion

The goal of the force sensor calibration was to obtain a reliable estimate of the force applied by the module during operation, and to establish a threshold at which the tool should cease pressing to avoid damaging itself or the object being dismantled. The origin constrained fit yielded a calibration slope of $a_1 = 26860.97$. Independent verification against fresh measurements gave a mean residual of $\bar{\epsilon} = -0.72$ N and a standard deviation of $\sigma_{\epsilon} = 2.75$ N, confirming that the sensor output closely tracks the true applied force across the full measurement range. The calibration is considered successful.

A.2 Laser Sensor Calibration

Purpose

A large part of the module's positional precision relies on the accuracy of the laser distance sensor. It is therefore essential to ensure that the raw voltage output read from the sensor is converted correctly into millimetres. This section describes the calibration procedure carried out to establish this conversion.

Test Setup

The laser sensor operates within a measurement range of 40 mm to 60 mm, measured from the sensor's glass plate to the target surface. To calibrate it, the sensor was mounted on the gantry and two reference boxes of known height were placed beneath it. The initial standoff distance was set just outside the lower bound of the sensor's range at 39 mm. The z axis was then stepped away from the boxes until a standoff distance of 61 mm was reached, covering the full operating range of the sensor with a small margin on either side. The reference heights were verified using a calibre, measured multiple times to reduce uncertainty. During the experiment the laser voltage output was recorded by the Arduino Mega board at every 0.5 mm step of the z axis, yielding 36 data points in total. A photograph of the test setup is shown in Figure 49.

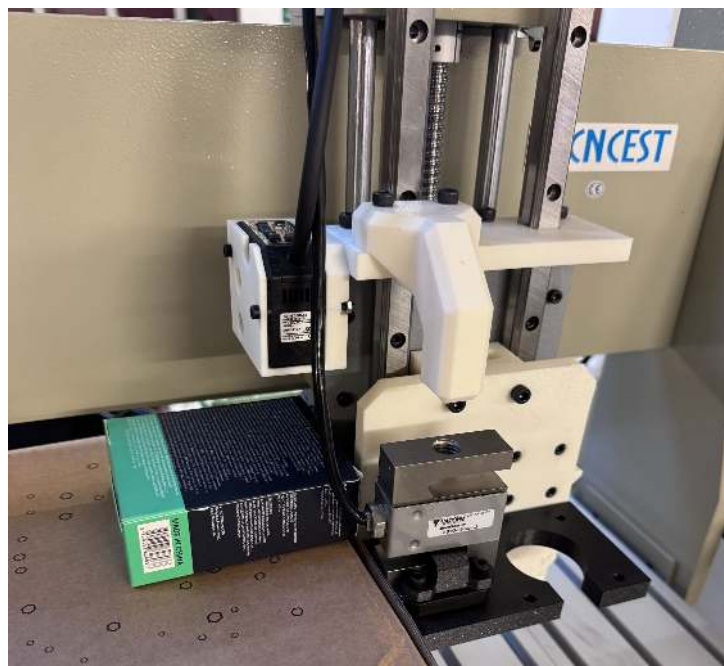


Figure 49: Test setup for the laser sensor calibration, showing the sensor mounted on the gantry above the two reference boxes.

Raw Data

Table 11: Laser sensor calibration measurements.

Distance	Raw avg.	Arduino V	Box V
43,5	936,70	4,5782	5,4938
44,0	917,15	4,4826	5,3792
44,5	898,35	4,3908	5,2689
45,0	878,80	4,2952	5,1543
45,5	858,50	4,1960	5,0352
46,0	838,90	4,1002	4,9202
46,5	818,85	4,0022	4,8026
47,0	799,40	3,9071	4,6886
47,5	779,65	3,8106	4,5727
48,0	759,95	3,7143	4,4572
48,5	740,05	3,6171	4,3405
49,0	720,80	3,5230	4,2276
49,5	701,25	3,4274	4,1129
50,0	681,95	3,3331	3,9997
50,5	661,10	3,2312	3,8774
51,0	641,55	3,1356	3,7628
51,5	622,70	3,0435	3,6522
52,0	603,10	2,9477	3,5372
52,5	582,60	2,8475	3,4170
53,0	563,55	2,7544	3,3053
53,5	545,05	2,6640	3,1968
54,0	525,15	2,5667	3,0801
54,5	506,40	2,4751	2,9701
55,0	485,60	2,3734	2,8481
55,5	464,75	2,2715	2,7258
56,0	446,75	2,1835	2,6202
56,5	425,05	2,0775	2,4930
57,0	406,05	1,9846	2,3815
57,5	386,90	1,8910	2,2692
58,0	366,30	1,7903	2,1484
58,5	347,55	1,6987	2,0384
59,0	328,40	1,6051	1,9261
59,5	308,85	1,5095	1,8114
60,0	289,15	1,4132	1,6959
60,5	268,50	1,3123	1,5748
61,0	251,35	1,2285	1,4742

Data Processing

A first order polynomial was fitted to the known distance versus measured voltage data using `numpy.polyfit`, establishing the linear relationship

$$V = a \cdot d + b, \quad (6)$$

where V is the voltage measured at the Arduino analog input in volts, d is the distance in millimetres, and a and b are the fitted slope and intercept respectively. Unlike the force sensor, the laser sensor is not tared before use and operates over an absolute distance range, so the intercept b is physically meaningful and is retained in the fit. The fitted coefficients were found to be

$$a = -0.19 \text{ V/mm}, \quad (7)$$

$$b = 12.93 \text{ V}. \quad (8)$$

The inverse of Equation 6,

$$d = \frac{V - b}{a} = \frac{V - 12.93}{-0.19}, \quad (9)$$

is used in the firmware to convert a measured voltage into a distance reading in real time. Figure 50 shows the calibration data together with the fitted line.

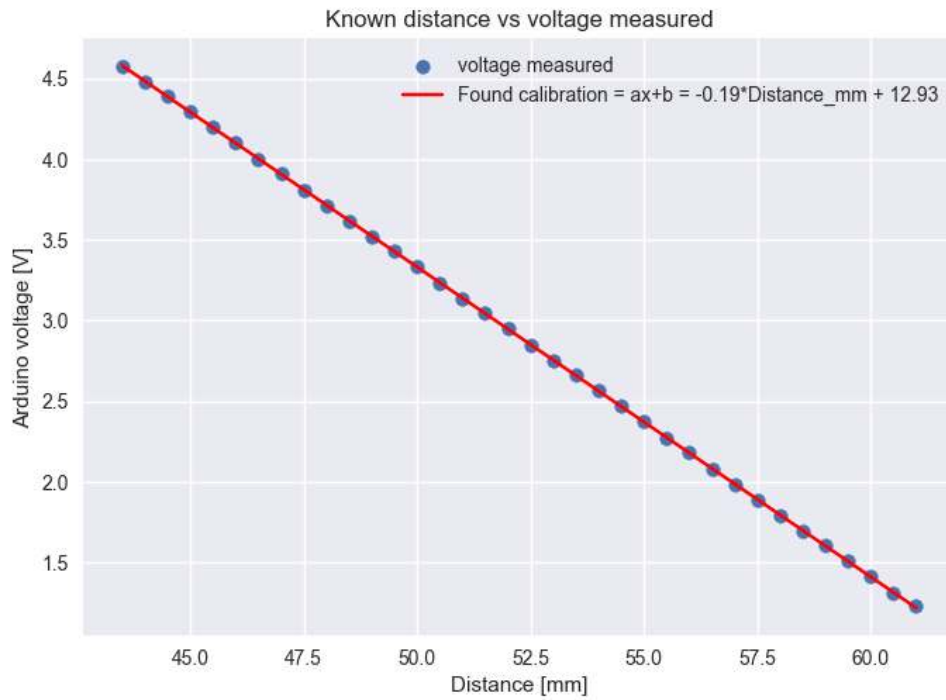


Figure 50: Voltage measured at the Arduino analog input as a function of known distance. The red line shows the first order polynomial fit with slope $a = -0.19 \text{ V/mm}$ and intercept $b = 12.93 \text{ V}$.

Conclusion

Through a linear regression of the measured voltage against the known standoff distance, the conversion coefficients $a = -0.19 \text{ V/mm}$ and $b = 12.93 \text{ V}$ were determined. The negative slope is physically consistent with the sensor's behaviour voltage decreases as distance increases. These values have been embedded in the firmware to translate raw analog readings into distance measurements within the sensor's operating window of 40 mm to 60 mm, enabling accurate positional feedback for the module during operation.

A.3 Camera Calibration

Purpose of Camera Calibration

The vision system was implemented to enable automated localization and identification of screws and operation targets within the workspace. In order for image-based detections to be usable by the gantry system, the relationship between image coordinates and physical workspace coordinates must be established through a camera calibration procedure. Without calibration, lens distortion, perspective effects, and unknown image scaling would introduce positional inaccuracies that reduce the precision of the automated screw engagement and press-operation processes.

The purpose of the calibration procedure was therefore to estimate the intrinsic camera parameters, correct lens distortion, establish pixel-to-millimeter scaling relationships, and relate image coordinates to the physical coordinate system of the machine. The calibrated camera model was subsequently used throughout the image processing pipeline for geometric interpretation, screw localization, and conversion from detected image features to gantry coordinates.

Since the camera is only used at a single Z-axis position during normal operation, the calibration procedure was focused on this operational height. The Z-axis is kept fully retracted during all image acquisitions, providing a fixed working distance and a consistent field of view over the workspace.

Intrinsic Camera Calibration

The intrinsic camera parameters were estimated using an OpenCV-based checkerboard calibration procedure. Multiple images of a checkerboard pattern were captured at different positions and orientations within the field of view, allowing estimation of the camera matrix and lens distortion coefficients based on the known checkerboard geometry.

The calibration process was used to estimate parameters including focal length, principal point, radial distortion coefficients, and tangential distortion coefficients. The obtained intrinsic parameters were subsequently used to undistort captured images prior to further image processing and object localization. All geometric measurements and coordinate conversions were therefore performed on undistorted images.

Manual Focus Calibration

Since image sharpness significantly affects feature detection accuracy, manual focus calibration was performed at the operational camera height used within the system, with the Z-axis fully retracted. Automatic focus control was disabled during operation in order to ensure repeatable image characteristics and avoid focus changes between captured images. A range of manual focus values was experimentally evaluated by capturing images at different focus settings. The captured images were then compared visually based on image sharpness, edge clarity, and feature visibility. The focus value that provided the clearest image at the operational height was selected for use during system operation. The selected manual focus value was:

$$F = 4.5 \quad (10)$$

Figure 51 shows examples of captured images obtained using different manual focus settings at the operational scanning height.

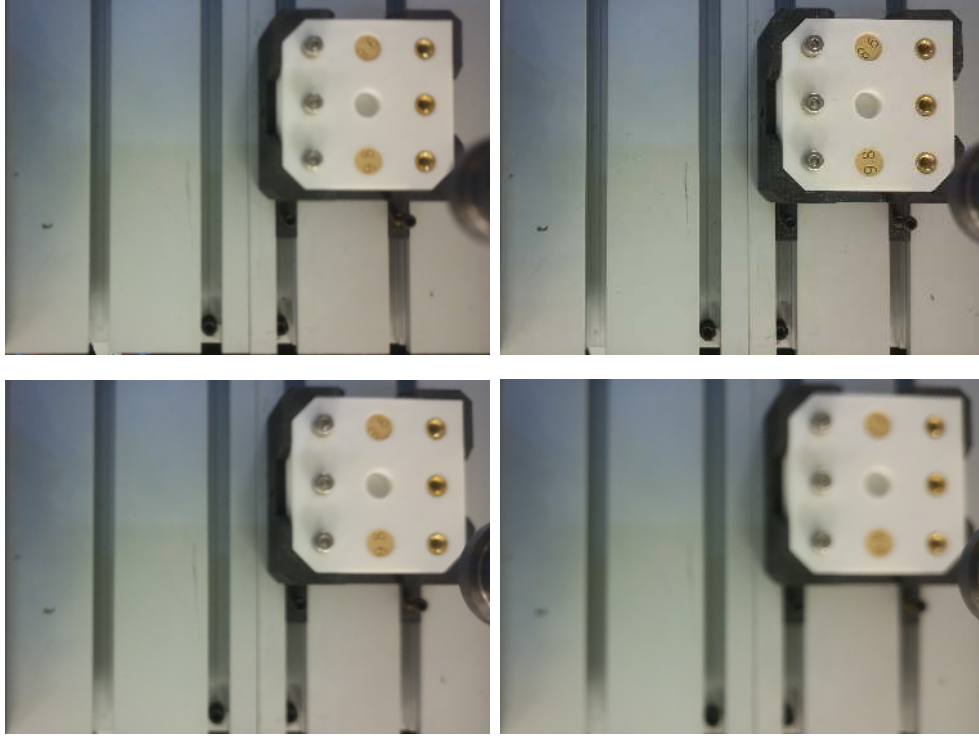


Figure 51: Comparison of captured images obtained using different manual focus settings at the operational scanning height.

A.3.1 Pixel-to-Millimeter Calibration

Following intrinsic calibration, a pixel-to-millimeter calibration procedure was performed to establish the physical scaling relationship between image coordinates and real-world dimensions. The calibration was performed at the single operational height used during system operation, with the Z-axis fully retracted.

The pixel-to-millimeter calibration was performed using a checkerboard pattern with a known square size of 20.0 mm. The average corner spacing measured across the checkerboard in both image directions was used to derive the scaling factors. Separate scaling factors were determined for the x - and y -directions in order to account for possible differences in image scaling between the two image axes.

At the operational height, the obtained scaling factors were:

$$S_x = \frac{20.0 \text{ mm}}{365.517 \text{ pixels}} = 0.054717 \text{ mm/pixel} \quad (11)$$

$$S_y = \frac{20.0 \text{ mm}}{365.569 \text{ pixels}} = 0.054709 \text{ mm/pixel} \quad (12)$$

These scaling factors were subsequently used to convert detected image coordinates into corresponding physical workspace coordinates during system operation. Figure 52 illustrates the pixel-to-millimeter calibration procedure.

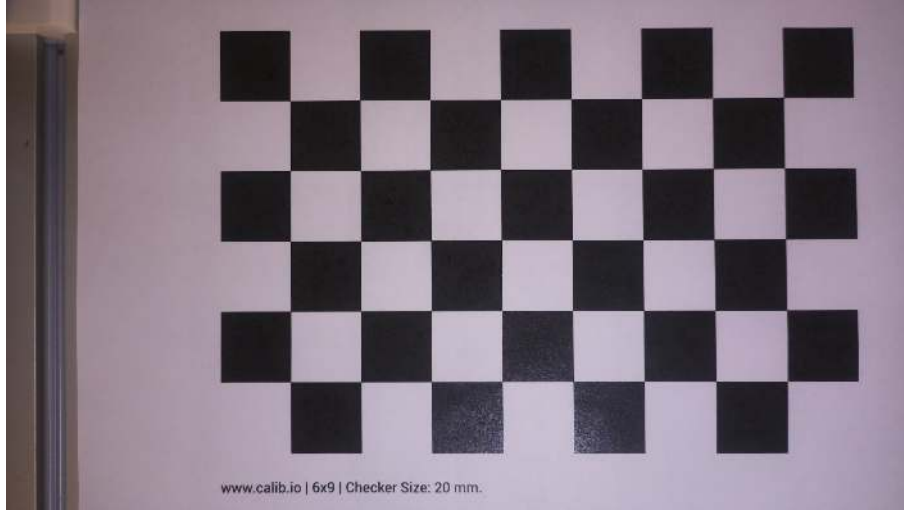


Figure 52: Pixel-to-millimeter calibration using known checkerboard reference dimensions at the operational scanning height.

A.3.2 Coordinate Relationship Between Camera and Machine Coordinates

Following the intrinsic and scaling calibration procedures, the image coordinate system could be related to the machine coordinate system. The camera was rigidly mounted relative to the toolhead in order to maintain a constant geometric relationship between the camera coordinate system and the machine coordinate system throughout operation.

Detected image coordinates could therefore be converted into corresponding physical coordinates by combining the pixel displacement relative to the image center, the pixel-to-millimeter scaling factors, the known camera offset, and the current gantry position. This established the geometric relationship required for transforming detected image features into physical workspace coordinates usable by the positioning system.

If a feature is detected at image coordinates (u, v) , its displacement from the image center can be expressed as:

$$\Delta u = u - u_0 \quad (13)$$

$$\Delta v = v - v_0 \quad (14)$$

where (u_0, v_0) is the image center. The corresponding physical displacement in the camera frame can then be estimated using the scaling factors:

$$\Delta x_{\text{cam}} = \Delta u S_x \quad (15)$$

$$\Delta y_{\text{cam}} = \Delta v S_y \quad (16)$$

The detected feature position in the gantry coordinate system can then be estimated as:

$$x_{\text{target}} = x_{\text{gantry}} + x_{\text{offset}} + \Delta x_{\text{cam}} \quad (17)$$

$$y_{\text{target}} = y_{\text{gantry}} + y_{\text{offset}} + \Delta y_{\text{cam}} \quad (18)$$

where x_{gantry} and y_{gantry} are the known gantry coordinates at the time of image acquisition, and x_{offset} and y_{offset} are the fixed offsets between the camera center and the toolhead reference point.

Since all workspace images are acquired at predefined gantry positions with known machine coordinates, localized image features can be transformed into global machine coordinates after conversion from pixels to physical distances. This allows detected screw locations and operation targets to be referenced directly within the gantry coordinate system used by the positioning controller.

B Technical Drawings

B.1 Housing

This Appendix contains supplementary technical drawings and manufacturing documentation referenced throughout the report.

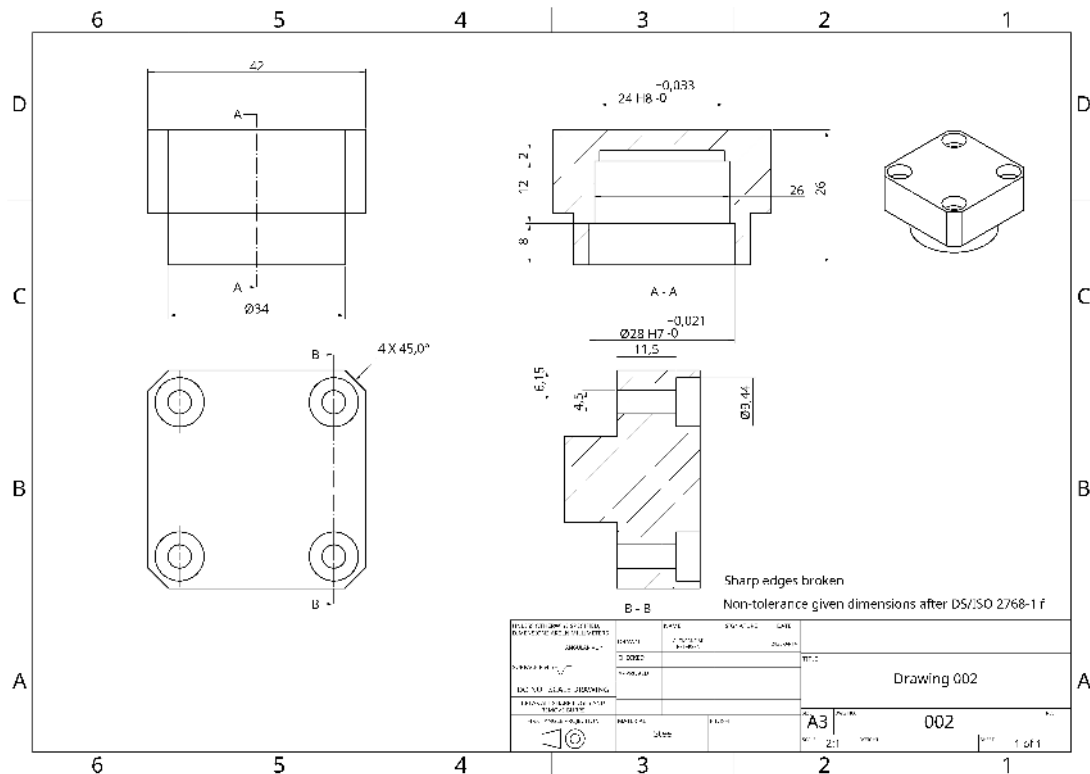


Figure 53: Technical drawing of the driven shaft housing component used in the driven shaft assembly. The component was manufactured using a CNC machine at the DTU workshop in Building 427.

B.2 Bit Adapter

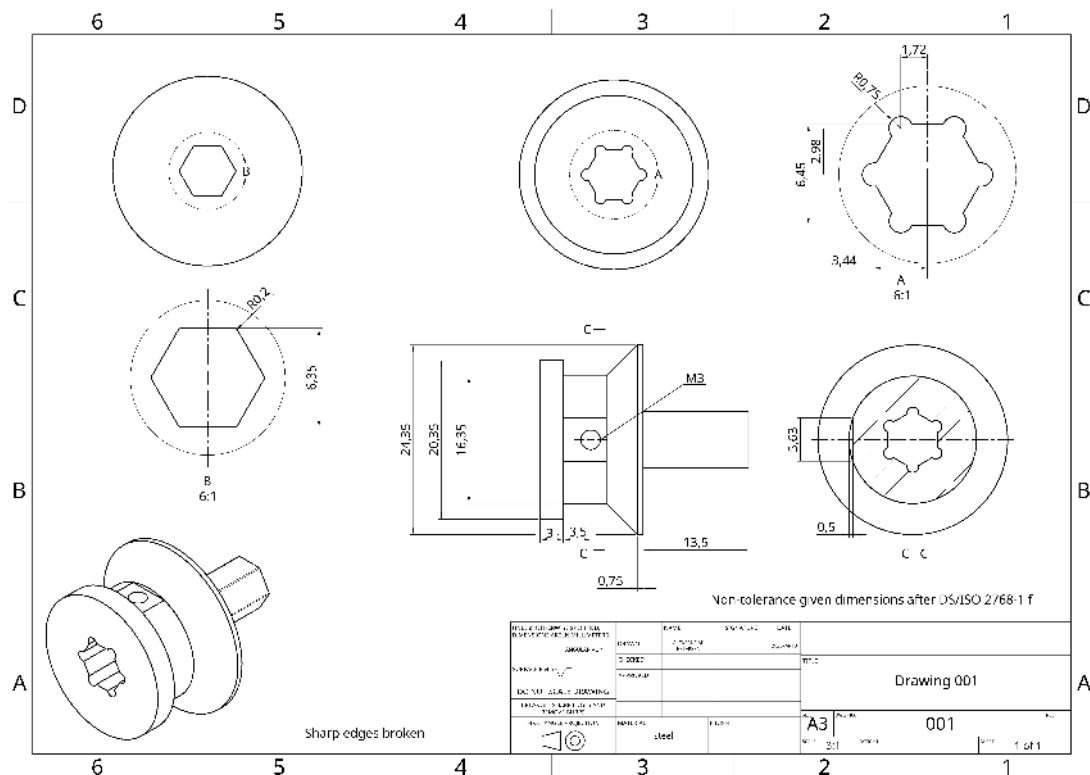


Figure 54: Technical drawing of the custom CNC-machined bit adapter used for passive tool retention and exchange within the bit changer station. The drawing was used for workshop manufacturing of the component.

C Flow-Charts

C.1 Unscrew Flow-chart

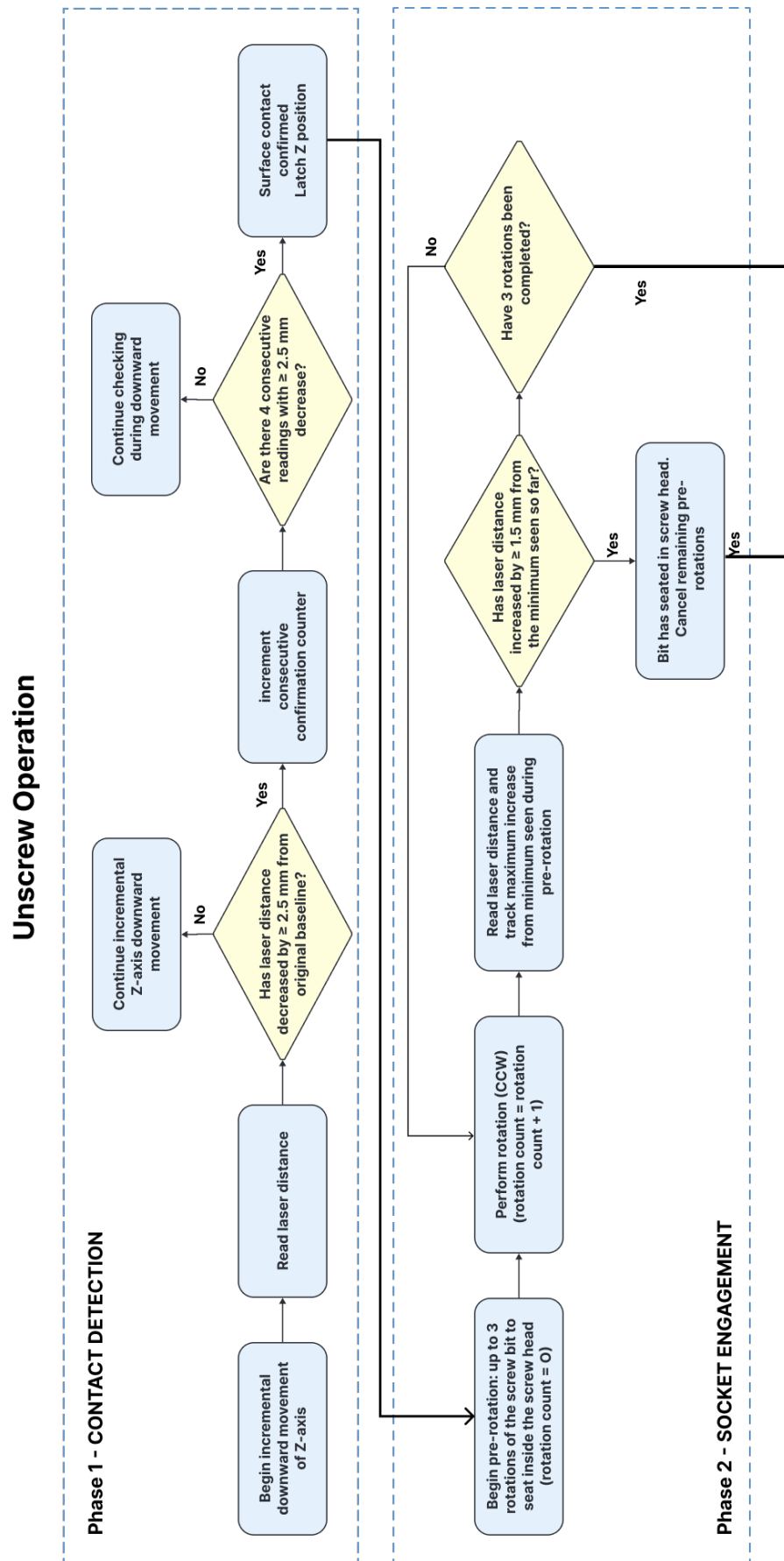


Figure 55: Upper section of the automated unscrewing logic flowchart.

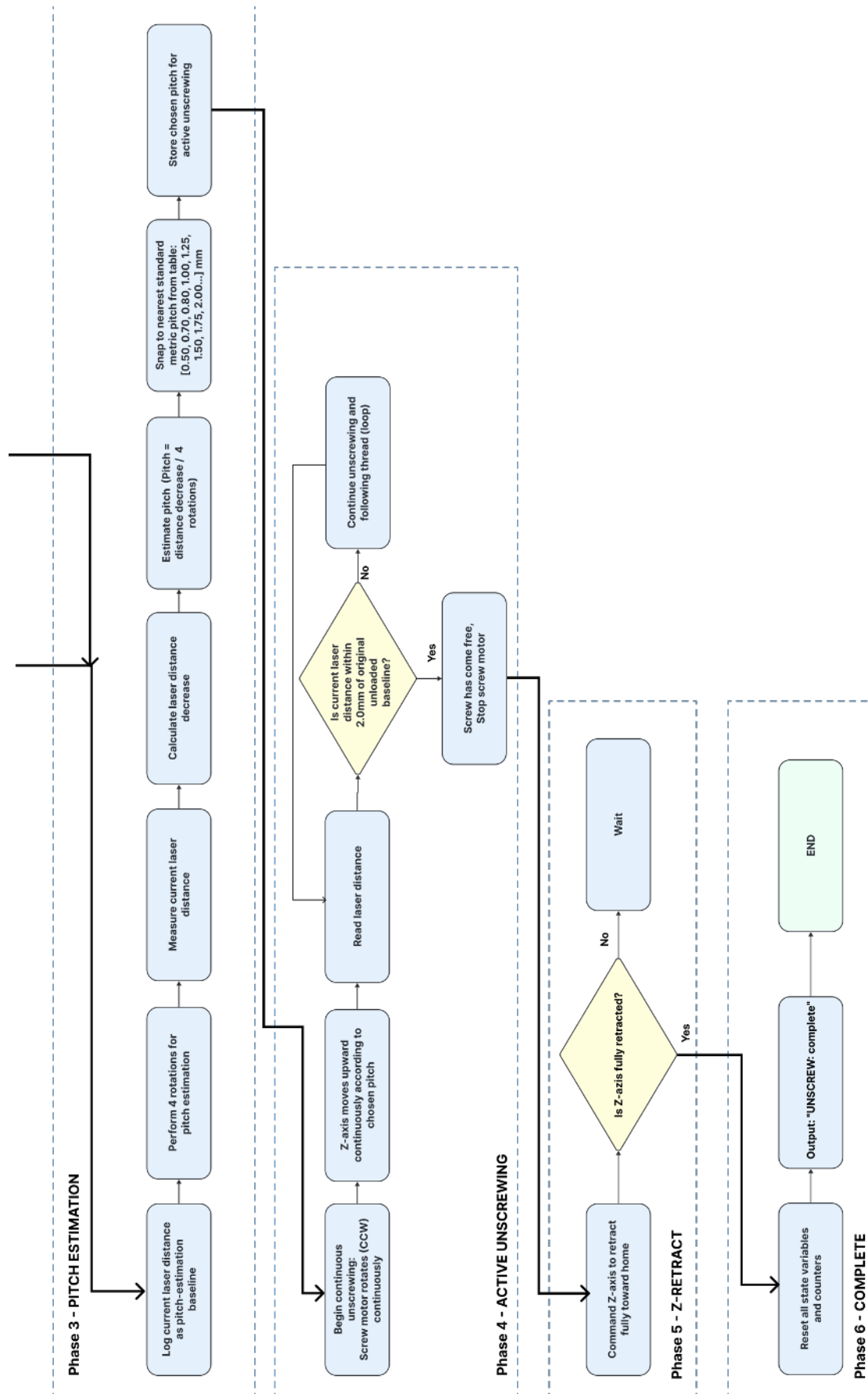


Figure 56: Lower section of the automated unscrewing logic flowchart.

C.2 Vision Flow-Chart

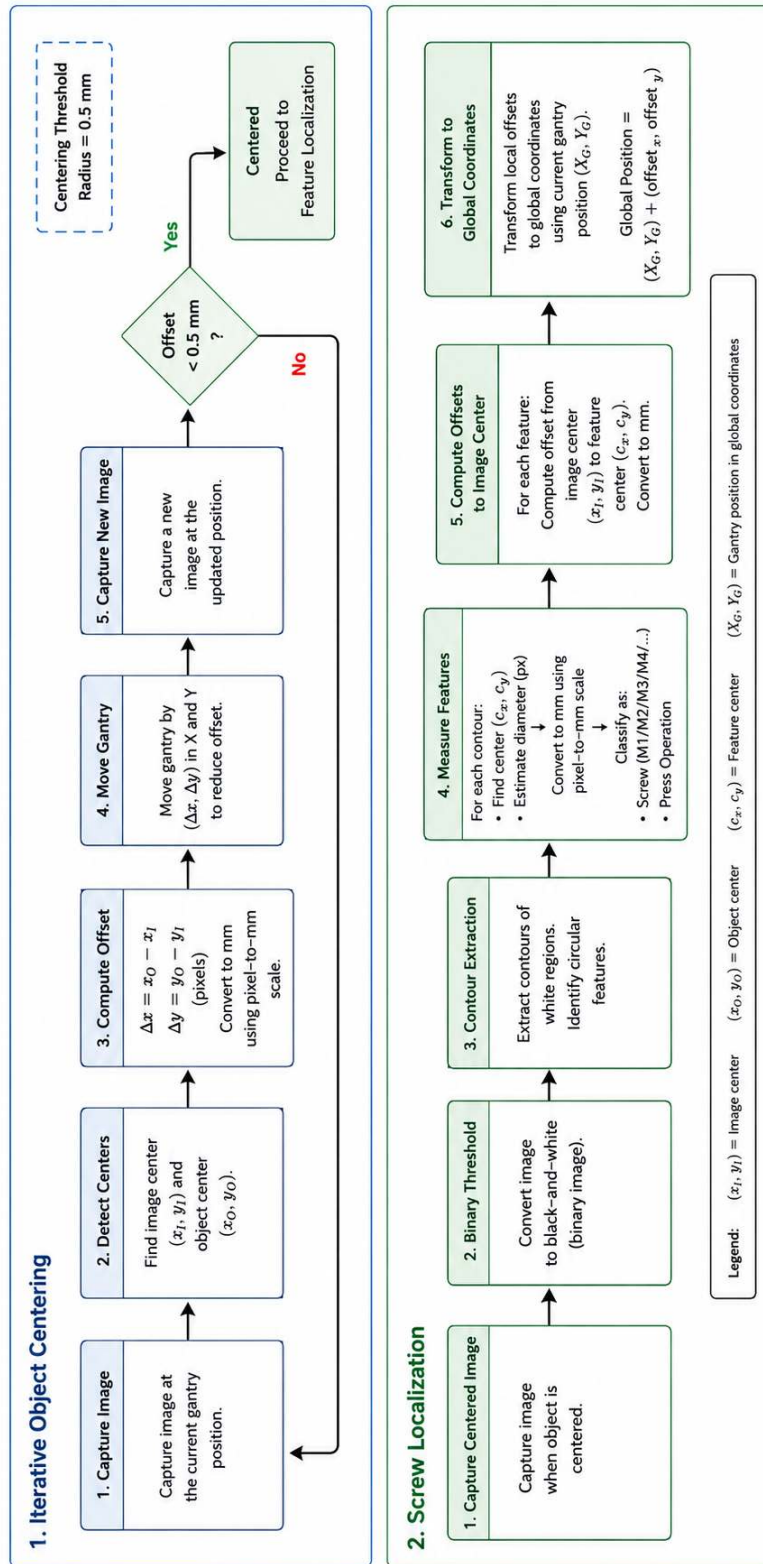


Figure 57: Overview of the vision based localization pipeline used by the system. The upper section illustrates the iterative object centering loop, while the lower section illustrates the screw and press operation localization process used to determine global feature coordinates.

D Additional CAD renderings

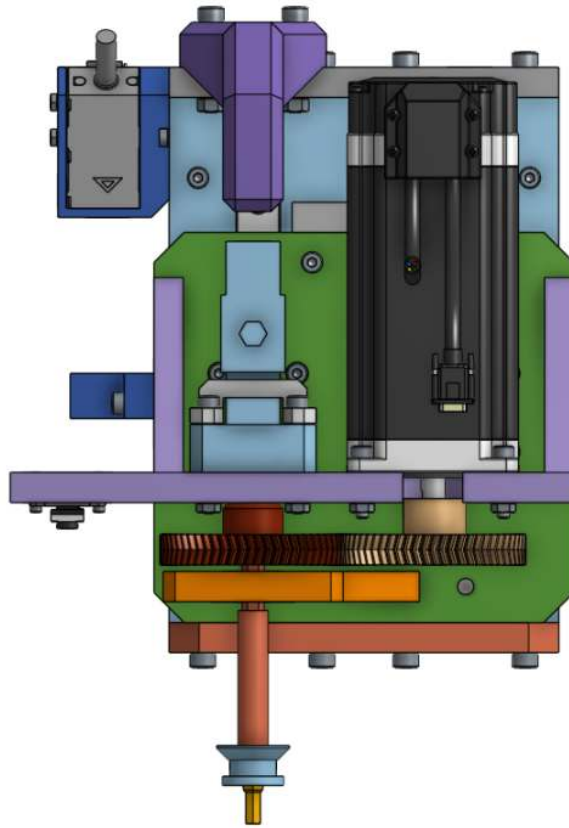


Figure 58: CAD render of the frontal view of the main assembly.

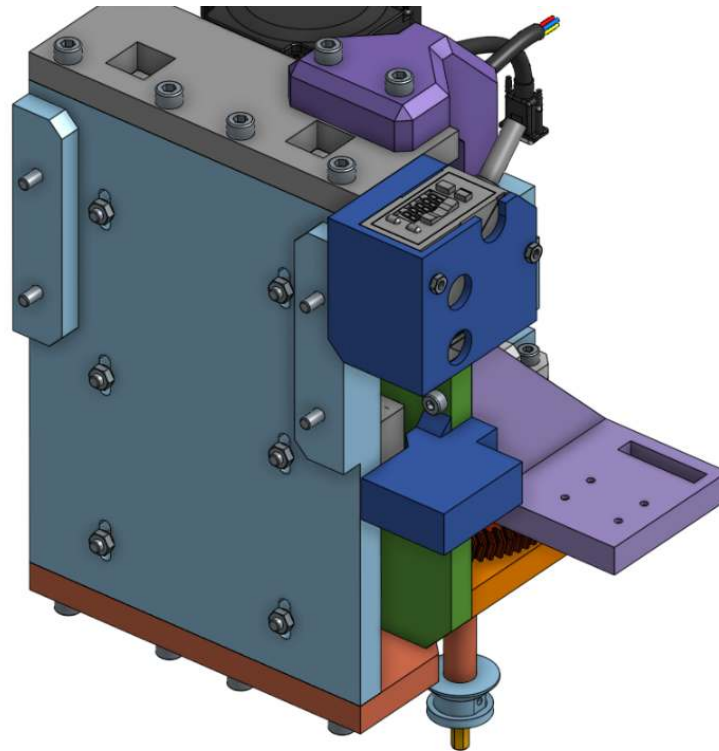


Figure 59: Main assembly viewed from a back angle.

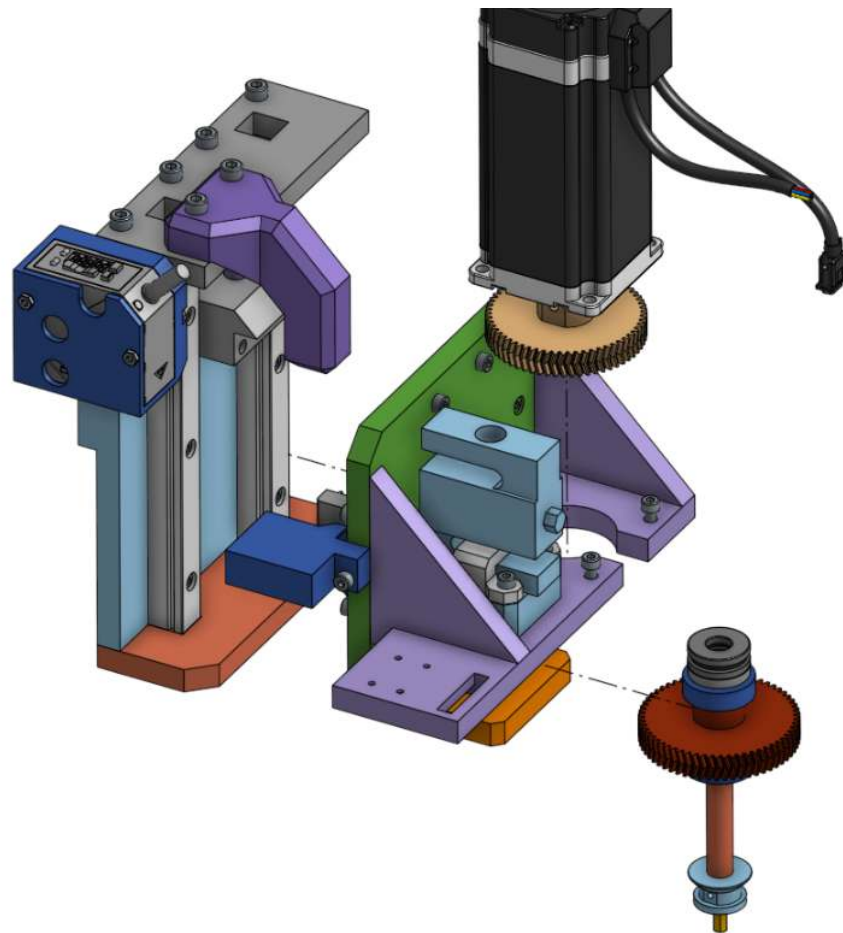


Figure 60: Exploded view unannotated.

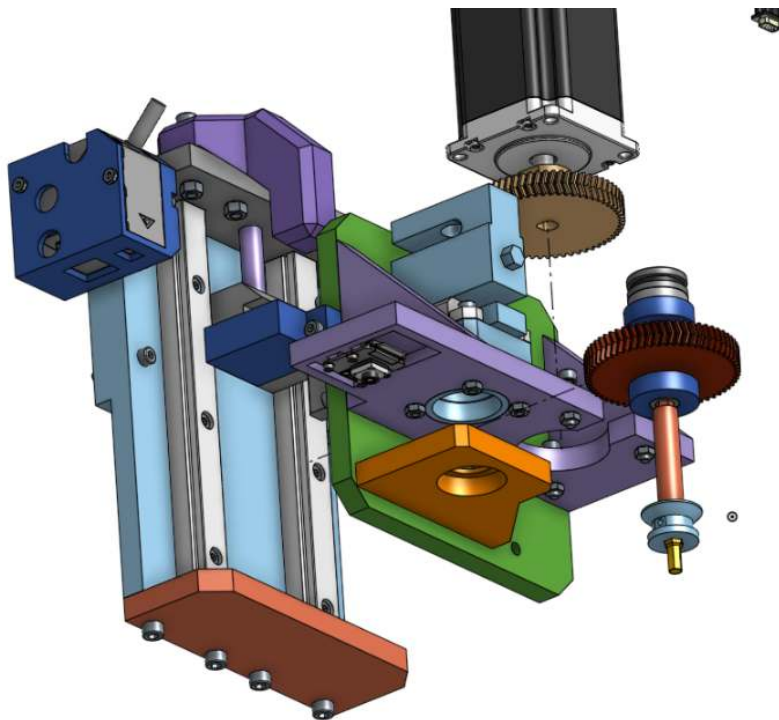


Figure 61: Assembly viewed from an angle below.

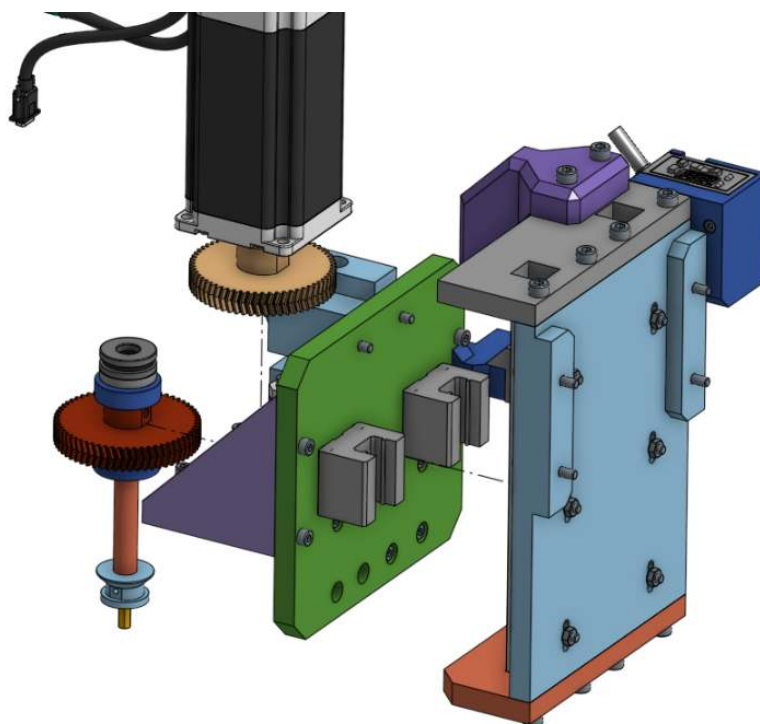


Figure 62: Assembly viewed from a back angle, showing the two sleds used.

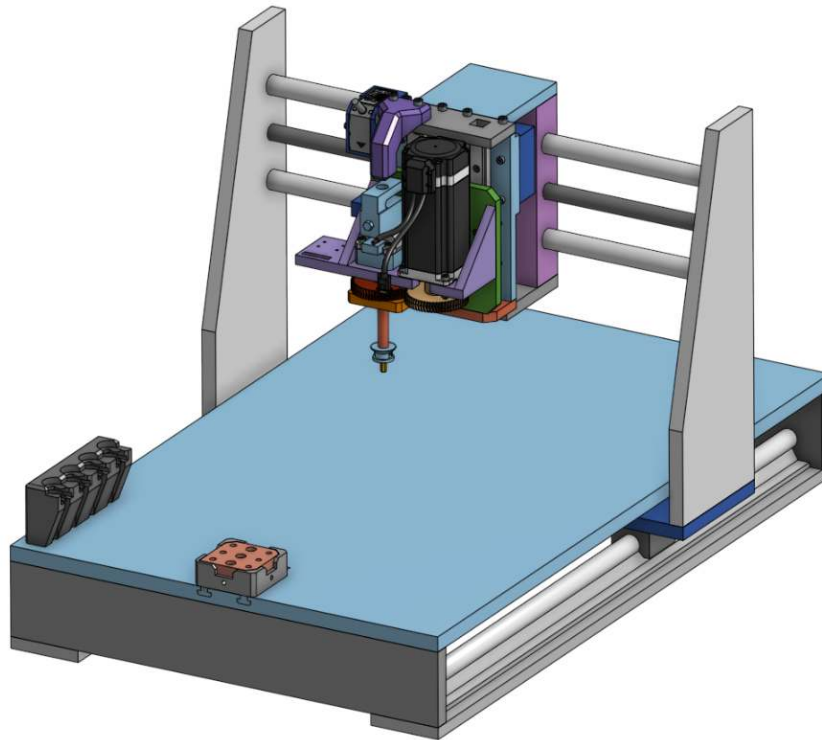
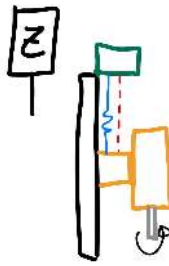


Figure 63: Full Assembly, including test objects and bit station

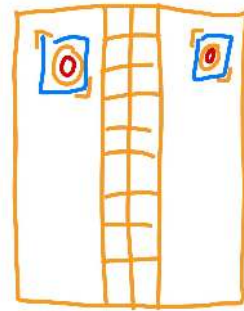
E Sketches from the design phase



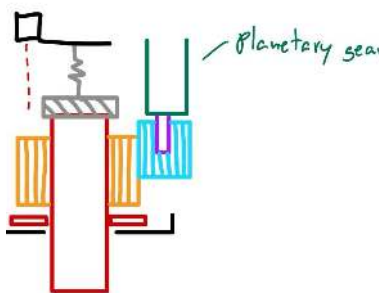
(a) Sketch 1



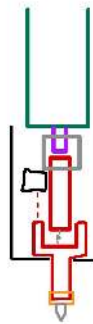
(b) Sketch 2



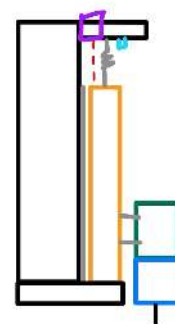
(c) Sketch 3



(d) Sketch 4



(e) Sketch 5



(f) Sketch 6

Figure 64: Initial sketches of the test object and unified module design.

